

Putnam 1985–2025

知识顺序试题集

按数学系学习路径重排 范畴标注 年份索引

范畴为章，题目成册，便于循序训练。

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编排说明

本试题集按照“完成题目通常需要先掌握的主要知识”从低阶到高阶编排。Putnam 题往往跨越多个领域；本书中每题所列“范畴”和“入口”取最自然的解题起点，而不是唯一可能的解法标签。

每章对应一个知识范畴；题目标题保留原始年份和 A/B 题号。书末附按年份索引，便于从原卷回查到本书中的范畴位置。

第 1 章

基础证明、集合与初等逻辑

范畴要点：集合、函数、基础构造、抽屉原理、递推入门

2004 年 A1 · 基础证明、集合与初等逻辑

年份 2004 题号 A1 入口集合

Basketball star Shanille O'Keal's team statistician keeps track of the number, $S(N)$, of successful free throws she has made in her first N attempts of the season. Early in the season, $S(N)$ was less than 80% of N , but by the end of the season, $S(N)$ was more than 80% of N . Was there necessarily a moment in between when $S(N)$ was exactly 80% of N ?

1986 年 A2 · 基础证明、集合与初等逻辑

年份 1986 题号 A2 入口集合

What is the units (i.e., rightmost) digit of

$$\left\lfloor \frac{10^{20000}}{10^{100} + 3} \right\rfloor ?$$

1996 年 A3 · 基础证明、集合与初等逻辑

年份 1996 题号 A3 入口集合

Suppose that each of 20 students has made a choice of anywhere from 0 to 6 courses from a total of 6 courses offered. Prove or disprove: there are 5 students and 2 courses such that all 5 have chosen both courses or all 5 have chosen neither course.

2014 年 A3 · 基础证明、集合与初等逻辑

年份 2014 题号 A3 入口集合

Let $a_0 = 5/2$ and $a_k = a_{k-1}^2 - 2$ for $k \geq 1$. Compute

$$\prod_{k=0}^{\infty} \left(1 - \frac{1}{a_k} \right)$$

in closed form.

2009 年 A4 · 基础证明、集合与初等逻辑

年份 2009 题号 A4 入口集合/构造

Let S be a set of rational numbers such that

- (a) $0 \in S$;
- (b) If $x \in S$ then $x + 1 \in S$ and $x - 1 \in S$; and
- (c) If $x \in S$ and $x \notin \{0, 1\}$, then $\frac{1}{x(x-1)} \in S$.

Must S contain all rational numbers?**1996 年 B3 · 基础证明、集合与初等逻辑**

年份 1996 题号 B3 入口函数与不等式

Given that $\{x_1, x_2, \dots, x_n\} = \{1, 2, \dots, n\}$, find, with proof, the largest possible value, as a function of n (with $n \geq 2$), of

$$x_1x_2 + x_2x_3 + \cdots + x_{n-1}x_n + x_nx_1.$$

2001 年 B4 · 基础证明、集合与初等逻辑

年份 2001 题号 B4 入口集合/构造

Let S denote the set of rational numbers different from $\{-1, 0, 1\}$. Define $f : S \rightarrow S$ by $f(x) = x - 1/x$. Prove or disprove that

$$\bigcap_{n=1}^{\infty} f^{(n)}(S) = \emptyset,$$

where $f^{(n)}$ denotes f composed with itself n times.

2004 年 B5 · 基础证明、集合与初等逻辑

年份 2004 题号 B5 入口集合

Evaluate

$$\lim_{x \rightarrow 1^-} \prod_{n=0}^{\infty} \left(\frac{1 + x^{n+1}}{1 + x^n} \right)^{x^n}.$$

2013 年 B5 · 基础证明、集合与初等逻辑

年份 2013 题号 B5 入口函数与不等式

Let $X = \{1, 2, \dots, n\}$, and let $k \in X$. Show that there are exactly $k \cdot n^{n-1}$ functions $f : X \rightarrow X$ such that for every $x \in X$ there is a $j \geq 0$ such that $f^{(j)}(x) \leq k$. [Here $f^{(j)}$ denotes the j^{th} iterate of f , so that $f^{(0)}(x) = x$ and $f^{(j+1)}(x) = f(f^{(j)}(x))$.]

第 2 章

初等组合与离散数学

范畴要点：计数、排列组合、二项式、初等图论、染色、博弈

1995 年 A1 · 初等组合与离散数学

年份 1995 题号 A1 入口积分

Let S be a set of real numbers which is closed under multiplication (that is, if a and b are in S , then so is ab). Let T and U be disjoint subsets of S whose union is S . Given that the product of any *three* (not necessarily distinct) elements of T is in T and that the product of any three elements of U is in U , show that at least one of the two subsets T, U is closed under multiplication.

1993 年 A3 · 初等组合与离散数学

年份 1993 题号 A3 入口级数/求和

Let $\mathcal{P}_n$ be the set of subsets of $\{1, 2, \dots, n\}$. Let $c(n, m)$ be the number of functions $f: \mathcal{P}_n \rightarrow \{1, 2, \dots, m\}$ such that $f(A \cap B) = \min\{f(A), f(B)\}$. Prove that

$$c(n, m) = \sum_{j=1}^m j^n.$$

2002 年 A3 · 初等组合与离散数学

年份 2002 题号 A3 入口积分

Let $n \geq 2$ be an integer and T_n be the number of non-empty subsets S of $\{1, 2, 3, \dots, n\}$ with the property that the average of the elements of S is an integer. Prove that $T_n - n$ is always even.

2024 年 A3 · 初等组合与离散数学

年份 2024 题号 A3 入口集合/构造

Let S be the set of bijections

$$T: \{1, 2, 3\} \times \{1, 2, \dots, 2024\} \rightarrow \{1, 2, \dots, 6072\}$$

such that $T(1, j) < T(2, j) < T(3, j)$ for all $j \in \{1, 2, \dots, 2024\}$ and $T(i, j) < T(i, j+1)$ for all $i \in \{1, 2, 3\}$ and $j \in \{1, 2, \dots, 2023\}$. Do there exist a and c in $\{1, 2, 3\}$ and b and d in $\{1, 2, \dots, 2024\}$ such that the fraction of elements T in S for which $T(a, b) < T(c, d)$ is at least $1/3$ and at most $2/3$?

1996 年 A4 · 初等组合与离散数学

年份 1996 题号 A4 入口函数与不等式

Let S be the set of ordered triples (a, b, c) of distinct elements of a finite set A . Suppose that

1. $(a, b, c) \in S$ if and only if $(b, c, a) \in S$;
2. $(a, b, c) \in S$ if and only if $(c, b, a) \notin S$;
3. (a, b, c) and (c, d, a) are both in S if and only if (b, c, d) and (d, a, b) are both in S .

Prove that there exists a one-to-one function g from A to R such that $g(a) < g(b) < g(c)$ implies $(a, b, c) \in S$. Note: R is the set of real numbers.

2006 年 A4 · 初等组合与离散数学

年份 2006 题号 A4 入口积分

Let $S = \{1, 2, \dots, n\}$ for some integer $n > 1$. Say a permutation π of S has a *local maximum* at $k \in S$ if

- (i) $\pi(k) > \pi(k+1)$ for $k = 1$;
- (ii) $\pi(k-1) < \pi(k)$ and $\pi(k) > \pi(k+1)$ for $1 < k < n$;
- (iii) $\pi(k-1) < \pi(k)$ for $k = n$.

(For example, if $n = 5$ and π takes values at $1, 2, 3, 4, 5$ of $2, 1, 4, 5, 3$, then π has a local maximum of 2 at $k = 1$, and a local maximum of 5 at $k = 4$.) What is the average number of local maxima of a permutation of S , averaging over all permutations of S ?

第 3 章

初等数论

范畴要点：整除、同余、素数、最大公因数、进位制、丢番图

1989 年 A1 · 初等数论

年份 1989 题号 A1 入口整除与同余

How many primes among the positive integers, written as usual in base 10, are alternating 1's and 0's, beginning and ending with 1?

2014 年 A1 · 初等数论

年份 2014 题号 A1 入口整除与同余

Prove that every nonzero coefficient of the Taylor series of

$$(1 - x + x^2)e^x$$

about $x = 0$ is a rational number whose numerator (in lowest terms) is either 1 or a prime number.

2016 年 A1 · 初等数论

年份 2016 题号 A1 入口多项式

Find the smallest positive integer j such that for every polynomial $p(x)$ with integer coefficients and for every integer k , the integer

$$p^{(j)}(k) = \left. \frac{d^j}{dx^j} p(x) \right|_{x=k}$$

(the j -th derivative of $p(x)$ at k) is divisible by 2016.

2020 年 A1 · 初等数论

年份 2020 题号 A1 入口环/形式幂级数

How many positive integers N satisfy all of the following three conditions?

- (i) N is divisible by 2020.
- (ii) N has at most 2020 decimal digits.
- (iii) The decimal digits of N are a string of consecutive ones followed by a string of consecutive zeros.

2025 年 A1 · 初等数论

年份 2025 题号 A1 入口整除与同余

Let m_0 and n_0 be distinct positive integers. For every positive integer k , define m_k and n_k to be the relatively prime positive integers such that

$$\frac{m_k}{n_k} = \frac{2m_{k-1} + 1}{2n_{k-1} + 1}.$$

Prove that $2m_k + 1$ and $2n_k + 1$ are relatively prime for all but finitely many positive integers k .

1987 年 A2 · 初等数论

年份 1987 题号 A2 入口进位制与数字

The sequence of digits

123456789101112131415161718192021...

is obtained by writing the positive integers in order. If the 10^n -th digit in this sequence occurs in the part of the sequence in which the m -digit numbers are placed, define $f(n)$ to be m . For example, $f(2) = 2$ because the 100th digit enters the sequence in the placement of the two-digit integer 55. Find, with proof, $f(1987)$.

2006 年 A2 · 初等数论

年份 2006 题号 A2 入口整除与同余

Alice and Bob play a game in which they take turns removing stones from a heap that initially has n stones. The number of stones removed at each turn must be one less than a prime number. The winner is the player who takes the last stone. Alice plays first. Prove that there are infinitely many n such that Bob has a winning strategy. (For example, if $n = 17$, then Alice might take 6 leaving 11; then Bob might take 1 leaving 10; then Alice can take the remaining stones to win.)

2015 年 A2 · 初等数论

年份 2015 题号 A2 入口整除与同余

Let $a_0 = 1$, $a_1 = 2$, and $a_n = 4a_{n-1} - a_{n-2}$ for $n \geq 2$. Find an odd prime factor of a_{2015} .

2017 年 A2 · 初等数论

年份 2017 题号 A2 入口多项式

Let $Q_0(x) = 1$, $Q_1(x) = x$, and

$$Q_n(x) = \frac{(Q_{n-1}(x))^2 - 1}{Q_{n-2}(x)}$$

for all $n \geq 2$. Show that, whenever n is a positive integer, $Q_n(x)$ is equal to a polynomial with integer coefficients.

1992 年 A3 · 初等数论

年份 1992 题号 A3 入口整除与同余

For a given positive integer m , find all triples (n, x, y) of positive integers, with n relatively prime to m , which satisfy

$$(x^2 + y^2)^m = (xy)^n.$$

1995 年 A3 · 初等数论

年份 1995 题号 A3 入口整除与同余

The number $d_1d_2 \dots d_9$ has nine (not necessarily distinct) decimal digits. The number $e_1e_2 \dots e_9$ is such that each of the nine 9-digit numbers formed by replacing just one of the digits d_i is $d_1d_2 \dots d_9$ by the corresponding digit e_i ($1 \leq i \leq 9$) is divisible by 7. The number $f_1f_2 \dots f_9$ is related to $e_1e_2 \dots e_9$ in the same way: that is, each of the nine numbers formed by replacing one of the e_i by the corresponding f_i is divisible by 7. Show that, for each i , $d_i - f_i$ is divisible by 7. [For example, if $d_1d_2 \dots d_9 = 199501996$, then e_6 may be 2 or 9, since 199502996 and 199509996 are multiples of 7.]

2001 年 A3 · 初等数论

年份 2001 题号 A3 入口多项式

For each integer m , consider the polynomial

$$P_m(x) = x^4 - (2m + 4)x^2 + (m - 2)^2.$$

For what values of m is $P_m(x)$ the product of two non-constant polynomials with integer coefficients?

2006 年 A3 · 初等数论

年份 2006 题号 A3 入口整除与同余

Let $1, 2, 3, \dots, 2005, 2006, 2007, 2009, 2012, 2016, \dots$ be a sequence defined by $x_k = k$ for $k = 1, 2, \dots, 2006$ and $x_{k+1} = x_k + x_{k-2005}$ for $k \geq 2006$. Show that the sequence

has 2005 consecutive terms each divisible by 2006.

2022 年 A3 · 初等数论

年份 2022 题号 A3 入口整除与同余

Let p be a prime number greater than 5. Let $f(p)$ denote the number of infinite sequences $a_1, a_2, a_3, \dots$ such that $a_n \in \{1, 2, \dots, p-1\}$ and $a_n a_{n+2} \equiv 1 + a_{n+1} \pmod{p}$ for all $n \geq 1$. Prove that $f(p)$ is congruent to 0 or 2 $\pmod{5}$.

2025 年 A3 · 初等数论

年份 2025 题号 A3 入口环/形式幂级数

Alice and Bob play a game with a string of n digits, each of which is restricted to be 0, 1, or 2. Initially all the digits are 0. A legal move is to add or subtract 1 from one digit to create a new string that has not appeared before. A player with no legal move loses, and the other player wins. Alice goes first, and the players alternate moves. For each $n \geq 1$, determine which player has a strategy that guarantees winning.

1985 年 A4 · 初等数论

年份 1985 题号 A4 入口进位制与数字

Define a sequence $\{a_i\}$ by $a_1 = 3$ and $a_{i+1} = 3^{a_i}$ for $i \geq 1$. Which integers between 00 and 99 inclusive occur as the last two digits in the decimal expansion of infinitely many a_i ?

1998 年 A4 · 初等数论

年份 1998 题号 A4 入口进位制与数字

Let $A_1 = 0$ and $A_2 = 1$. For $n > 2$, the number A_n is defined by concatenating the decimal expansions of A_{n-1} and A_{n-2} from left to right. For example $A_3 = A_2 A_1 = 10$, $A_4 = A_3 A_2 = 101$, $A_5 = A_4 A_3 = 10110$, and so forth. Determine all n such that 11 divides A_n .

2007 年 A4 · 初等数论

年份 2007 题号 A4 入口多项式

A *repunit* is a positive integer whose digits in base 10 are all ones. Find all polynomials f with real coefficients such that if n is a repunit, then so is $f(n)$.

2010 年 A4 · 初等数论

年份 2010 题号 A4 入口整除与同余

Prove that for each positive integer n , the number $10^{10^{10^n}} + 10^{10^n} + 10^n - 1$ is not prime.

2024 年 A4 · 初等数论

年份 2024 题号 A4 入口整除与同余

Find all primes $p > 5$ for which there exists an integer a and an integer r satisfying $1 \leq r \leq p - 1$ with the following property: the sequence $1, a, a^2, \dots, a^{p-5}$ can be rearranged to form a sequence $b_0, b_1, b_2, \dots, b_{p-5}$ such that $b_n - b_{n-1} - r$ is divisible by p for $1 \leq n \leq p - 5$.

1996 年 A5 · 初等数论

年份 1996 题号 A5 入口多项式

If p is a prime number greater than 3 and $k = \lfloor 2p/3 \rfloor$, prove that the sum

$$\binom{p}{1} + \binom{p}{2} + \dots + \binom{p}{k}$$

of binomial coefficients is divisible by p^2 .

2014 年 A5 · 初等数论

年份 2014 题号 A5 入口多项式

Let

$$P_n(x) = 1 + 2x + 3x^2 + \dots + nx^{n-1}.$$

Prove that the polynomials $P_j(x)$ and $P_k(x)$ are relatively prime for all positive integers j and k with $j \neq k$.

2015 年 A5 · 初等数论

年份 2015 题号 A5 入口整除与同余

Let q be an odd positive integer, and let N_q denote the number of integers a such that $0 < a < q/4$ and $\gcd(a, q) = 1$. Show that N_q is odd if and only if q is of the form p^k with k a positive integer and p a prime congruent to 5 or 7 modulo 8.

2021 年 A5 · 初等数论

年份 2021 题号 A5 入口整除与同余

Let A be the set of all integers n such that $1 \leq n \leq 2021$ and $\gcd(n, 2021) = 1$. For every nonnegative integer j , let

$$S(j) = \sum_{n \in A} n^j.$$

Determine all values of j such that $S(j)$ is a multiple of 2021.

1987 年 A6 · 初等数论

年份 1987 题号 A6 入口进位制与数字

For each positive integer n , let $a(n)$ be the number of zeroes in the base 3 representation of n . For which positive real numbers x does the series

$$\sum_{n=1}^{\infty} \frac{x^{a(n)}}{n^3}$$

converge?

2000 年 A6 · 初等数论

年份 2000 题号 A6 入口多项式

Let $f(x)$ be a polynomial with integer coefficients. Define a sequence $a_0, a_1, \dots$ of integers such that $a_0 = 0$ and $a_{n+1} = f(a_n)$ for all $n \geq 0$. Prove that if there exists a positive integer m for which $a_m = 0$ then either $a_1 = 0$ or $a_2 = 0$.

2002 年 A6 · 初等数论

年份 2002 题号 A6 入口进位制与数字

Fix an integer $b \geq 2$. Let $f(1) = 1$, $f(2) = 2$, and for each $n \geq 3$, define $f(n) = nf(d)$, where d is the number of base- b digits of n . For which values of b does

$$\sum_{n=1}^{\infty} \frac{1}{f(n)}$$

converge?

2021 年 A6 · 初等数论

年份 2021 题号 A6 入口多项式

Let $P(x)$ be a polynomial whose coefficients are all either 0 or 1. Suppose that $P(x)$ can be written as a product of two nonconstant polynomials with integer coefficients. Does it follow that $P(2)$ is a composite integer?

2025 年 A6 · 初等数论

年份 2025 题号 A6 入口整除与同余

Let $b_0 = 0$ and, for $n \geq 0$, define $b_{n+1} = 2b_n^2 + b_n + 1$. For each $k \geq 1$, show that $b_{2^{k+1}} - 2b_{2^k}$ is divisible by 2^{2k+2} but not by 2^{2k+3} .

2004 年 B1 · 初等数论

年份 2004 题号 B1 入口多项式

Let $P(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_0$ be a polynomial with integer coefficients. Suppose that r is a rational number such that $P(r) = 0$. Show that the n numbers

$$c_n r, c_n r^2 + c_{n-1} r, c_n r^3 + c_{n-1} r^2 + c_{n-2} r, \\ \dots, c_n r^n + c_{n-1} r^{n-1} + \cdots + c_1 r$$

are integers.

2007 年 B1 · 初等数论

年份 2007 题号 B1 入口多项式

Let f be a polynomial with positive integer coefficients. Prove that if n is a positive integer, then $f(n)$ divides $f(f(n) + 1)$ if and only if $n = 1$. [Editor's note: one must assume f is nonconstant.]

2009 年 B1 · 初等数论

年份 2009 题号 B1 入口整除与同余

Show that every positive rational number can be written as a quotient of products of factorials of (not necessarily distinct) primes. For example,

$$\frac{10}{9} = \frac{2! \cdot 5!}{3! \cdot 3! \cdot 3!}.$$

2014 年 B1 · 初等数论

年份 2014 题号 B1 入口进位制与数字

A *base 10 over-expansion* of a positive integer N is an expression of the form

$$N = d_k 10^k + d_{k-1} 10^{k-1} + \cdots + d_0 10^0$$

with $d_k \neq 0$ and $d_i \in \{0, 1, 2, \dots, 10\}$ for all i . For instance, the integer $N = 10$ has two base 10 over-expansions: $10 = 10 \cdot 10^0$ and the usual base 10 expansion $10 = 1 \cdot 10^1 + 0 \cdot 10^0$. Which positive integers have a unique base 10 over-expansion?

2020 年 B1 · 初等数论

年份 2020 题号 B1 入口整除与同余

For a positive integer n , define $d(n)$ to be the sum of the digits of n when written in binary (for example, $d(13) = 1 + 1 + 0 + 1 = 3$). Let

$$S = \sum_{k=1}^{2020} (-1)^{d(k)} k^3.$$

Determine S modulo 2020.

2022 年 B1 · 初等数论

年份 2022 题号 B1 入口多项式

Suppose that $P(x) = a_1x + a_2x^2 + \cdots + a_nx^n$ is a polynomial with integer coefficients, with a_1 odd. Suppose that $e^{P(x)} = b_0 + b_1x + b_2x^2 + \cdots$ for all x . Prove that b_k is nonzero for all $k \geq 0$.

1985 年 B2 · 初等数论

年份 1985 题号 B2 入口多项式

Define polynomials $f_n(x)$ for $n \geq 0$ by $f_0(x) = 1$, $f_n(0) = 0$ for $n \geq 1$, and

$$\frac{d}{dx} f_{n+1}(x) = (n+1)f_n(x+1)$$

for $n \geq 0$. Find, with proof, the explicit factorization of $f_{100}(1)$ into powers of distinct primes.

2000 年 B2 · 初等数论

年份 2000 题号 B2 入口整除与同余

Prove that the expression

$$\frac{\gcd(m, n)}{n} \binom{n}{m}$$

is an integer for all pairs of integers $n \geq m \geq 1$.

2011 年 B2 · 初等数论

年份 2011 题号 B2 入口整除与同余

Let S be the set of all ordered triples (p, q, r) of prime numbers for which at least one rational number x satisfies $px^2 + qx + r = 0$. Which primes appear in seven or more elements of S ?

2015 年 B2 · 初等数论

年份 2015 题号 B2 入口进位制与数字

Given a list of the positive integers $1, 2, 3, 4, \dots$, take the first three numbers $1, 2, 3$ and their sum 6 and cross all four numbers off the list. Repeat with the three smallest remaining numbers $4, 5, 7$ and their sum 16 . Continue in this way, crossing off the three smallest remaining numbers and their sum, and consider the sequence of sums produced: $6, 16, 27, 36, \dots$. Prove or disprove that there is some number in the sequence whose base 10 representation ends with 2015.

1986 年 B3 · 初等数论

年份 1986 题号 B3 入口多项式

Let Γ consist of all polynomials in x with integer coefficients. For f and g in Γ and m a positive integer, let $f \equiv g \pmod{m}$ mean that every coefficient of $f - g$ is an integral multiple of m . Let n and p be positive integers with p prime. Given that f, g, h, r and s are in Γ with $rf + sg \equiv 1 \pmod{p}$ and $fg \equiv h \pmod{p}$, prove that there exist F and G in Γ with $F \equiv f \pmod{p}$, $G \equiv g \pmod{p}$, and $FG \equiv h \pmod{p^n}$.

1997 年 B3 · 初等数论

年份 1997 题号 B3 入口整除与同余

For each positive integer n , write the sum $\sum_{m=1}^n 1/m$ in the form p_n/q_n , where p_n and q_n are relatively prime positive integers. Determine all n such that 5 does not divide q_n .

2003 年 B3 · 初等数论

年份 2003 题号 B3 入口积分

Show that for each positive integer n ,

$$n! = \prod_{i=1}^n \text{lcm}\{1, 2, \dots, \lfloor n/i \rfloor\}.$$

(Here lcm denotes the least common multiple, and $\lfloor x \rfloor$ denotes the greatest integer $\leq x$.)

2025 年 B3 · 初等数论

年份 2025 题号 B3 入口整除与同余

Suppose S is a nonempty set of positive integers with the property that if n is in S , then every positive divisor of $2025^n - 15^n$ is in S . Must S contain all positive integers?

1991 年 B4 · 初等数论

年份 1991 题号 B4 入口整除与同余

Suppose p is an odd prime. Prove that

$$\sum_{j=0}^p \binom{p}{j} \binom{p+j}{j} \equiv 2^p + 1 \pmod{p^2}.$$

2008 年 B4 · 初等数论

年份 2008 题号 B4 入口多项式

Let p be a prime number. Let $h(x)$ be a polynomial with integer coefficients such that $h(0), h(1), \dots, h(p^2 - 1)$ are distinct modulo p^2 . Show that $h(0), h(1), \dots, h(p^3 - 1)$ are distinct modulo p^3 .

1997 年 B5 · 初等数论

年份 1997 题号 B5 入口整除与同余

Prove that for $n \geq 2$,

$$\underbrace{2^{2^{\cdots 2}}}_{n \text{ terms}} \equiv \underbrace{2^{2^{\cdots 2}}}_{n-1 \text{ terms}} \pmod{n}.$$

1998 年 B5 · 初等数论

年份 1998 题号 B5 入口进位制与数字

Let N be the positive integer with 1998 decimal digits, all of them 1; that is,

$$N = 1111 \cdots 11.$$

Find the thousandth digit after the decimal point of $\sqrt{N}$.

2002 年 B5 · 初等数论

年份 2002 题号 B5 入口进位制与数字

A palindrome in base b is a positive integer whose base- b digits read the same backwards and forwards; for example, 2002 is a 4-digit palindrome in base 10. Note that 200 is not a palindrome in base 10, but it is the 3-digit palindrome 242 in base

9, and 404 in base 7. Prove that there is an integer which is a 3-digit palindrome in base b for at least 2002 different values of b .

2023 年 B5 · 初等数论

年份 2023 题号 B5 入口整除与同余

Determine which positive integers n have the following property: For all integers m that are relatively prime to n , there exists a permutation $\pi: \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ such that $\pi(\pi(k)) \equiv mk \pmod{n}$ for all $k \in \{1, 2, \dots, n\}$.

2025 年 B5 · 初等数论

年份 2025 题号 B5 入口整除与同余

Let p be a prime number greater than 3. For each $k \in \{1, \dots, p-1\}$, let $I(k) \in \{1, 2, \dots, p-1\}$ be such that $k \cdot I(k) \equiv 1 \pmod{p}$. Prove that the number of integers $k \in \{1, \dots, p-2\}$ such that $I(k+1) < I(k)$ is greater than $p/4 - 1$.

1994 年 B6 · 初等数论

年份 1994 题号 B6 入口整除与同余

For any integer n , set

$$n_a = 101a - 100 \cdot 2^a.$$

Show that for $0 \leq a, b, c, d \leq 99$, $n_a + n_b \equiv n_c + n_d \pmod{10100}$ implies $\{a, b\} = \{c, d\}$.

1999 年 B6 · 初等数论

年份 1999 题号 B6 入口整除与同余

Let S be a finite set of integers, each greater than 1. Suppose that for each integer n there is some $s \in S$ such that $\gcd(s, n) = 1$ or $\gcd(s, n) = s$. Show that there exist $s, t \in S$ such that $\gcd(s, t)$ is prime.

2011 年 B6 · 初等数论

年份 2011 题号 B6 入口整除与同余

Let p be an odd prime. Show that for at least $(p+1)/2$ values of n in $\{0, 1, 2, \dots, p-1\}$,

$$\sum_{k=0}^{p-1} k!n^k \quad \text{is not divisible by } p.$$

2015 年 B6 · 初等数论

年份 2015 题号 B6 入口整除与同余

For each positive integer k , let $A(k)$ be the number of odd divisors of k in the interval $[1, \sqrt{2k})$. Evaluate

$$\sum_{k=1}^{\infty} (-1)^{k-1} \frac{A(k)}{k}.$$

2017 年 B6 · 初等数论

年份 2017 题号 B6 入口整除与同余

Find the number of ordered 64-tuples $(x_0, x_1, \dots, x_{63})$ such that $x_0, x_1, \dots, x_{63}$ are distinct elements of $\{1, 2, \dots, 2017\}$ and

$$x_0 + x_1 + 2x_2 + 3x_3 + \cdots + 63x_{63}$$

is divisible by 2017.

第 4 章

单变量微积分与初等分析

范畴要点：极限、导数、积分、三角函数、初等不等式、计算型级数

1985 年 A1 · 单变量微积分与初等分析

年份 1985 题号 A1 入口积分

Determine, with proof, the number of ordered triples (A_1, A_2, A_3) of sets which have the property that

(i) $A_1 \cup A_2 \cup A_3 = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and

(ii) $A_1 \cap A_2 \cap A_3 = \emptyset$.

Express your answer in the form $2^a 3^b 5^c 7^d$, where a, b, c, d are nonnegative integers.

1986 年 A1 · 单变量微积分与初等分析

年份 1986 题号 A1 入口集合/构造

Find, with explanation, the maximum value of $f(x) = x^3 - 3x$ on the set of all real numbers x satisfying $x^4 + 36 \leq 13x^2$.

1990 年 A1 · 单变量微积分与初等分析

年份 1990 题号 A1 入口递推/序列

Let

$$T_0 = 2, T_1 = 3, T_2 = 6,$$

and for $n \geq 3$,

$$T_n = (n + 4)T_{n-1} - 4nT_{n-2} + (4n - 8)T_{n-3}.$$

The first few terms are

$$2, 3, 6, 14, 40, 152, 784, 5168, 40576.$$

Find, with proof, a formula for T_n of the form $T_n = A_n + B_n$, where $\{A_n\}$ and $\{B_n\}$ are well-known sequences.

1992 年 A1 · 单变量微积分与初等分析

年份 1992 题号 A1 入口积分

Prove that $f(n) = 1 - n$ is the only integer-valued function defined on the integers that satisfies the following conditions.

- (i) $f(f(n)) = n$, for all integers n ;
- (ii) $f(f(n+2)+2) = n$ for all integers n ;
- (iii) $f(0) = 1$.

2000 年 A1 · 单变量微积分与初等分析

年份 2000 题号 A1 入口级数/求和

Let A be a positive real number. What are the possible values of $\sum_{j=0}^{\infty} x_j^2$, given that $x_0, x_1, \dots$ are positive numbers for which $\sum_{j=0}^{\infty} x_j = A$?

2003 年 A1 · 单变量微积分与初等分析

年份 2003 题号 A1 入口积分

Let n be a fixed positive integer. How many ways are there to write n as a sum of positive integers,

$$n = a_1 + a_2 + \cdots + a_k,$$

with k an arbitrary positive integer and $a_1 \leq a_2 \leq \cdots \leq a_k \leq a_1 + 1$? For example, with $n = 4$ there are four ways: 4, 2+2, 1+1+2, 1+1+1+1.

2005 年 A1 · 单变量微积分与初等分析

年份 2005 题号 A1 入口积分

Show that every positive integer is a sum of one or more numbers of the form $2^r 3^s$, where r and s are nonnegative integers and no summand divides another. (For example, $23 = 9 + 8 + 6$.)

2007 年 A1 · 单变量微积分与初等分析

年份 2007 题号 A1 入口三角函数

Find all values of α for which the curves $y = \alpha x^2 + \alpha x + \frac{1}{24}$ and $x = \alpha y^2 + \alpha y + \frac{1}{24}$ are tangent to each other.

2008 年 A1 · 单变量微积分与初等分析

年份 2008 题号 A1 入口函数与不等式

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that $f(x, y) + f(y, z) + f(z, x) = 0$ for all real numbers x, y , and z . Prove that there exists a function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x, y) = g(x) - g(y)$ for all real numbers x and y .

2010 年 A1 · 单变量微积分与初等分析

年份 2010 题号 A1 入口积分

Given a positive integer n , what is the largest k such that the numbers $1, 2, \dots, n$ can be put into k boxes so that the sum of the numbers in each box is the same? [When $n = 8$, the example $\{1, 2, 3, 6\}, \{4, 8\}, \{5, 7\}$ shows that the largest k is at least 3.]

2017 年 A1 · 单变量微积分与初等分析

年份 2017 题号 A1 入口积分

Let S be the smallest set of positive integers such that

- (a) 2 is in S ,
- (b) n is in S whenever n^2 is in S , and
- (c) $(n + 5)^2$ is in S whenever n is in S .

Which positive integers are not in S ?

(The set S is “smallest” in the sense that S is contained in any other such set.)

2018 年 A1 · 单变量微积分与初等分析

年份 2018 题号 A1 入口积分

Find all ordered pairs (a, b) of positive integers for which

$$\frac{1}{a} + \frac{1}{b} = \frac{3}{2018}.$$

2019 年 A1 · 单变量微积分与初等分析

年份 2019 题号 A1 入口积分

Determine all possible values of the expression

$$A^3 + B^3 + C^3 - 3ABC$$

where A, B , and C are nonnegative integers.

2022 年 A1 · 单变量微积分与初等分析

年份 2022 题号 A1 入口积分

Determine all ordered pairs of real numbers (a, b) such that the line $y = ax + b$ intersects the curve $y = \ln(1 + x^2)$ in exactly one point.

2023 年 A1 · 单变量微积分与初等分析

年份 2023 题号 A1 入口积分

For a positive integer n , let $f_n(x) = \cos(x) \cos(2x) \cos(3x) \cdots \cos(nx)$. Find the smallest n such that $|f_n''(0)| > 2023$.

2024 年 A1 · 单变量微积分与初等分析

年份 2024 题号 A1 入口积分

Determine all positive integers n for which there exist positive integers a , b , and c satisfying

$$2a^n + 3b^n = 4c^n.$$

1989 年 A2 · 单变量微积分与初等分析

年份 1989 题号 A2 入口积分

Evaluate $\int_0^a \int_0^b e^{\max\{b^2x^2, a^2y^2\}} dy dx$ where a and b are positive.

1990 年 A2 · 单变量微积分与初等分析

年份 1990 题号 A2 入口极限

Is $\sqrt{2}$ the limit of a sequence of numbers of the form $\sqrt[3]{n} - \sqrt[3]{m}$ ($n, m = 0, 1, 2, \dots$)?

1992 年 A2 · 单变量微积分与初等分析

年份 1992 题号 A2 入口积分

Define $C(\alpha)$ to be the coefficient of x^{1992} in the power series about $x = 0$ of $(1+x)^\alpha$. Evaluate

$$\int_0^1 \left(C(-y-1) \sum_{k=1}^{1992} \frac{1}{y+k} \right) dy.$$

1993 年 A2 · 单变量微积分与初等分析

年份 1993 题号 A2 入口递推/序列

Let $(x_n)_{n \geq 0}$ be a sequence of nonzero real numbers such that $x_n^2 - x_{n-1}x_{n+1} = 1$ for $n = 1, 2, 3, \dots$. Prove there exists a real number a such that $x_{n+1} = ax_n - x_{n-1}$ for all $n \geq 1$.

2000 年 A2 · 单变量微积分与初等分析

年份 2000 题号 A2 入口积分

Prove that there exist infinitely many integers n such that $n, n+1, n+2$ are each the sum of the squares of two integers. [Example: $0 = 0^2 + 0^2$, $1 = 0^2 + 1^2$, $2 = 1^2 + 1^2$.]

2003 年 A2 · 单变量微积分与初等分析

年份 2003 题号 A2 入口极限

Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be nonnegative real numbers. Show that

$$\begin{aligned} (a_1 a_2 \cdots a_n)^{1/n} + (b_1 b_2 \cdots b_n)^{1/n} \\ \leq [(a_1 + b_1)(a_2 + b_2) \cdots (a_n + b_n)]^{1/n}. \end{aligned}$$

2010 年 A2 · 单变量微积分与初等分析

年份 2010 题号 A2 入口积分

Find all differentiable functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f'(x) = \frac{f(x+n) - f(x)}{n}$$

for all real numbers x and all positive integers n .

2013 年 A2 · 单变量微积分与初等分析

年份 2013 题号 A2 入口积分

Let S be the set of all positive integers that are *not* perfect squares. For n in S , consider choices of integers $a_1, a_2, \dots, a_r$ such that $n < a_1 < a_2 < \cdots < a_r$ and $n \cdot a_1 \cdot a_2 \cdots a_r$ is a perfect square, and let $f(n)$ be the minimum of a_r over all such choices. For example, $2 \cdot 3 \cdot 6$ is a perfect square, while $2 \cdot 3$, $2 \cdot 4$, $2 \cdot 5$, $2 \cdot 3 \cdot 4$, $2 \cdot 3 \cdot 5$, $2 \cdot 4 \cdot 5$, and $2 \cdot 3 \cdot 4 \cdot 5$ are not, and so $f(2) = 6$. Show that the function f from S to the integers is one-to-one.

2016 年 A2 · 单变量微积分与初等分析

年份 2016 题号 A2 入口积分

Given a positive integer n , let $M(n)$ be the largest integer m such that

$$\binom{m}{n-1} > \binom{m-1}{n}.$$

Evaluate

$$\lim_{n \rightarrow \infty} \frac{M(n)}{n}.$$

2020 年 A2 · 单变量微积分与初等分析

年份 2020 题号 A2 入口积分

Let k be a nonnegative integer. Evaluate

$$\sum_{j=0}^k 2^{k-j} \binom{k+j}{j}.$$

2021 年 A2 · 单变量微积分与初等分析

年份 2021 题号 A2 入口极限

For every positive real number x , let

$$g(x) = \lim_{r \rightarrow 0} ((x+1)^{r+1} - x^{r+1})^{\frac{1}{r}}.$$

Find $\lim_{x \rightarrow \infty} \frac{g(x)}{x}$.

2025 年 A2 · 单变量微积分与初等分析

年份 2025 题号 A2 入口积分

Find the largest real number a and the smallest real number b such that

$$ax(\pi - x) \leq \sin x \leq bx(\pi - x)$$

for all x in the interval $[0, \pi]$.

1985 年 A3 · 单变量微积分与初等分析

年份 1985 题号 A3 入口积分

Let d be a real number. For each integer $m \geq 0$, define a sequence $\{a_m(j)\}$, $j = 0, 1, 2, \dots$ by the condition

$$\begin{aligned} a_m(0) &= d/2^m, \\ a_m(j+1) &= (a_m(j))^2 + 2a_m(j), \quad j \geq 0. \end{aligned}$$

Evaluate $\lim_{n \rightarrow \infty} a_n(n)$.

1986 年 A3 · 单变量微积分与初等分析

年份 1986 题号 A3 入口积分

Evaluate $\sum_{n=0}^{\infty} \operatorname{Arccot}(n^2 + n + 1)$, where $\operatorname{Arccot} t$ for $t \geq 0$ denotes the number θ in the interval $0 < \theta \leq \pi/2$ with $\cot \theta = t$.

1997 年 A3 · 单变量微积分与初等分析

年份 1997 题号 A3 入口积分

Evaluate

$$\int_0^\infty \left(x - \frac{x^3}{2} + \frac{x^5}{2 \cdot 4} - \frac{x^7}{2 \cdot 4 \cdot 6} + \cdots \right) \left(1 + \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} + \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \cdots \right) dx.$$

1998 年 A3 · 单变量微积分与初等分析

年份 1998 题号 A3 入口积分

Let f be a real function on the real line with continuous third derivative. Prove that there exists a point a such that

$$f(a) \cdot f'(a) \cdot f''(a) \cdot f'''(a) \geq 0.$$

1999 年 A3 · 单变量微积分与初等分析

年份 1999 题号 A3 入口积分

Consider the power series expansion

$$\frac{1}{1 - 2x - x^2} = \sum_{n=0}^{\infty} a_n x^n.$$

Prove that, for each integer $n \geq 0$, there is an integer m such that

$$a_n^2 + a_{n+1}^2 = a_m.$$

2003 年 A3 · 单变量微积分与初等分析

年份 2003 题号 A3 入口三角函数

Find the minimum value of

$$|\sin x + \cos x + \tan x + \cot x + \sec x + \csc x|$$

for real numbers x .

2010 年 A3 · 单变量微积分与初等分析

年份 2010 题号 A3 入口导数/可微

Suppose that the function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ has continuous partial derivatives and satisfies the equation

$$h(x, y) = a \frac{\partial h}{\partial x}(x, y) + b \frac{\partial h}{\partial y}(x, y)$$

for some constants a, b . Prove that if there is a constant M such that $|h(x, y)| \leq M$ for all $(x, y) \in \mathbb{R}^2$, then h is identically zero.

2011 年 A3 · 单变量微积分与初等分析

年份 2011 题号 A3 入口积分

Find a real number c and a positive number L for which

$$\lim_{r \rightarrow \infty} \frac{r^c \int_0^{\pi/2} x^r \sin x \, dx}{\int_0^{\pi/2} x^r \cos x \, dx} = L.$$

2013 年 A3 · 单变量微积分与初等分析

年份 2013 题号 A3 入口极限

Suppose that the real numbers $a_0, a_1, \dots, a_n$ and x , with $0 < x < 1$, satisfy

$$\frac{a_0}{1-x} + \frac{a_1}{1-x^2} + \dots + \frac{a_n}{1-x^{n+1}} = 0.$$

Prove that there exists a real number y with $0 < y < 1$ such that

$$a_0 + a_1 y + \dots + a_n y^n = 0.$$

2018 年 A3 · 单变量微积分与初等分析

年份 2018 题号 A3 入口级数/求和

Determine the greatest possible value of $\sum_{i=1}^{10} \cos(3x_i)$ for real numbers $x_1, x_2, \dots, x_{10}$ satisfying $\sum_{i=1}^{10} \cos(x_i) = 0$.

1993 年 A4 · 单变量微积分与初等分析

年份 1993 题号 A4 入口积分

Let $x_1, x_2, \dots, x_{19}$ be positive integers each of which is less than or equal to 93. Let $y_1, y_2, \dots, y_{93}$ be positive integers each of which is less than or equal to 19. Prove that there exists a (nonempty) sum of some x_i 's equal to a sum of some y_j 's.

1995 年 A4 · 单变量微积分与初等分析

年份 1995 题号 A4 入口环/形式幂级数

Suppose we have a necklace of n beads. Each bead is labeled with an integer and the sum of all these labels is $n - 1$. Prove that we can cut the necklace to form a string whose consecutive labels $x_1, x_2, \dots, x_n$ satisfy

$$\sum_{i=1}^k x_i \leq k - 1 \quad \text{for } k = 1, 2, \dots, n.$$

1999 年 A4 · 单变量微积分与初等分析

年份 1999 题号 A4 入口级数/求和

Sum the series

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{m^2 n}{3^m (n 3^m + m 3^n)}.$$

2003 年 A4 · 单变量微积分与初等分析

年份 2003 题号 A4 入口极限

Suppose that a, b, c, A, B, C are real numbers, $a \neq 0$ and $A \neq 0$, such that

$$|ax^2 + bx + c| \leq |Ax^2 + Bx + C|$$

for all real numbers x . Show that

$$|b^2 - 4ac| \leq |B^2 - 4AC|.$$

2004 年 A4 · 单变量微积分与初等分析

年份 2004 题号 A4 入口积分

Show that for any positive integer n there is an integer N such that the product $x_1 x_2 \cdots x_n$ can be expressed identically in the form

$$x_1 x_2 \cdots x_n = \sum_{i=1}^N c_i (a_{i1} x_1 + a_{i2} x_2 + \cdots + a_{in} x_n)^n$$

where the c_i are rational numbers and each a_{ij} is one of the numbers $-1, 0, 1$.

2008 年 A4 · 单变量微积分与初等分析

年份 2008 题号 A4 入口级数/求和

Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} x & \text{if } x \leq e \\ xf(\ln x) & \text{if } x > e. \end{cases}$$

Does $\sum_{n=1}^{\infty} \frac{1}{f(n)}$ converge?

2012 年 A4 · 单变量微积分与初等分析

年份 2012 题号 A4 入口积分

Let q and r be integers with $q > 0$, and let A and B be intervals on the real line. Let T be the set of all $b + mq$ where b and m are integers with b in B , and let S be the set of all integers a in A such that ra is in T . Show that if the product of the lengths of A and B is less than q , then S is the intersection of A with some arithmetic progression.

2015 年 A4 · 单变量微积分与初等分析

年份 2015 题号 A4 入口积分

For each real number x , let

$$f(x) = \sum_{n \in S_x} \frac{1}{2^n},$$

where S_x is the set of positive integers n for which $\lfloor nx \rfloor$ is even. What is the largest real number L such that $f(x) \geq L$ for all $x \in [0, 1)$? (As usual, $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z .)

2021 年 A4 · 单变量微积分与初等分析

年份 2021 题号 A4 入口积分

Let

$$I(R) = \iint_{x^2+y^2 \leq R^2} \left(\frac{1+2x^2}{1+x^4+6x^2y^2+y^4} - \frac{1+y^2}{2+x^4+y^4} \right) dx dy.$$

Find

$$\lim_{R \rightarrow \infty} I(R),$$

or show that this limit does not exist.

1985 年 A5 · 单变量微积分与初等分析

年份 1985 题号 A5 入口积分

Let $I_m = \int_0^{2\pi} \cos(x) \cos(2x) \cdots \cos(mx) dx$. For which integers m , $1 \leq m \leq 10$ is $I_m \neq 0$?

1986 年 A5 · 单变量微积分与初等分析

年份 1986 题号 A5 入口导数/可微

Suppose $f_1(x), f_2(x), \dots, f_n(x)$ are functions of n real variables $x = (x_1, \dots, x_n)$ with continuous second-order partial derivatives everywhere on $\mathbb{R}^n$. Suppose further that there are constants c_{ij} such that

$$\frac{\partial f_i}{\partial x_j} - \frac{\partial f_j}{\partial x_i} = c_{ij}$$

for all i and j , $1 \leq i \leq n$, $1 \leq j \leq n$. Prove that there is a function $g(x)$ on $\mathbb{R}^n$ such that $f_i + \partial g / \partial x_i$ is linear for all i , $1 \leq i \leq n$. (A linear function is one of the form

$$a_0 + a_1 x_1 + a_2 x_2 + \dots + a_n x_n.)$$

1987 年 A5 · 单变量微积分与初等分析

年份 1987 题号 A5 入口导数/可微

Let

$$\vec{G}(x, y) = \left(\frac{-y}{x^2 + 4y^2}, \frac{x}{x^2 + 4y^2}, 0 \right).$$

Prove or disprove that there is a vector-valued function

$$\vec{F}(x, y, z) = (M(x, y, z), N(x, y, z), P(x, y, z))$$

with the following properties:

- (i) M, N, P have continuous partial derivatives for all $(x, y, z) \neq (0, 0, 0)$;
- (ii) $\text{Curl } \vec{F} = \vec{0}$ for all $(x, y, z) \neq (0, 0, 0)$;
- (iii) $\vec{F}(x, y, 0) = \vec{G}(x, y)$.

1988 年 A5 · 单变量微积分与初等分析

年份 1988 题号 A5 入口函数与不等式

Prove that there exists a *unique* function f from the set $\mathbb{R}^+$ of positive real numbers to $\mathbb{R}^+$ such that

$$f(f(x)) = 6x - f(x)$$

and

$$f(x) > 0$$

for all $x > 0$.

1991 年 A5 · 单变量微积分与初等分析

年份 1991 题号 A5 入口积分

Find the maximum value of

$$\int_0^y \sqrt{x^4 + (y - y^2)^2} dx$$

for $0 \leq y \leq 1$.

1992 年 A5 · 单变量微积分与初等分析

年份 1992 题号 A5 入口积分

For each positive integer n , let $a_n = 0$ (or 1) if the number of 1's in the binary representation of n is even (or odd), respectively. Show that there do not exist positive integers k and m such that

$$a_{k+j} = a_{k+m+j} = a_{k+2m+j},$$

for $0 \leq j \leq m-1$.

1993 年 A5 · 单变量微积分与初等分析

年份 1993 题号 A5 入口积分

Show that

$$\begin{aligned} & \int_{-100}^{-10} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx + \\ & \int_{\frac{1}{101}}^{\frac{1}{11}} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx + \\ & \int_{\frac{101}{100}}^{\frac{11}{10}} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx \end{aligned}$$

is a rational number.

1994 年 A5 · 单变量微积分与初等分析

年份 1994 题号 A5 入口积分

Let $(r_n)_{n \geq 0}$ be a sequence of positive real numbers such that $\lim_{n \rightarrow \infty} r_n = 0$. Let S be the set of numbers representable as a sum

$$r_{i_1} + r_{i_2} + \cdots + r_{i_{1994}},$$

with $i_1 < i_2 < \cdots < i_{1994}$. Show that every nonempty interval (a, b) contains a nonempty subinterval (c, d) that does not intersect S .

1995 年 A5 · 单变量微积分与初等分析

年份 1995 题号 A5 入口导数/可微

Let $x_1, x_2, \dots, x_n$ be differentiable (real-valued) functions of a single variable t which satisfy

$$\begin{aligned}\frac{dx_1}{dt} &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ \frac{dx_2}{dt} &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ &\vdots \\ \frac{dx_n}{dt} &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n\end{aligned}$$

for some constants $a_{ij} > 0$. Suppose that for all i , $x_i(t) \rightarrow 0$ as $t \rightarrow \infty$. Are the functions $x_1, x_2, \dots, x_n$ necessarily linearly dependent?

1997 年 A5 · 单变量微积分与初等分析

年份 1997 题号 A5 入口积分

Let N_n denote the number of ordered n -tuples of positive integers $(a_1, a_2, \dots, a_n)$ such that $1/a_1 + 1/a_2 + \dots + 1/a_n = 1$. Determine whether N_{10} is even or odd.

1998 年 A5 · 单变量微积分与初等分析

年份 1998 题号 A5 入口积分

Let $\mathcal{F}$ be a finite collection of open discs in $\mathbb{R}^2$ whose union contains a set $E \subseteq \mathbb{R}^2$. Show that there is a pairwise disjoint subcollection $D_1, \dots, D_n$ in $\mathcal{F}$ such that

$$E \subseteq \cup_{j=1}^n 3D_j.$$

Here, if D is the disc of radius r and center P , then $3D$ is the disc of radius $3r$ and center P .

2001 年 A5 · 单变量微积分与初等分析

年份 2001 题号 A5 入口积分

Prove that there are unique positive integers a, n such that $a^{n+1} - (a+1)^n = 2001$.

2002 年 A5 · 单变量微积分与初等分析

年份 2002 题号 A5 入口积分

Define a sequence by $a_0 = 1$, together with the rules $a_{2n+1} = a_n$ and $a_{2n+2} = a_n + a_{n+1}$ for each integer $n \geq 0$. Prove that every positive rational number appears in the set

$$\left\{ \frac{a_{n-1}}{a_n} : n \geq 1 \right\} = \left\{ \frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{3}{2}, \dots \right\}.$$

2005 年 A5 · 单变量微积分与初等分析

年份 2005 题号 A5 入口积分

Evaluate $\int_0^1 \frac{\ln(x+1)}{x^2+1} dx$.

2006 年 A5 · 单变量微积分与初等分析

年份 2006 题号 A5 入口积分

Let n be a positive odd integer and let θ be a real number such that θ/π is irrational. Set $a_k = \tan(\theta + k\pi/n)$, $k = 1, 2, \dots, n$. Prove that

$$\frac{a_1 + a_2 + \dots + a_n}{a_1 a_2 \dots a_n}$$

is an integer, and determine its value.

2020 年 A5 · 单变量微积分与初等分析

年份 2020 题号 A5 入口积分

Let a_n be the number of sets S of positive integers for which

$$\sum_{k \in S} F_k = n,$$

where the Fibonacci sequence $(F_k)_{k \geq 1}$ satisfies $F_{k+2} = F_{k+1} + F_k$ and begins $F_1 = 1, F_2 = 1, F_3 = 2, F_4 = 3$. Find the largest integer n such that $a_n = 2020$.

1991 年 A6 · 单变量微积分与初等分析

年份 1991 题号 A6 入口积分

Let $A(n)$ denote the number of sums of positive integers

$$a_1 + a_2 + \dots + a_r$$

which add up to n with

$$\begin{aligned} a_1 &> a_2 + a_3, a_2 > a_3 + a_4, \dots, \\ a_{r-2} &> a_{r-1} + a_r, a_{r-1} > a_r. \end{aligned}$$

Let $B(n)$ denote the number of $b_1 + b_2 + \dots + b_s$ which add up to n , with

1. $b_1 \geq b_2 \geq \cdots \geq b_s$,
2. each b_i is in the sequence $1, 2, 4, \dots, g_j, \dots$ defined by $g_1 = 1$, $g_2 = 2$, and $g_j = g_{j-1} + g_{j-2} + 1$, and
3. if $b_1 = g_k$ then every element in $\{1, 2, 4, \dots, g_k\}$ appears at least once as a b_i .

Prove that $A(n) = B(n)$ for each $n \geq 1$.

(For example, $A(7) = 5$ because the relevant sums are $7, 6+1, 5+2, 4+3, 4+2+1$, and $B(7) = 5$ because the relevant sums are $4+2+1, 2+2+2+1, 2+2+1+1+1, 2+1+1+1+1+1, 1+1+1+1+1+1+1$.)

1993 年 A6 · 单变量微积分与初等分析

年份 1993 题号 A6 入口积分

The infinite sequence of 2's and 3's

2, 3, 3, 2, 3, 3, 3, 2, 3, 3, 3, 2, 3, 3, 2, 3, 3,
3, 2, 3, 3, 3, 2, 3, 3, 3, 2, 3, 3, 2, 3, 3, 2, ...

has the property that, if one forms a second sequence that records the number of 3's between successive 2's, the result is identical to the given sequence. Show that there exists a real number r such that, for any n , the n th term of the sequence is 2 if and only if $n = 1 + \lfloor rm \rfloor$ for some nonnegative integer m . (Note: $\lfloor x \rfloor$ denotes the largest integer less than or equal to x .)

1997 年 A6 · 单变量微积分与初等分析

年份 1997 题号 A6 入口积分

For a positive integer n and any real number c , define x_k recursively by $x_0 = 0$, $x_1 = 1$, and for $k \geq 0$,

$$x_{k+2} = \frac{cx_{k+1} - (n-k)x_k}{k+1}.$$

Fix n and then take c to be the largest value for which $x_{n+1} = 0$. Find x_k in terms of n and k , $1 \leq k \leq n$.

1999 年 A6 · 单变量微积分与初等分析

年份 1999 题号 A6 入口积分

The sequence $(a_n)_{n \geq 1}$ is defined by $a_1 = 1, a_2 = 2, a_3 = 24$, and, for $n \geq 4$,

$$a_n = \frac{6a_{n-1}^2 a_{n-3} - 8a_{n-1} a_{n-2}^2}{a_{n-2} a_{n-3}}.$$

Show that, for all n , a_n is an integer multiple of n .

2013 年 A6 · 单变量微积分与初等分析

年份 2013 题号 A6 入口级数/求和

Define a function $w : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ as follows. For $|a|, |b| \leq 2$, let $w(a, b)$ be as in the table shown; otherwise, let $w(a, b) = 0$.

$w(a, b)$		b				
		-2	-1	0	1	2
a	-2	-1	-2	2	-2	-1
	-1	-2	4	-4	4	-2
	0	2	-4	12	-4	2
	1	-2	4	-4	4	-2
	2	-1	-2	2	-2	-1

For every finite subset S of $\mathbb{Z} \times \mathbb{Z}$, define

$$A(S) = \sum_{(s, s') \in S \times S} w(s - s').$$

Prove that if S is any finite nonempty subset of $\mathbb{Z} \times \mathbb{Z}$, then $A(S) > 0$. (For example, if $S = \{(0, 1), (0, 2), (2, 0), (3, 1)\}$, then the terms in $A(S)$ are 12, 12, 12, 12, 4, 4, 0, 0, 0, 0, -1, -1, -2, -2, -4, -4.)

2020 年 A6 · 单变量微积分与初等分析

年份 2020 题号 A6 入口积分

For a positive integer N , let f_N^a be the function defined by

$$f_N(x) = \sum_{n=0}^N \frac{N + 1/2 - n}{(N + 1)(2n + 1)} \sin((2n + 1)x).$$

Determine the smallest constant M such that $f_N(x) \leq M$ for all N and all real x .

^aCorrected from F_N in the source.

2022 年 A6 · 单变量微积分与初等分析

年份 2022 题号 A6 入口积分

Let n be a positive integer. Determine, in terms of n , the largest integer m with the following property: There exist real numbers $x_1, \dots, x_{2n}$ with $-1 < x_1 < x_2 < \dots < x_{2n} < 1$ such that the sum of the lengths of the n intervals

$$[x_1^{2k-1}, x_2^{2k-1}], [x_3^{2k-1}, x_4^{2k-1}], \dots, [x_{2n-1}^{2k-1}, x_{2n}^{2k-1}]$$

is equal to 1 for all integers k with $1 \leq k \leq m$.

1987 年 B1 · 单变量微积分与初等分析

年份 1987 题号 B1 入口积分

Evaluate

$$\int_2^4 \frac{\sqrt{\ln(9-x)} dx}{\sqrt{\ln(9-x)} + \sqrt{\ln(x+3)}}.$$

1988 年 B1 · 单变量微积分与初等分析

年份 1988 题号 B1 入口积分

A *composite* (positive integer) is a product ab with a and b not necessarily distinct integers in $\{2, 3, 4, \dots\}$. Show that every composite is expressible as $xy + xz + yz + 1$, with x, y, z positive integers.

1991 年 B1 · 单变量微积分与初等分析

年份 1991 题号 B1 入口积分

For each integer $n \geq 0$, let $S(n) = n - m^2$, where m is the greatest integer with $m^2 \leq n$. Define a sequence $(a_k)_{k=0}^\infty$ by $a_0 = A$ and $a_{k+1} = a_k + S(a_k)$ for $k \geq 0$. For what positive integers A is this sequence eventually constant?

1992 年 B1 · 单变量微积分与初等分析

年份 1992 题号 B1 入口集合/构造

Let S be a set of n distinct real numbers. Let A_S be the set of numbers that occur as averages of two distinct elements of S . For a given $n \geq 2$, what is the smallest possible number of elements in A_S ?

1993 年 B1 · 单变量微积分与初等分析

年份 1993 题号 B1 入口积分

Find the smallest positive integer n such that for every integer m with $0 < m < 1993$, there exists an integer k for which

$$\frac{m}{1993} < \frac{k}{n} < \frac{m+1}{1994}.$$

1994 年 B1 · 单变量微积分与初等分析

年份 1994 题号 B1 入口积分

Find all positive integers n that are within 250 of exactly 15 perfect squares.

1997 年 B1 · 单变量微积分与初等分析

年份 1997 题号 B1 入口积分

Let $\{x\}$ denote the distance between the real number x and the nearest integer. For each positive integer n , evaluate

$$F_n = \sum_{m=1}^{6n-1} \min\left(\left\{\frac{m}{6n}\right\}, \left\{\frac{m}{3n}\right\}\right).$$

(Here $\min(a, b)$ denotes the minimum of a and b .)

1998 年 B1 · 单变量微积分与初等分析

年份 1998 题号 B1 入口极限

Find the minimum value of

$$\frac{(x + 1/x)^6 - (x^6 + 1/x^6) - 2}{(x + 1/x)^3 + (x^3 + 1/x^3)}$$

for $x > 0$.

2000 年 B1 · 单变量微积分与初等分析

年份 2000 题号 B1 入口积分

Let a_j, b_j, c_j be integers for $1 \leq j \leq N$. Assume for each j , at least one of a_j, b_j, c_j is odd. Show that there exist integers r, s, t such that $ra_j + sb_j + tc_j$ is odd for at least $4N/7$ values of j , $1 \leq j \leq N$.

2010 年 B1 · 单变量微积分与初等分析

年份 2010 题号 B1 入口积分

Is there an infinite sequence of real numbers $a_1, a_2, a_3, \dots$ such that

$$a_1^m + a_2^m + a_3^m + \dots = m$$

for every positive integer m ?

2011 年 B1 · 单变量微积分与初等分析

年份 2011 题号 B1 入口积分

Let h and k be positive integers. Prove that for every $\epsilon > 0$, there are positive integers m and n such that

$$\epsilon < |h\sqrt{m} - k\sqrt{n}| < 2\epsilon.$$

2013 年 B1 · 单变量微积分与初等分析

年份 2013 题号 B1 入口积分

For positive integers n , let the numbers $c(n)$ be determined by the rules $c(1) = 1$, $c(2n) = c(n)$, and $c(2n+1) = (-1)^n c(n)$. Find the value of

$$\sum_{n=1}^{2013} c(n)c(n+2).$$

2024 年 B1 · 单变量微积分与初等分析

年份 2024 题号 B1 入口积分

Let n and k be positive integers. The square in the i th row and j th column of an n -by- n grid contains the number $i+j-k$. For which n and k is it possible to select n squares from the grid, no two in the same row or column, such that the numbers contained in the selected squares are exactly $1, 2, \dots, n$?

1987 年 B2 · 单变量微积分与初等分析

年份 1987 题号 B2 入口积分

Let r, s and t be integers with $0 \leq r$, $0 \leq s$ and $r+s \leq t$. Prove that

$$\frac{\binom{s}{0}}{\binom{t}{r}} + \frac{\binom{s}{1}}{\binom{t}{r+1}} + \cdots + \frac{\binom{s}{s}}{\binom{t}{r+s}} = \frac{t+1}{(t+1-s)\binom{t-s}{r}}.$$

1988 年 B2 · 单变量微积分与初等分析

年份 1988 题号 B2 入口极限

Prove or disprove: If x and y are real numbers with $y \geq 0$ and $y(y+1) \leq (x+1)^2$, then $y(y-1) \leq x^2$.

1990 年 B2 · 单变量微积分与初等分析

年份 1990 题号 B2 入口级数/求和

Prove that for $|x| < 1$, $|z| > 1$,

$$1 + \sum_{j=1}^{\infty} (1+x^j)P_j = 0,$$

where P_j is

$$\frac{(1-z)(1-zx)(1-zx^2)\cdots(1-zx^{j-1})}{(z-x)(z-x^2)(z-x^3)\cdots(z-x^j)}.$$

1992 年 B2 · 单变量微积分与初等分析

年份 1992 题号 B2 入口积分

For nonnegative integers n and k , define $Q(n, k)$ to be the coefficient of x^k in the expansion of $(1 + x + x^2 + x^3)^n$. Prove that

$$Q(n, k) = \sum_{j=0}^k \binom{n}{j} \binom{n}{k-2j},$$

where $\binom{a}{b}$ is the standard binomial coefficient. (Reminder: For integers a and b with $a \geq 0$, $\binom{a}{b} = \frac{a!}{b!(a-b)!}$ for $0 \leq b \leq a$, with $\binom{a}{b} = 0$ otherwise.)

1994 年 B2 · 单变量微积分与初等分析

年份 1994 题号 B2 入口积分

For which real numbers c is there a straight line that intersects the curve

$$x^4 + 9x^3 + cx^2 + 9x + 4$$

in four distinct points?

1996 年 B2 · 单变量微积分与初等分析

年份 1996 题号 B2 入口积分

Show that for every positive integer n ,

$$\left(\frac{2n-1}{e}\right)^{\frac{2n-1}{2}} < 1 \cdot 3 \cdot 5 \cdots (2n-1) < \left(\frac{2n+1}{e}\right)^{\frac{2n+1}{2}}.$$

2001 年 B2 · 单变量微积分与初等分析

年份 2001 题号 B2 入口极限

Find all pairs of real numbers (x, y) satisfying the system of equations

$$\begin{aligned} \frac{1}{x} + \frac{1}{2y} &= (x^2 + 3y^2)(3x^2 + y^2) \\ \frac{1}{x} - \frac{1}{2y} &= 2(y^4 - x^4). \end{aligned}$$

2003 年 B2 · 单变量微积分与初等分析

年份 2003 题号 B2 入口积分

Let n be a positive integer. Starting with the sequence $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}$, form a new sequence of $n-1$ entries $\frac{3}{4}, \frac{5}{12}, \dots, \frac{2n-1}{2n(n-1)}$ by taking the averages of two consecutive

entries in the first sequence. Repeat the averaging of neighbors on the second sequence to obtain a third sequence of $n - 2$ entries, and continue until the final sequence produced consists of a single number x_n . Show that $x_n < 2/n$.

2004 年 B2 · 单变量微积分与初等分析

年份 2004 题号 B2 入口积分

Let m and n be positive integers. Show that

$$\frac{(m+n)!}{(m+n)^{m+n}} < \frac{m!}{m^m} \frac{n!}{n^n}.$$

2005 年 B2 · 单变量微积分与初等分析

年份 2005 题号 B2 入口积分

Find all positive integers $n, k_1, \dots, k_n$ such that $k_1 + \dots + k_n = 5n - 4$ and

$$\frac{1}{k_1} + \dots + \frac{1}{k_n} = 1.$$

2006 年 B2 · 单变量微积分与初等分析

年份 2006 题号 B2 入口积分

Prove that, for every set $X = \{x_1, x_2, \dots, x_n\}$ of n real numbers, there exists a non-empty subset S of X and an integer m such that

$$\left| m + \sum_{s \in S} s \right| \leq \frac{1}{n+1}.$$

2008 年 B2 · 单变量微积分与初等分析

年份 2008 题号 B2 入口积分

Let $F_0(x) = \ln x$. For $n \geq 0$ and $x > 0$, let $F_{n+1}(x) = \int_0^x F_n(t) dt$. Evaluate

$$\lim_{n \rightarrow \infty} \frac{n! F_n(1)}{\ln n}.$$

2014 年 B2 · 单变量微积分与初等分析

年份 2014 题号 B2 入口积分

Suppose that f is a function on the interval $[1, 3]$ such that $-1 \leq f(x) \leq 1$ for all x and $\int_1^3 f(x) dx = 0$. How large can $\int_1^3 \frac{f(x)}{x} dx$ be?

2016 年 B2 · 单变量微积分与初等分析

年份 2016 题号 B2 入口积分

Define a positive integer n to be *squarish* if either n is itself a perfect square or the distance from n to the nearest perfect square is a perfect square. For example, 2016 is squarish, because the nearest perfect square to 2016 is $45^2 = 2025$ and $2025 - 2016 = 9$ is a perfect square. (Of the positive integers between 1 and 10, only 6 and 7 are not squarish.)

For a positive integer N , let $S(N)$ be the number of squarish integers between 1 and N , inclusive. Find positive constants α and β such that

$$\lim_{N \rightarrow \infty} \frac{S(N)}{N^\alpha} = \beta,$$

or show that no such constants exist.

2017 年 B2 · 单变量微积分与初等分析

年份 2017 题号 B2 入口环/形式幂级数

Suppose that a positive integer N can be expressed as the sum of k consecutive positive integers

$$N = a + (a + 1) + (a + 2) + \cdots + (a + k - 1)$$

for $k = 2017$ but for no other values of $k > 1$. Considering all positive integers N with this property, what is the smallest positive integer a that occurs in any of these expressions?

2019 年 B2 · 单变量微积分与初等分析

年份 2019 题号 B2 入口级数/求和

For all $n \geq 1$, let

$$a_n = \sum_{k=1}^{n-1} \frac{\sin\left(\frac{(2k-1)\pi}{2n}\right)}{\cos^2\left(\frac{(k-1)\pi}{2n}\right) \cos^2\left(\frac{k\pi}{2n}\right)}.$$

Determine

$$\lim_{n \rightarrow \infty} \frac{a_n}{n^3}.$$

2021 年 B2 · 单变量微积分与初等分析

年份 2021 题号 B2 入口级数/求和

Determine the maximum value of the sum

$$S = \sum_{n=1}^{\infty} \frac{n}{2^n} (a_1 a_2 \cdots a_n)^{1/n}$$

over all sequences $a_1, a_2, a_3, \dots$ of nonnegative real numbers satisfying

$$\sum_{k=1}^{\infty} a_k = 1.$$

2022 年 B2 · 单变量微积分与初等分析

年份 2022 题号 B2 入口积分

Let $\times$ represent the cross product in $\mathbb{R}^3$. For what positive integers n does there exist a set $S \subset \mathbb{R}^3$ with exactly n elements such that

$$S = \{v \times w : v, w \in S\}?$$

2023 年 B2 · 单变量微积分与初等分析

年份 2023 题号 B2 入口积分

For each positive integer n , let $k(n)$ be the number of ones in the binary representation of $2023 \cdot n$. What is the minimum value of $k(n)$?

1985 年 B3 · 单变量微积分与初等分析

年份 1985 题号 B3 入口积分

Let

$$\begin{array}{cccc} a_{1,1} & a_{1,2} & a_{1,3} & \dots \\ a_{2,1} & a_{2,2} & a_{2,3} & \dots \\ a_{3,1} & a_{3,2} & a_{3,3} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

be a doubly infinite array of positive integers, and suppose each positive integer appears exactly eight times in the array. Prove that $a_{m,n} > mn$ for some pair of positive integers (m, n) .

1988 年 B3 · 单变量微积分与初等分析

年份 1988 题号 B3 入口积分

For every n in the set $N = \{1, 2, \dots\}$ of positive integers, let r_n be the minimum value of $|c - d\sqrt{3}|$ for all nonnegative integers c and d with $c + d = n$. Find, with proof, the smallest positive real number g with $r_n \leq g$ for all $n \in N$.

1999 年 B3 · 单变量微积分与初等分析

年份 1999 题号 B3 入口积分

Let $A = \{(x, y) : 0 \leq x, y < 1\}$. For $(x, y) \in A$, let

$$S(x, y) = \sum_{\frac{1}{2} \leq \frac{m}{n} \leq 2} x^m y^n,$$

where the sum ranges over all pairs (m, n) of positive integers satisfying the indicated inequalities. Evaluate

$$\lim_{(x,y) \rightarrow (1,1), (x,y) \in A} (1 - xy^2)(1 - x^2y)S(x, y).$$

2000 年 B3 · 单变量微积分与初等分析

年份 2000 题号 B3 入口级数/求和

Let $f(t) = \sum_{j=1}^N a_j \sin(2\pi jt)$, where each a_j is real and a_N is not equal to 0. Let N_k denote the number of zeroes (including multiplicities) of $\frac{d^k f}{dt^k}$. Prove that

$$N_0 \leq N_1 \leq N_2 \leq \cdots \text{ and } \lim_{k \rightarrow \infty} N_k = 2N.$$

[Editorial clarification: only zeroes in $[0, 1)$ should be counted.]

2001 年 B3 · 单变量微积分与初等分析

年份 2001 题号 B3 入口积分

For any positive integer n , let $\langle n \rangle$ denote the closest integer to $\sqrt{n}$. Evaluate

$$\sum_{n=1}^{\infty} \frac{2^{\langle n \rangle} + 2^{-\langle n \rangle}}{2^n}.$$

2002 年 B3 · 单变量微积分与初等分析

年份 2002 题号 B3 入口积分

Show that, for all integers $n > 1$,

$$\frac{1}{2ne} < \frac{1}{e} - \left(1 - \frac{1}{n}\right)^n < \frac{1}{ne}.$$

2007 年 B3 · 单变量微积分与初等分析

年份 2007 题号 B3 入口积分

Let $x_0 = 1$ and for $n \geq 0$, let $x_{n+1} = 3x_n + \lfloor x_n \sqrt{5} \rfloor$. In particular, $x_1 = 5$, $x_2 = 26$, $x_3 = 136$, $x_4 = 712$. Find a closed-form expression for x_{2007} . ($\lfloor a \rfloor$ means the largest integer $\leq a$.)

2010 年 B3 · 单变量微积分与初等分析

年份 2010 题号 B3 入口积分

There are 2010 boxes labeled $B_1, B_2, \dots, B_{2010}$, and $2010n$ balls have been distributed among them, for some positive integer n . You may redistribute the balls by a sequence of moves, each of which consists of choosing an i and moving *exactly* i balls from box B_i into any one other box. For which values of n is it possible to reach the distribution with exactly n balls in each box, regardless of the initial distribution of balls?

2017 年 B3 · 单变量微积分与初等分析

年份 2017 题号 B3 入口级数/求和

Suppose that $f(x) = \sum_{i=0}^{\infty} c_i x^i$ is a power series for which each coefficient c_i is 0 or 1. Show that if $f(2/3) = 3/2$, then $f(1/2)$ must be irrational.

2018 年 B3 · 单变量微积分与初等分析

年份 2018 题号 B3 入口积分

Find all positive integers $n < 10^{100}$ for which simultaneously n divides 2^n , $n - 1$ divides $2^n - 1$, and $n - 2$ divides $2^n - 2$.

2024 年 B3 · 单变量微积分与初等分析

年份 2024 题号 B3 入口三角函数

Let r_n be the n th smallest positive solution to $\tan x = x$, where the argument of tangent is in radians. Prove that

$$0 < r_{n+1} - r_n - \pi < \frac{1}{(n^2 + n)\pi}$$

for $n \geq 1$.

1986 年 B4 · 单变量微积分与初等分析

年份 1986 题号 B4 入口积分

For a positive real number r , let $G(r)$ be the minimum value of $|r - \sqrt{m^2 + 2n^2}|$ for all integers m and n . Prove or disprove the assertion that $\lim_{r \rightarrow \infty} G(r)$ exists and

equals 0.

1987 年 B4 · 单变量微积分与初等分析

年份 1987 题号 B4 入口极限

Let $(x_1, y_1) = (0.8, 0.6)$ and let $x_{n+1} = x_n \cos y_n - y_n \sin y_n$ and $y_{n+1} = x_n \sin y_n + y_n \cos y_n$ for $n = 1, 2, 3, \dots$. For each of $\lim_{n \rightarrow \infty} x_n$ and $\lim_{n \rightarrow \infty} y_n$, prove that the limit exists and find it or prove that the limit does not exist.

1993 年 B4 · 单变量微积分与初等分析

年份 1993 题号 B4 入口积分

The function $K(x, y)$ is positive and continuous for $0 \leq x \leq 1, 0 \leq y \leq 1$, and the functions $f(x)$ and $g(x)$ are positive and continuous for $0 \leq x \leq 1$. Suppose that for all $x, 0 \leq x \leq 1$,

$$\int_0^1 f(y)K(x, y) dy = g(x)$$

and

$$\int_0^1 g(y)K(x, y) dy = f(x).$$

Show that $f(x) = g(x)$ for $0 \leq x \leq 1$.

1995 年 B4 · 单变量微积分与初等分析

年份 1995 题号 B4 入口积分

Evaluate

$$\sqrt[s]{2207 - \frac{1}{2207 - \frac{1}{2207 - \dots}}}.$$

Express your answer in the form $\frac{a+b\sqrt{c}}{d}$, where a, b, c, d are integers.

1997 年 B4 · 单变量微积分与初等分析

年份 1997 题号 B4 入口积分

Let $a_{m,n}$ denote the coefficient of x^n in the expansion of $(1 + x + x^2)^m$. Prove that for all [integers] $k \geq 0$,

$$0 \leq \sum_{i=0}^{\lfloor \frac{2k}{3} \rfloor} (-1)^i a_{k-i,i} \leq 1.$$

1998 年 B4 · 单变量微积分与初等分析

年份 1998 题号 B4 入口积分

Find necessary and sufficient conditions on positive integers m and n so that

$$\sum_{i=0}^{mn-1} (-1)^{\lfloor i/m \rfloor + \lfloor i/n \rfloor} = 0.$$

1999 年 B4 · 单变量微积分与初等分析

年份 1999 题号 B4 入口导数/可微

Let f be a real function with a continuous third derivative such that $f(x), f'(x), f''(x), f'''(x)$ are positive for all x . Suppose that $f'''(x) \leq f(x)$ for all x . Show that $f'(x) < 2f(x)$ for all x .

2003 年 B4 · 单变量微积分与初等分析

年份 2003 题号 B4 入口积分

Let $f(z) = az^4 + bz^3 + cz^2 + dz + e = a(z-r_1)(z-r_2)(z-r_3)(z-r_4)$ where a, b, c, d, e are integers, $a \neq 0$. Show that if $r_1 + r_2$ is a rational number and $r_1 + r_2 \neq r_3 + r_4$, then $r_1 r_2$ is a rational number.

2005 年 B4 · 单变量微积分与初等分析

年份 2005 题号 B4 入口积分

For positive integers m and n , let $f(m, n)$ denote the number of n -tuples $(x_1, x_2, \dots, x_n)$ of integers such that $|x_1| + |x_2| + \dots + |x_n| \leq m$. Show that $f(m, n) = f(n, m)$.

2012 年 B4 · 单变量微积分与初等分析

年份 2012 题号 B4 入口极限

Suppose that $a_0 = 1$ and that $a_{n+1} = a_n + e^{-a_n}$ for $n = 0, 1, 2, \dots$. Does $a_n - \log n$ have a finite limit as $n \rightarrow \infty$? (Here $\log n = \log_e n = \ln n$.)

2017 年 B4 · 单变量微积分与初等分析

年份 2017 题号 B4 入口级数/求和

Evaluate the sum

$$\begin{aligned} & \sum_{k=0}^{\infty} \left(3 \cdot \frac{\ln(4k+2)}{4k+2} - \frac{\ln(4k+3)}{4k+3} - \frac{\ln(4k+4)}{4k+4} - \frac{\ln(4k+5)}{4k+5} \right) \\ &= 3 \cdot \frac{\ln 2}{2} - \frac{\ln 3}{3} - \frac{\ln 4}{4} - \frac{\ln 5}{5} + 3 \cdot \frac{\ln 6}{6} - \frac{\ln 7}{7} \\ & \quad - \frac{\ln 8}{8} - \frac{\ln 9}{9} + 3 \cdot \frac{\ln 10}{10} - \cdots \end{aligned}$$

(As usual, $\ln x$ denotes the natural logarithm of x .)

2018 年 B4 · 单变量微积分与初等分析

年份 2018 题号 B4 入口递推/序列

Given a real number a , we define a sequence by $x_0 = 1$, $x_1 = x_2 = a$, and $x_{n+1} = 2x_n x_{n-1} - x_{n-2}$ for $n \geq 2$. Prove that if $x_n = 0$ for some n , then the sequence is periodic.

2020 年 B4 · 单变量微积分与初等分析

年份 2020 题号 B4 入口积分

Let n be a positive integer, and let V_n be the set of integer $(2n+1)$ -tuples $\mathbf{v} = (s_0, s_1, \dots, s_{2n-1}, s_{2n})$ for which $s_0 = s_{2n} = 0$ and $|s_j - s_{j-1}| = 1$ for $j = 1, 2, \dots, 2n$. Define

$$q(\mathbf{v}) = 1 + \sum_{j=1}^{2n-1} 3^{s_j},$$

and let $M(n)$ be the average of $\frac{1}{q(\mathbf{v})}$ over all $\mathbf{v} \in V_n$. Evaluate $M(2020)$.

2021 年 B4 · 单变量微积分与初等分析

年份 2021 题号 B4 入口递推/序列

Let $F_0, F_1, \dots$ be the sequence of Fibonacci numbers, with $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. For $m > 2$, let R_m be the remainder when the product $\prod_{k=1}^{F_m-1} k^k$ is divided by F_m . Prove that R_m is also a Fibonacci number.

2022 年 B4 · 单变量微积分与初等分析

年份 2022 题号 B4 入口积分

Find all integers n with $n \geq 4$ for which there exists a sequence of distinct real numbers $x_1, \dots, x_n$ such that each of the sets

$$\{x_1, x_2, x_3\}, \{x_2, x_3, x_4\}, \dots, \\ \{x_{n-2}, x_{n-1}, x_n\}, \{x_{n-1}, x_n, x_1\}, \text{ and } \{x_n, x_1, x_2\}$$

forms a 3-term arithmetic progression when arranged in increasing order.

1985 年 B5 · 单变量微积分与初等分析

年份 1985 题号 B5 入口积分

Evaluate $\int_0^\infty t^{-1/2} e^{-1985(t+t^{-1})} dt$. You may assume that $\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}$.

1994 年 B5 · 单变量微积分与初等分析

年份 1994 题号 B5 入口积分

For any real number α , define the function $f_\alpha(x) = \lfloor \alpha x \rfloor$. Let n be a positive integer. Show that there exists an α such that for $1 \leq k \leq n$,

$$f_\alpha^k(n^2) = n^2 - k = f_{\alpha^k}(n^2).$$

1996 年 B5 · 单变量微积分与初等分析

年份 1996 题号 B5 入口环/形式幂级数

Given a finite string S of symbols X and O , we write $\Delta(S)$ for the number of X 's in S minus the number of O 's. For example, $\Delta(XOOXOOX) = -1$. We call a string S **balanced** if every substring T of (consecutive symbols of) S has $-2 \leq \Delta(T) \leq 2$. Thus, $XOOXOOX$ is not balanced, since it contains the substring $OOXOO$. Find, with proof, the number of balanced strings of length n .

2010 年 B5 · 单变量微积分与初等分析

年份 2010 题号 B5 入口三角函数

Is there a strictly increasing function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f'(x) = f(f(x))$ for all x ?

2011 年 B5 · 单变量微积分与初等分析

年份 2011 题号 B5 入口积分

Let $a_1, a_2, \dots$ be real numbers. Suppose that there is a constant A such that for all n ,

$$\int_{-\infty}^{\infty} \left(\sum_{i=1}^n \frac{1}{1 + (x - a_i)^2} \right)^2 dx \leq An.$$

Prove there is a constant $B > 0$ such that for all n ,

$$\sum_{i,j=1}^n (1 + (a_i - a_j)^2) \geq Bn^3.$$

2016 年 B5 · 单变量微积分与初等分析

年份 2016 题号 B5 入口积分

Find all functions f from the interval $(1, \infty)$ to $(1, \infty)$ with the following property: if $x, y \in (1, \infty)$ and $x^2 \leq y \leq x^3$, then $(f(x))^2 \leq f(y) \leq (f(x))^3$.

1988 年 B6 · 单变量微积分与初等分析

年份 1988 题号 B6 入口积分

Prove that there exist an infinite number of ordered pairs (a, b) of integers such that for every positive integer t , the number $at + b$ is a triangular number if and only if t is a triangular number. (The triangular numbers are the $t_n = n(n+1)/2$ with n in $\{0, 1, 2, \dots\}$.)

1991 年 B6 · 单变量微积分与初等分析

年份 1991 题号 B6 入口三角函数

Let a and b be positive numbers. Find the largest number c , in terms of a and b , such that

$$a^x b^{1-x} \leq a \frac{\sinh ux}{\sinh u} + b \frac{\sinh u(1-x)}{\sinh u}$$

for all u with $0 < |u| \leq c$ and for all x , $0 < x < 1$. (Note: $\sinh u = (e^u - e^{-u})/2$.)

1993 年 B6 · 单变量微积分与初等分析

年份 1993 题号 B6 入口积分

Let S be a set of three, not necessarily distinct, positive integers. Show that one can transform S into a set containing 0 by a finite number of applications of the following rule: Select two of the three integers, say x and y , where $x \leq y$ and replace them with $2x$ and $y - x$.

1995 年 B6 · 单变量微积分与初等分析

年份 1995 题号 B6 入口积分

For a positive real number α , define

$$S(\alpha) = \{\lfloor n\alpha \rfloor : n = 1, 2, 3, \dots\}.$$

Prove that $\{1, 2, 3, \dots\}$ cannot be expressed as the disjoint union of three sets $S(\alpha)$, $S(\beta)$ and $S(\gamma)$. [As usual, $\lfloor x \rfloor$ is the greatest integer $\leq x$.]

1998 年 B6 · 单变量微积分与初等分析

年份 1998 题号 B6 入口积分

Prove that, for any integers a, b, c , there exists a positive integer n such that $\sqrt{n^3 + an^2 + bn + c}$ is not an integer.

2001 年 B6 · 单变量微积分与初等分析

年份 2001 题号 B6 入口积分

Assume that $(a_n)_{n \geq 1}$ is an increasing sequence of positive real numbers such that $\lim a_n/n = 0$. Must there exist infinitely many positive integers n such that $a_{n-i} + a_{n+i} < 2a_n$ for $i = 1, 2, \dots, n-1$?

2006 年 B6 · 单变量微积分与初等分析

年份 2006 题号 B6 入口积分

Let k be an integer greater than 1. Suppose $a_0 > 0$, and define

$$a_{n+1} = a_n + \frac{1}{\sqrt[k]{a_n}}$$

for $n > 0$. Evaluate

$$\lim_{n \rightarrow \infty} \frac{a_n^{k+1}}{n^k}.$$

2009 年 B6 · 单变量微积分与初等分析

年份 2009 题号 B6 入口积分

Prove that for every positive integer n , there is a sequence of integers $a_0, a_1, \dots, a_{2009}$ with $a_0 = 0$ and $a_{2009} = n$ such that each term after a_0 is either an earlier term plus 2^k for some nonnegative integer k , or of the form $b \bmod c$ for some earlier positive terms b and c . [Here $b \bmod c$ denotes the remainder when b is divided by c , so $0 \leq (b \bmod c) < c$.]

2016 年 B6 · 单变量微积分与初等分析

年份 2016 题号 B6 入口级数/求和

Evaluate

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \sum_{n=0}^{\infty} \frac{1}{k2^n + 1}.$$

2018 年 B6 · 单变量微积分与初等分析

年份 2018 题号 B6 入口级数/求和

Let S be the set of sequences of length 2018 whose terms are in the set $\{1, 2, 3, 4, 5, 6, 10\}$ and sum to 3860. Prove that the cardinality of S is at most

$$2^{3860} \cdot \left(\frac{2018}{2048} \right)^{2018}.$$

2020 年 B6 · 单变量微积分与初等分析

年份 2020 题号 B6 入口积分

Let n be a positive integer. Prove that

$$\sum_{k=1}^n (-1)^{\lfloor k(\sqrt{2}-1) \rfloor} \geq 0.$$

(As usual, $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x .)

2024 年 B6 · 单变量微积分与初等分析

年份 2024 题号 B6 入口级数/求和

For a real number a , let $F_a(x) = \sum_{n \geq 1} n^a e^{2n} x^{n^2}$ for $0 \leq x < 1$. Find a real number c such that

$$\begin{aligned} \lim_{x \rightarrow 1^-} F_a(x) e^{-1/(1-x)} &= 0 && \text{for all } a < c, \text{ and} \\ \lim_{x \rightarrow 1^-} F_a(x) e^{-1/(1-x)} &= \infty && \text{for all } a > c. \end{aligned}$$

2025 年 B6 · 单变量微积分与初等分析

年份 2025 题号 B6 入口三角函数

Let $\mathbb{N} = \{1, 2, 3, \dots\}$. Find the largest real constant r such that there exists a function $g: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$g(n+1) - g(n) \geq (g(g(n)))^r$$

for all $n \in \mathbb{N}$.

第 5 章

线性代数

范畴要点： 矩阵、行列式、特征值、向量空间、秩、线性变换

1991 年 A1 · 线性代数

年份 1991 题号 A1 入口积分

A 2×3 rectangle has vertices as $(0, 0)$, $(2, 0)$, $(0, 3)$, and $(2, 3)$. It rotates 90° clockwise about the point $(2, 0)$. It then rotates 90° clockwise about the point $(5, 0)$, then 90° clockwise about the point $(7, 0)$, and finally, 90° clockwise about the point $(10, 0)$. (The side originally on the x -axis is now back on the x -axis.) Find the area of the region above the x -axis and below the curve traced out by the point whose initial position is $(1, 1)$.

1991 年 A2 · 线性代数

年份 1991 题号 A2 入口矩阵/线性算子

Let $\mathbf{A}$ and $\mathbf{B}$ be different $n \times n$ matrices with real entries. If $\mathbf{A}^3 = \mathbf{B}^3$ and $\mathbf{A}^2\mathbf{B} = \mathbf{B}^2\mathbf{A}$, can $\mathbf{A}^2 + \mathbf{B}^2$ be invertible?

2008 年 A2 · 线性代数

年份 2008 题号 A2 入口矩阵/线性算子

Alan and Barbara play a game in which they take turns filling entries of an initially empty 2008×2008 array. Alan plays first. At each turn, a player chooses a real number and places it in a vacant entry. The game ends when all the entries are filled. Alan wins if the determinant of the resulting matrix is nonzero; Barbara wins if it is zero. Which player has a winning strategy?

2014 年 A2 · 线性代数

年份 2014 题号 A2 入口矩阵/线性算子

Let A be the $n \times n$ matrix whose entry in the i -th row and j -th column is

$$\frac{1}{\min(i, j)}$$

for $1 \leq i, j \leq n$. Compute $\det(A)$.

2018 年 A2 · 线性代数

年份 2018 题号 A2 入口矩阵/线性算子

Let $S_1, S_2, \dots, S_{2^n-1}$ be the nonempty subsets of $\{1, 2, \dots, n\}$ in some order, and let M be the $(2^n - 1) \times (2^n - 1)$ matrix whose (i, j) entry is

$$m_{ij} = \begin{cases} 0 & \text{if } S_i \cap S_j = \emptyset; \\ 1 & \text{otherwise.} \end{cases}$$

Calculate the determinant of M .

2004 年 A3 · 线性代数

年份 2004 题号 A3 入口矩阵/线性算子

Define a sequence $\{u_n\}_{n=0}^\infty$ by $u_0 = u_1 = u_2 = 1$, and thereafter by the condition that

$$\det \begin{pmatrix} u_n & u_{n+1} \\ u_{n+2} & u_{n+3} \end{pmatrix} = n!$$

for all $n \geq 0$. Show that u_n is an integer for all n . (By convention, $0! = 1$.)

2009 年 A3 · 线性代数

年份 2009 题号 A3 入口矩阵/线性算子

Let d_n be the determinant of the $n \times n$ matrix whose entries, from left to right and then from top to bottom, are $\cos 1, \cos 2, \dots, \cos n^2$. (For example,

$$d_3 = \begin{vmatrix} \cos 1 & \cos 2 & \cos 3 \\ \cos 4 & \cos 5 & \cos 6 \\ \cos 7 & \cos 8 & \cos 9 \end{vmatrix}.$$

The argument of $\cos$ is always in radians, not degrees.) Evaluate $\lim_{n \rightarrow \infty} d_n$.

1986 年 A4 · 线性代数

年份 1986 题号 A4 入口矩阵/线性算子

A *transversal* of an $n \times n$ matrix A consists of n entries of A , no two in the same row or column. Let $f(n)$ be the number of $n \times n$ matrices A satisfying the following two conditions:

- Each entry $\alpha_{i,j}$ of A is in the set $\{-1, 0, 1\}$.
- The sum of the n entries of a transversal is the same for all transversals of A .

An example of such a matrix A is

$$A = \begin{pmatrix} -1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Determine with proof a formula for $f(n)$ of the form

$$f(n) = a_1 b_1^n + a_2 b_2^n + a_3 b_3^n + a_4,$$

where the a_i 's and b_i 's are rational numbers.

1994 年 A4 · 线性代数

年份 1994 题号 A4 入口矩阵/线性算子

Let A and B be 2×2 matrices with integer entries such that $A, A+B, A+2B, A+3B$, and $A+4B$ are all invertible matrices whose inverses have integer entries. Show that $A+5B$ is invertible and that its inverse has integer entries.

2002 年 A4 · 线性代数

年份 2002 题号 A4 入口矩阵/线性算子

In Determinant Tic-Tac-Toe, Player 1 enters a 1 in an empty 3×3 matrix. Player 0 counters with a 0 in a vacant position, and play continues in turn until the 3×3 matrix is completed with five 1's and four 0's. Player 0 wins if the determinant is 0 and player 1 wins otherwise. Assuming both players pursue optimal strategies, who will win and how?

2005 年 A4 · 线性代数

年份 2005 题号 A4 入口矩阵/线性算子

Let H be an $n \times n$ matrix all of whose entries are ± 1 and whose rows are mutually orthogonal. Suppose H has an $a \times b$ submatrix whose entries are all 1. Show that $ab \leq n$.

2011 年 A4 · 线性代数

年份 2011 题号 A4 入口矩阵/线性算子

For which positive integers n is there an $n \times n$ matrix with integer entries such that every dot product of a row with itself is even, while every dot product of two different rows is odd?

2025 年 A4 · 线性代数

年份 2025 题号 A4 入口矩阵/线性算子

Find the minimal value of k such that there exist k -by- k real matrices $A_1, \dots, A_{2025}$ with the property that $A_i A_j = A_j A_i$ if and only if $|i - j| \in \{0, 1, 2024\}$.

1990 年 A5 · 线性代数

年份 1990 题号 A5 入口矩阵/线性算子

If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same size such that $\mathbf{ABAB} = \mathbf{0}$, does it follow that $\mathbf{BABA} = \mathbf{0}$?

1988 年 A6 · 线性代数

年份 1988 题号 A6 入口矩阵

If a linear transformation A on an n -dimensional vector space has $n + 1$ eigenvectors such that any n of them are linearly independent, does it follow that A is a scalar multiple of the identity? Prove your answer.

1995 年 A6 · 线性代数

年份 1995 题号 A6 入口矩阵/线性算子

Suppose that each of n people writes down the numbers 1,2,3 in random order in one column of a $3 \times n$ matrix, with all orders equally likely and with the orders for different columns independent of each other. Let the row sums a, b, c of the resulting matrix be rearranged (if necessary) so that $a \leq b \leq c$. Show that for some $n \geq 1995$, it is at least four times as likely that both $b = a + 1$ and $c = a + 2$ as that $a = b = c$.

2014 年 A6 · 线性代数

年份 2014 题号 A6 入口矩阵/线性算子

Let n be a positive integer. What is the largest k for which there exist $n \times n$ matrices $M_1, \dots, M_k$ and $N_1, \dots, N_k$ with real entries such that for all i and j , the matrix product $M_i N_j$ has a zero entry somewhere on its diagonal if and only if $i \neq j$?

2015 年 A6 · 线性代数

年份 2015 题号 A6 入口多项式

Let n be a positive integer. Suppose that A, B , and M are $n \times n$ matrices with real entries such that $AM = MB$, and such that A and B have the same characteristic polynomial. Prove that $\det(A - MX) = \det(B - XM)$ for every $n \times n$ matrix X with real entries.

2024 年 A6 · 线性代数

年份 2024 题号 A6 入口矩阵/线性算子

Let $c_0, c_1, c_2, \dots$ be the sequence defined so that

$$\frac{1 - 3x - \sqrt{1 - 14x + 9x^2}}{4} = \sum_{k=0}^{\infty} c_k x^k$$

for sufficiently small x . For a positive integer n , let A be the n -by- n matrix with i, j -entry c_{i+j-1} for i and j in $\{1, \dots, n\}$. Find the determinant of A .

2018 年 B1 · 线性代数

年份 2018 题号 B1 入口矩阵/线性算子

Let $\mathcal{P}$ be the set of vectors defined by

$$\mathcal{P} = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \middle| 0 \leq a \leq 2, 0 \leq b \leq 100, \text{ and } a, b \in \mathbb{Z} \right\}.$$

Find all $\mathbf{v} \in \mathcal{P}$ such that the set $\mathcal{P} \setminus \{\mathbf{v}\}$ obtained by omitting vector $\mathbf{v}$ from $\mathcal{P}$ can be partitioned into two sets of equal size and equal sum.

1990 年 B3 · 线性代数

年份 1990 题号 B3 入口矩阵/线性算子

Let S be a set of 2×2 integer matrices whose entries a_{ij} (1) are all squares of integers and, (2) satisfy $a_{ij} \leq 200$. Show that if S has more than 50387 ($= 15^4 - 15^2 - 15 + 2$) elements, then it has two elements that commute.

1995 年 B3 · 线性代数

年份 1995 题号 B3 入口矩阵/线性算子

To each positive integer with n^2 decimal digits, we associate the determinant of the matrix obtained by writing the digits in order across the rows. For example, for $n = 2$, to the integer 8617 we associate $\det \begin{pmatrix} 8 & 6 \\ 1 & 7 \end{pmatrix} = 50$. Find, as a function of n , the sum of all the determinants associated with n^2 -digit integers. (Leading digits are assumed to be nonzero; for example, for $n = 2$, there are 9000 determinants.)

2014 年 B3 · 线性代数

年份 2014 题号 B3 入口矩阵/线性算子

Let A be an $m \times n$ matrix with rational entries. Suppose that there are at least $m + n$ distinct prime numbers among the absolute values of the entries of A . Show

that the rank of A is at least 2.

2015 年 B3 · 线性代数

年份 2015 题号 B3 入口矩阵/线性算子

Let S be the set of all 2×2 real matrices

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

whose entries a, b, c, d (in that order) form an arithmetic progression. Find all matrices M in S for which there is some integer $k > 1$ such that M^k is also in S .

2019 年 B3 · 线性代数

年份 2019 题号 B3 入口矩阵/线性算子

Let Q be an n -by- n real orthogonal matrix, and let $u \in \mathbb{R}^n$ be a unit column vector (that is, $u^T u = 1$). Let $P = I - 2uu^T$, where I is the n -by- n identity matrix. Show that if 1 is not an eigenvalue of Q , then 1 is an eigenvalue of PQ .

1994 年 B4 · 线性代数

年份 1994 题号 B4 入口矩阵/线性算子

For $n \geq 1$, let d_n be the greatest common divisor of the entries of $A^n - I$, where

$$A = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix} \quad \text{and} \quad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Show that $\lim_{n \rightarrow \infty} d_n = \infty$.

1996 年 B4 · 线性代数

年份 1996 题号 B4 入口矩阵/线性算子

For any square matrix A , we can define $\sin A$ by the usual power series:

$$\sin A = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} A^{2n+1}.$$

Prove or disprove: there exists a 2×2 matrix A with real entries such that

$$\sin A = \begin{pmatrix} 1 & 1996 \\ 0 & 1 \end{pmatrix}.$$

2009 年 B4 · 线性代数

年份 2009 题号 B4 入口多项式

Say that a polynomial with real coefficients in two variables, x, y , is *balanced* if the average value of the polynomial on each circle centered at the origin is 0. The balanced polynomials of degree at most 2009 form a vector space V over $\mathbb{R}$. Find the dimension of V .

2011 年 B4 · 线性代数

年份 2011 题号 B4 入口矩阵/线性算子

In a tournament, 2011 players meet 2011 times to play a multiplayer game. Every game is played by all 2011 players together and ends with each of the players either winning or losing. The standings are kept in two 2011×2011 matrices, $T = (T_{hk})$ and $W = (W_{hk})$. Initially, $T = W = 0$. After every game, for every (h, k) (including for $h = k$), if players h and k tied (that is, both won or both lost), the entry T_{hk} is increased by 1, while if player h won and player k lost, the entry W_{hk} is increased by 1 and W_{kh} is decreased by 1.

Prove that at the end of the tournament, $\det(T + iW)$ is a non-negative integer divisible by 2^{2010} .

2016 年 B4 · 线性代数

年份 2016 题号 B4 入口矩阵/线性算子

Let A be a $2n \times 2n$ matrix, with entries chosen independently at random. Every entry is chosen to be 0 or 1, each with probability $1/2$. Find the expected value of $\det(A - A^t)$ (as a function of n), where A^t is the transpose of A .

2025 年 B4 · 线性代数

年份 2025 题号 B4 入口矩阵/线性算子

For $n \geq 2$, let $A = [a_{i,j}]_{i,j=1}^n$ be an n -by- n matrix of nonnegative integers such that

- (a) $a_{i,j} = 0$ when $i + j \leq n$;
- (b) $a_{i+1,j} \in \{a_{i,j}, a_{i,j} + 1\}$ when $1 \leq i \leq n - 1$ and $1 \leq j \leq n$; and
- (c) $a_{i,j+1} \in \{a_{i,j}, a_{i,j} + 1\}$ when $1 \leq i \leq n$ and $1 \leq j \leq n - 1$.

Let S be the sum of the entries of A , and let N be the number of nonzero entries of A . Prove that

$$S \leq \frac{(n+2)N}{3}.$$

1987 年 B5 · 线性代数

年份 1987 题号 B5 入口复数函数

Let O_n be the n -dimensional vector $(0, 0, \dots, 0)$. Let M be a $2n \times n$ matrix of complex numbers such that whenever $(z_1, z_2, \dots, z_{2n})M = O_n$, with complex z_i , not all zero, then at least one of the z_i is not real. Prove that for arbitrary real numbers $r_1, r_2, \dots, r_{2n}$, there are complex numbers $w_1, w_2, \dots, w_n$ such that

$$\operatorname{re} \left[M \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \right] = \begin{pmatrix} r_1 \\ \vdots \\ r_n \end{pmatrix}.$$

(Note: if C is a matrix of complex numbers, $\operatorname{re}(C)$ is the matrix whose entries are the real parts of the entries of C .)

1988 年 B5 · 线性代数

年份 1988 题号 B5 入口矩阵/线性算子

For positive integers n , let M_n be the $2n+1$ by $2n+1$ skew-symmetric matrix for which each entry in the first n subdiagonals below the main diagonal is 1 and each of the remaining entries below the main diagonal is -1. Find, with proof, the rank of M_n . (According to one definition, the rank of a matrix is the largest k such that there is a $k \times k$ submatrix with nonzero determinant.)

One may note that

$$M_1 = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 0 & -1 & -1 & 1 & 1 \\ 1 & 0 & -1 & -1 & 1 \\ 1 & 1 & 0 & -1 & -1 \\ -1 & 1 & 1 & 0 & -1 \\ -1 & -1 & 1 & 1 & 0 \end{pmatrix}.$$

1992 年 B5 · 线性代数

年份 1992 题号 B5 入口行列式

Let D_n denote the value of the $(n-1) \times (n-1)$ determinant

$$\begin{bmatrix} 3 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 4 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 5 & 1 & \cdots & 1 \\ 1 & 1 & 1 & 6 & \cdots & 1 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & 1 & \cdots & n+1 \end{bmatrix}.$$

Is the set $\left\{\frac{D_n}{n!}\right\}_{n \geq 2}$ bounded?

1999 年 B5 · 线性代数

年份 1999 题号 B5 入口矩阵/线性算子

For an integer $n \geq 3$, let $\theta = 2\pi/n$. Evaluate the determinant of the $n \times n$ matrix $I+A$, where I is the $n \times n$ identity matrix and $A = (a_{jk})$ has entries $a_{jk} = \cos(j\theta + k\theta)$ for all j, k .

2021 年 B5 · 线性代数

年份 2021 题号 B5 入口矩阵/线性算子

Say that an n -by- n matrix $A = (a_{ij})_{1 \leq i, j \leq n}$ with integer entries is *very odd* if, for every nonempty subset S of $\{1, 2, \dots, n\}$, the $|S|$ -by- $|S|$ submatrix $(a_{ij})_{i, j \in S}$ has odd determinant. Prove that if A is very odd, then A^k is very odd for every $k \geq 1$.

1992 年 B6 · 线性代数

年份 1992 题号 B6 入口矩阵/线性算子

Let $\mathcal{M}$ be a set of real $n \times n$ matrices such that

- (i) $I \in \mathcal{M}$, where I is the $n \times n$ identity matrix;
- (ii) if $A \in \mathcal{M}$ and $B \in \mathcal{M}$, then either $AB \in \mathcal{M}$ or $-AB \in \mathcal{M}$, but not both;
- (iii) if $A \in \mathcal{M}$ and $B \in \mathcal{M}$, then either $AB = BA$ or $AB = -BA$;
- (iv) if $A \in \mathcal{M}$ and $A \neq I$, there is at least one $B \in \mathcal{M}$ such that $AB = -BA$.

Prove that $\mathcal{M}$ contains at most n^2 matrices.

2002 年 B6 · 线性代数

年份 2002 题号 B6 入口多项式

Let p be a prime number. Prove that the determinant of the matrix

$$\begin{pmatrix} x & y & z \\ x^p & y^p & z^p \\ x^{p^2} & y^{p^2} & z^{p^2} \end{pmatrix}$$

is congruent modulo p to a product of polynomials of the form $ax + by + cz$, where a, b, c are integers. (We say two integer polynomials are congruent modulo p if corresponding coefficients are congruent modulo p .)

2010 年 B6 · 线性代数

年份 2010 题号 B6 入口矩阵/线性算子

Let A be an $n \times n$ matrix of real numbers for some $n \geq 1$. For each positive integer k , let $A^{[k]}$ be the matrix obtained by raising each entry to the k th power. Show that if $A^k = A^{[k]}$ for $k = 1, 2, \dots, n+1$, then $A^k = A^{[k]}$ for all $k \geq 1$.

2023 年 B6 · 线性代数

年份 2023 题号 B6 入口矩阵/线性算子

Let n be a positive integer. For i and j in $\{1, 2, \dots, n\}$, let $s(i, j)$ be the number of pairs (a, b) of nonnegative integers satisfying $ai + bj = n$. Let S be the n -by- n matrix

whose (i, j) entry is $s(i, j)$. For example, when $n = 5$, we have $S =$

$$\begin{bmatrix} 6 & 3 & 2 & 2 & 2 \\ 3 & 0 & 1 & 0 & 1 \\ 2 & 1 & 0 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 \\ 2 & 1 & 1 & 1 & 2 \end{bmatrix}.$$

Compute the determinant of S .

第 6 章

几何与多变量微积分

范畴要点：平面/空间几何、凸性、坐标法、面积体积、质心、多元函数

1987 年 A1 · 几何与多变量微积分

年份 1987 题号 A1 入口平面/空间几何

Curves A, B, C and D are defined in the plane as follows:

$$\begin{aligned} A &= \left\{ (x, y) : x^2 - y^2 = \frac{x}{x^2 + y^2} \right\}, \\ B &= \left\{ (x, y) : 2xy + \frac{y}{x^2 + y^2} = 3 \right\}, \\ C &= \{ (x, y) : x^3 - 3xy^2 + 3y = 1 \}, \\ D &= \{ (x, y) : 3x^2y - 3x - y^3 = 0 \}. \end{aligned}$$

Prove that $A \cap B = C \cap D$.

1988 年 A1 · 几何与多变量微积分

年份 1988 题号 A1 入口积分

Let R be the region consisting of the points (x, y) of the cartesian plane satisfying both $|x| - |y| \leq 1$ and $|y| \leq 1$. Sketch the region R and find its area.

1993 年 A1 · 几何与多变量微积分

年份 1993 题号 A1 入口积分

The horizontal line $y = c$ intersects the curve $y = 2x - 3x^3$ in the first quadrant as in the figure. Find c so that the areas of the two shaded regions are equal. [Figure not included. The first region is bounded by the y -axis, the line $y = c$ and the curve; the other lies under the curve and above the line $y = c$ between their two points of intersection.]

1996 年 A1 · 几何与多变量微积分

年份 1996 题号 A1 入口积分

Find the least number A such that for any two squares of combined area 1, a rectangle of area A exists such that the two squares can be packed in the rectangle (without interior overlap). You may assume that the sides of the squares are parallel

to the sides of the rectangle.

1997 年 A1 · 几何与多变量微积分

年份 1997 题号 A1 入口积分

A rectangle, $HOMF$, has sides $HO = 11$ and $OM = 5$. A triangle ABC has H as the intersection of the altitudes, O the center of the circumscribed circle, M the midpoint of BC , and F the foot of the altitude from A . What is the length of BC ?

1998 年 A1 · 几何与多变量微积分

年份 1998 题号 A1 入口进位制与数字

A right circular cone has base of radius 1 and height 3. A cube is inscribed in the cone so that one face of the cube is contained in the base of the cone. What is the side-length of the cube?

2006 年 A1 · 几何与多变量微积分

年份 2006 题号 A1 入口积分

Find the volume of the region of points (x, y, z) such that

$$(x^2 + y^2 + z^2 + 8)^2 \leq 36(x^2 + y^2).$$

2009 年 A1 · 几何与多变量微积分

年份 2009 题号 A1 入口积分

Let f be a real-valued function on the plane such that for every square $ABCD$ in the plane, $f(A) + f(B) + f(C) + f(D) = 0$. Does it follow that $f(P) = 0$ for all points P in the plane?

2011 年 A1 · 几何与多变量微积分

年份 2011 题号 A1 入口积分

Define a *growing spiral* in the plane to be a sequence of points with integer coordinates $P_0 = (0, 0), P_1, \dots, P_n$ such that $n \geq 2$ and:

- the directed line segments $P_0P_1, P_1P_2, \dots, P_{n-1}P_n$ are in the successive coordinate directions east (for P_0P_1), north, west, south, east, etc.;
- the lengths of these line segments are positive and strictly increasing.

[Picture omitted.] How many of the points (x, y) with integer coordinates $0 \leq x \leq 2011, 0 \leq y \leq 2011$ cannot be the last point, P_n of any growing spiral?

2013 年 A1 · 几何与多变量微积分

年份 2013 题号 A1 入口整除与同余

Recall that a regular icosahedron is a convex polyhedron having 12 vertices and 20 faces; the faces are congruent equilateral triangles. On each face of a regular icosahedron is written a nonnegative integer such that the sum of all 20 integers is 39. Show that there are two faces that share a vertex and have the same integer written on them.

2015 年 A1 · 几何与多变量微积分

年份 2015 题号 A1 入口积分

Let A and B be points on the same branch of the hyperbola $xy = 1$. Suppose that P is a point lying between A and B on this hyperbola, such that the area of the triangle APB is as large as possible. Show that the region bounded by the hyperbola and the chord AP has the same area as the region bounded by the hyperbola and the chord PB .

2021 年 A1 · 几何与多变量微积分

年份 2021 题号 A1 入口积分

A grasshopper starts at the origin in the coordinate plane and makes a sequence of hops. Each hop has length 5, and after each hop the grasshopper is at a point whose coordinates are both integers; thus, there are 12 possible locations for the grasshopper after the first hop. What is the smallest number of hops needed for the grasshopper to reach the point $(2021, 2021)$?

1985 年 A2 · 几何与多变量微积分

年份 1985 题号 A2 入口三角函数

Let T be an acute triangle. Inscribe a rectangle R in T with one side along a side of T . Then inscribe a rectangle S in the triangle formed by the side of R opposite the side on the boundary of T , and the other two sides of T , with one side along the side of R . For any polygon X , let $A(X)$ denote the area of X . Find the maximum value, or show that no maximum exists, of $\frac{A(R)+A(S)}{A(T)}$, where T ranges over all triangles and R, S over all rectangles as above.

1994 年 A2 · 几何与多变量微积分

年份 1994 题号 A2 入口平面/空间几何

Let A be the area of the region in the first quadrant bounded by the line $y = \frac{1}{2}x$, the x -axis, and the ellipse $\frac{1}{9}x^2 + y^2 = 1$. Find the positive number m such that A is equal to the area of the region in the first quadrant bounded by the line $y = mx$,

the y -axis, and the ellipse $\frac{1}{9}x^2 + y^2 = 1$.

1996 年 A2 · 几何与多变量微积分

年份 1996 题号 A2 入口积分

Let C_1 and C_2 be circles whose centers are 10 units apart, and whose radii are 1 and 3. Find, with proof, the locus of all points M for which there exists points X on C_1 and Y on C_2 such that M is the midpoint of the line segment XY .

2002 年 A2 · 几何与多变量微积分

年份 2002 题号 A2 入口积分

Given any five points on a sphere, show that some four of them must lie on a closed hemisphere.

2004 年 A2 · 几何与多变量微积分

年份 2004 题号 A2 入口平面几何

For $i = 1, 2$ let T_i be a triangle with side lengths a_i, b_i, c_i , and area A_i . Suppose that $a_1 \leq a_2, b_1 \leq b_2, c_1 \leq c_2$, and that T_2 is an acute triangle. Does it follow that $A_1 \leq A_2$?

2005 年 A2 · 几何与多变量微积分

年份 2005 题号 A2 入口积分

Let $\mathbf{S} = \{(a, b) | a = 1, 2, \dots, n, b = 1, 2, 3\}$. A *rook tour* of $\mathbf{S}$ is a polygonal path made up of line segments connecting points $p_1, p_2, \dots, p_{3n}$ in sequence such that

- (i) $p_i \in \mathbf{S}$,
- (ii) p_i and p_{i+1} are a unit distance apart, for $1 \leq i < 3n$,
- (iii) for each $p \in \mathbf{S}$ there is a unique i such that $p_i = p$. How many rook tours are there that begin at $(1, 1)$ and end at $(n, 1)$?

(An example of such a rook tour for $n = 5$ was depicted in the original.)

2007 年 A2 · 几何与多变量微积分

年份 2007 题号 A2 入口积分

Find the least possible area of a convex set in the plane that intersects both branches of the hyperbola $xy = 1$ and both branches of the hyperbola $xy = -1$. (A set S in the plane is called *convex* if for any two points in S the line segment connecting them is contained in S .)

2019 年 A2 · 几何与多变量微积分

年份 2019 题号 A2 入口三角函数

In the triangle $\triangle ABC$, let G be the centroid, and let I be the center of the inscribed circle. Let α and β be the angles at the vertices A and B , respectively. Suppose that the segment IG is parallel to AB and that $\beta = 2 \tan^{-1}(1/3)$. Find α .

1990 年 A3 · 几何与多变量微积分

年份 1990 题号 A3 入口积分

Prove that any convex pentagon whose vertices (no three of which are collinear) have integer coordinates must have area greater than or equal to $5/2$.

1994 年 A3 · 几何与多变量微积分

年份 1994 题号 A3 入口积分

Show that if the points of an isosceles right triangle of side length 1 are each colored with one of four colors, then there must be two points of the same color which are at least a distance $2 - \sqrt{2}$ apart.

2000 年 A3 · 几何与多变量微积分

年份 2000 题号 A3 入口三角函数

The octagon $P_1P_2P_3P_4P_5P_6P_7P_8$ is inscribed in a circle, with the vertices around the circumference in the given order. Given that the polygon $P_1P_3P_5P_7$ is a square of area 5, and the polygon $P_2P_4P_6P_8$ is a rectangle of area 4, find the maximum possible area of the octagon.

2021 年 A3 · 几何与多变量微积分

年份 2021 题号 A3 入口积分

Determine all positive integers N for which the sphere

$$x^2 + y^2 + z^2 = N$$

has an inscribed regular tetrahedron whose vertices have integer coordinates.

1988 年 A4 · 几何与多变量微积分

年份 1988 题号 A4 入口积分

- (a) If every point of the plane is painted one of three colors, do there necessarily exist two points of the same color exactly one inch apart?

(b) What if “three” is replaced by “nine”?

1990 年 A4 · 几何与多变量微积分

年份 1990 题号 A4 入口积分

Consider a paper punch that can be centered at any point of the plane and that, when operated, removes from the plane precisely those points whose distance from the center is irrational. How many punches are needed to remove every point?

1991 年 A4 · 几何与多变量微积分

年份 1991 题号 A4 入口积分

Does there exist an infinite sequence of closed discs $D_1, D_2, D_3, \dots$ in the plane, with centers $c_1, c_2, c_3, \dots$, respectively, such that

1. the c_i have no limit point in the finite plane,
2. the sum of the areas of the D_i is finite, and
3. every line in the plane intersects at least one of the D_i ?

2001 年 A4 · 几何与多变量微积分

年份 2001 题号 A4 入口积分

Triangle ABC has an area 1. Points E, F, G lie, respectively, on sides BC, CA, AB such that AE bisects BF at point R , BF bisects CG at point S , and CG bisects AE at point T . Find the area of the triangle RST .

2013 年 A4 · 几何与多变量微积分

年份 2013 题号 A4 入口进位制与数字

A finite collection of digits 0 and 1 is written around a circle. An *arc* of length $L \geq 0$ consists of L consecutive digits around the circle. For each arc w , let $Z(w)$ and $N(w)$ denote the number of 0's in w and the number of 1's in w , respectively. Assume that $|Z(w) - Z(w')| \leq 1$ for any two arcs w, w' of the same length. Suppose that some arcs $w_1, \dots, w_k$ have the property that

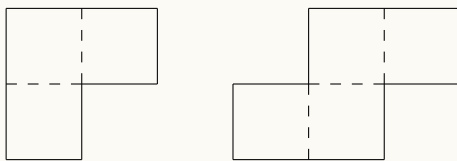
$$Z = \frac{1}{k} \sum_{j=1}^k Z(w_j) \text{ and } N = \frac{1}{k} \sum_{j=1}^k N(w_j)$$

are both integers. Prove that there exists an arc w with $Z(w) = Z$ and $N(w) = N$.

2016 年 A4 · 几何与多变量微积分

年份 2016 题号 A4 入口群论

Consider a $(2m-1) \times (2n-1)$ rectangular region, where m and n are integers such that $m, n \geq 4$. This region is to be tiled using tiles of the two types shown:



(The dotted lines divide the tiles into 1×1 squares.) The tiles may be rotated and reflected, as long as their sides are parallel to the sides of the rectangular region. They must all fit within the region, and they must cover it completely without overlapping.

What is the minimum number of tiles required to tile the region?

2019 年 A4 · 几何与多变量微积分

年份 2019 题号 A4 入口积分

Let f be a continuous real-valued function on $\mathbb{R}^3$. Suppose that for every sphere S of radius 1, the integral of $f(x, y, z)$ over the surface of S equals 0. Must $f(x, y, z)$ be identically 0?

2000 年 A5 · 几何与多变量微积分

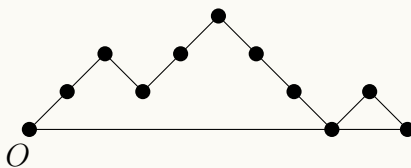
年份 2000 题号 A5 入口积分

Three distinct points with integer coordinates lie in the plane on a circle of radius $r > 0$. Show that two of these points are separated by a distance of at least $r^{1/3}$.

2003 年 A5 · 几何与多变量微积分

年份 2003 题号 A5 入口平面几何

A Dyck n -path is a lattice path of n upsteps $(1, 1)$ and n downsteps $(1, -1)$ that starts at the origin O and never dips below the x -axis. A return is a maximal sequence of contiguous downsteps that terminates on the x -axis. For example, the Dyck 5-path illustrated has two returns, of length 3 and 1 respectively.



Show that there is a one-to-one correspondence between the Dyck n -paths with no return of even length and the Dyck $(n-1)$ -paths.

2013 年 A5 · 几何与多变量微积分

年份 2013 题号 A5 入口积分

For $m \geq 3$, a list of $\binom{m}{3}$ real numbers a_{ijk} ($1 \leq i < j < k \leq m$) is said to be *area definite* for $\mathbb{R}^n$ if the inequality

$$\sum_{1 \leq i < j < k \leq m} a_{ijk} \cdot \text{Area}(\Delta A_i A_j A_k) \geq 0$$

holds for every choice of m points $A_1, \dots, A_m$ in $\mathbb{R}^n$. For example, the list of four numbers $a_{123} = a_{124} = a_{134} = 1$, $a_{234} = -1$ is area definite for $\mathbb{R}^2$. Prove that if a list of $\binom{m}{3}$ numbers is area definite for $\mathbb{R}^2$, then it is area definite for $\mathbb{R}^3$.

1998 年 A6 · 几何与多变量微积分

年份 1998 题号 A6 入口积分

Let A, B, C denote distinct points with integer coordinates in $\mathbb{R}^2$. Prove that if

$$(|AB| + |BC|)^2 < 8 \cdot [ABC] + 1$$

then A, B, C are three vertices of a square. Here $|XY|$ is the length of segment XY and $[ABC]$ is the area of triangle ABC .

2001 年 A6 · 几何与多变量微积分

年份 2001 题号 A6 入口平面几何

Can an arc of a parabola inside a circle of radius 1 have a length greater than 4?

2007 年 A6 · 几何与多变量微积分

年份 2007 题号 A6 入口积分

A *triangulation* $\mathcal{T}$ of a polygon P is a finite collection of triangles whose union is P , and such that the intersection of any two triangles is either empty, or a shared vertex, or a shared side. Moreover, each side is a side of exactly one triangle in $\mathcal{T}$. Say that $\mathcal{T}$ is *admissible* if every internal vertex is shared by 6 or more triangles. For example, [figure omitted.] Prove that there is an integer M_n , depending only on n , such that any admissible triangulation of a polygon P with n sides has at most M_n triangles.

2012 年 A6 · 几何与多变量微积分

年份 2012 题号 A6 入口积分

Let $f(x, y)$ be a continuous, real-valued function on $\mathbb{R}^2$. Suppose that, for every rectangular region R of area 1, the double integral of $f(x, y)$ over R equals 0. Must

$f(x, y)$ be identically 0?

2018 年 A6 · 几何与多变量微积分

年份 2018 题号 A6 入口积分

Suppose that A, B, C , and D are distinct points, no three of which lie on a line, in the Euclidean plane. Show that if the squares of the lengths of the line segments AB , AC , AD , BC , BD , and CD are rational numbers, then the quotient

$$\frac{\text{area}(\triangle ABC)}{\text{area}(\triangle ABD)}$$

is a rational number.

1986 年 B1 · 几何与多变量微积分

年份 1986 题号 B1 入口进位制与数字

Inscribe a rectangle of base b and height h in a circle of radius one, and inscribe an isosceles triangle in the region of the circle cut off by one base of the rectangle (with that side as the base of the triangle). For what value of h do the rectangle and triangle have the same area?

1999 年 B1 · 几何与多变量微积分

年份 1999 题号 B1 入口积分

Right triangle ABC has right angle at C and $\angle BAC = \theta$; the point D is chosen on AB so that $|AC| = |AD| = 1$; the point E is chosen on BC so that $\angle CDE = \theta$. The perpendicular to BC at E meets AB at F . Evaluate $\lim_{\theta \rightarrow 0} |EF|$.

2006 年 B1 · 几何与多变量微积分

年份 2006 题号 B1 入口积分

Show that the curve $x^3 + 3xy + y^3 = 1$ contains only one set of three distinct points, A , B , and C , which are vertices of an equilateral triangle, and find its area.

2008 年 B1 · 几何与多变量微积分

年份 2008 题号 B1 入口积分

What is the maximum number of rational points that can lie on a circle in $\mathbb{R}^2$ whose center is not a rational point? (A *rational point* is a point both of whose coordinates are rational numbers.)

2017 年 B1 · 几何与多变量微积分

年份 2017 题号 B1 入口积分

Let L_1 and L_2 be distinct lines in the plane. Prove that L_1 and L_2 intersect if and only if, for every real number $\lambda \neq 0$ and every point P not on L_1 or L_2 , there exist points A_1 on L_1 and A_2 on L_2 such that $\overrightarrow{PA_2} = \lambda \overrightarrow{PA_1}$.

2019 年 B1 · 几何与多变量微积分

年份 2019 题号 B1 入口积分

Denote by $\mathbb{Z}^2$ the set of all points (x, y) in the plane with integer coordinates. For each integer $n \geq 0$, let P_n be the subset of $\mathbb{Z}^2$ consisting of the point $(0, 0)$ together with all points (x, y) such that $x^2 + y^2 = 2^k$ for some integer $k \leq n$. Determine, as a function of n , the number of four-point subsets of P_n whose elements are the vertices of a square.

2025 年 B1 · 几何与多变量微积分

年份 2025 题号 B1 入口积分

Suppose that each point in the plane is colored either red or green, subject to the following condition: For every three noncollinear points A, B, C of the same color, the center of the circle passing through A, B and C is also this color. Prove that all points of the plane are the same color.

1995 年 B2 · 几何与多变量微积分

年份 1995 题号 B2 入口三角函数

An ellipse, whose semi-axes have lengths a and b , rolls without slipping on the curve $y = c \sin\left(\frac{x}{a}\right)$. How are a, b, c related, given that the ellipse completes one revolution when it traverses one period of the curve?

1998 年 B2 · 几何与多变量微积分

年份 1998 题号 B2 入口积分

Given a point (a, b) with $0 < b < a$, determine the minimum perimeter of a triangle with one vertex at (a, b) , one on the x -axis, and one on the line $y = x$. You may assume that a triangle of minimum perimeter exists.

2010 年 B2 · 几何与多变量微积分

年份 2010 题号 B2 入口积分

Given that A, B , and C are noncollinear points in the plane with integer coordinates such that the distances AB, AC , and BC are integers, what is the smallest possible

value of AB ?

2024 年 B2 · 几何与多变量微积分

年份 2024 题号 B2 入口整除与同余

Two convex quadrilaterals are called *partners* if they have three vertices in common and they can be labeled $ABCD$ and $ABCE$ so that E is the reflection of D across the perpendicular bisector of the diagonal $\overline{AC}$. Is there an infinite sequence of convex quadrilaterals such that each quadrilateral is a partner of its successor and no two elements of the sequence are congruent? [A diagram has been omitted.]

2025 年 B2 · 几何与多变量微积分

年份 2025 题号 B2 入口三角函数

Let $f: [0, 1] \rightarrow [0, \infty)$ be strictly increasing and continuous. Let R be the region bounded by $x = 0$, $x = 1$, $y = 0$, and $y = f(x)$. Let x_1 be the x -coordinate of the centroid of R . Let x_2 be the x -coordinate of the centroid of the solid generated by rotating R around the x -axis. Prove that $x_1 < x_2$.

1991 年 B3 · 几何与多变量微积分

年份 1991 题号 B3 入口积分

Does there exist a real number L such that, if m and n are integers greater than L , then an $m \times n$ rectangle may be expressed as a union of 4×6 and 5×7 rectangles, any two of which intersect at most along their boundaries?

1992 年 B3 · 几何与多变量微积分

年份 1992 题号 B3 入口级数/求和

For any pair (x, y) of real numbers, a sequence $(a_n(x, y))_{n \geq 0}$ is defined as follows:

$$\begin{aligned} a_0(x, y) &= x, \\ a_{n+1}(x, y) &= \frac{(a_n(x, y))^2 + y^2}{2}, \quad \text{for } n \geq 0. \end{aligned}$$

Find the area of the region

$$\{(x, y) | (a_n(x, y))_{n \geq 0} \text{ converges}\}.$$

1998 年 B3 · 几何与多变量微积分

年份 1998 题号 B3 入口三角函数

let H be the unit hemisphere $\{(x, y, z) : x^2 + y^2 + z^2 = 1, z \geq 0\}$, C the unit circle $\{(x, y, 0) : x^2 + y^2 = 1\}$, and P the regular pentagon inscribed in C . Determine the surface area of that portion of H lying over the planar region inside P , and write your answer in the form $A \sin \alpha + B \cos \beta$, where A, B, α, β are real numbers.

2004 年 B3 · 几何与多变量微积分

年份 2004 题号 B3 入口函数与不等式

Determine all real numbers $a > 0$ for which there exists a nonnegative continuous function $f(x)$ defined on $[0, a]$ with the property that the region

$$R = \{(x, y); 0 \leq x \leq a, 0 \leq y \leq f(x)\}$$

has perimeter k units and area k square units for some real number k .

2006 年 B3 · 几何与多变量微积分

年份 2006 题号 B3 入口积分

Let S be a finite set of points in the plane. A linear partition of S is an unordered pair $\{A, B\}$ of subsets of S such that $A \cup B = S$, $A \cap B = \emptyset$, and A and B lie on opposite sides of some straight line disjoint from S (A or B may be empty). Let L_S be the number of linear partitions of S . For each positive integer n , find the maximum of L_S over all sets S of n points.

2008 年 B3 · 几何与多变量微积分

年份 2008 题号 B3 入口平面几何

What is the largest possible radius of a circle contained in a 4-dimensional hypercube of side length 1?

2016 年 B3 · 几何与多变量微积分

年份 2016 题号 B3 入口积分

Suppose that S is a finite set of points in the plane such that the area of triangle $\triangle ABC$ is at most 1 whenever A, B , and C are in S . Show that there exists a triangle of area 4 that (together with its interior) covers the set S .

2021 年 B3 · 几何与多变量微积分

年份 2021 题号 B3 入口积分

Let $h(x, y)$ be a real-valued function that is twice continuously differentiable throughout $\mathbb{R}^2$, and define

$$\rho(x, y) = yh_x - xh_y.$$

Prove or disprove: For any positive constants d and r with $d > r$, there is a circle $\mathcal{S}$ of radius r whose center is a distance d away from the origin such that the integral of ρ over the interior of $\mathcal{S}$ is zero.

2004 年 B4 · 几何与多变量微积分

年份 2004 题号 B4 入口积分

Let n be a positive integer, $n \geq 2$, and put $\theta = 2\pi/n$. Define points $P_k = (k, 0)$ in the xy -plane, for $k = 1, 2, \dots, n$. Let R_k be the map that rotates the plane counterclockwise by the angle θ about the point P_k . Let R denote the map obtained by applying, in order, R_1 , then $R_2, \dots$, then R_n . For an arbitrary point (x, y) , find, and simplify, the coordinates of $R(x, y)$.

2006 年 B4 · 几何与多变量微积分

年份 2006 题号 B4 入口积分

Let Z denote the set of points in $\mathbb{R}^n$ whose coordinates are 0 or 1. (Thus Z has 2^n elements, which are the vertices of a unit hypercube in $\mathbb{R}^n$.) Given a vector subspace V of $\mathbb{R}^n$, let $Z(V)$ denote the number of members of Z that lie in V . Let k be given, $0 \leq k \leq n$. Find the maximum, over all vector subspaces $V \subseteq \mathbb{R}^n$ of dimension k , of the number of points in $V \cap Z$. [Editorial note: the proposers probably intended to write $Z(V)$ instead of “the number of points in $V \cap Z$ ”, but this changes nothing.]

2015 年 B4 · 几何与多变量微积分

年份 2015 题号 B4 入口积分

Let T be the set of all triples (a, b, c) of positive integers for which there exist triangles with side lengths a, b, c . Express

$$\sum_{(a,b,c) \in T} \frac{2^a}{3^b 5^c}$$

as a rational number in lowest terms.

1989 年 B5 · 几何与多变量微积分

年份 1989 题号 B5 入口积分

Label the vertices of a trapezoid T (quadrilateral with two parallel sides) inscribed in the unit circle as A, B, C, D so that AB is parallel to CD and A, B, C, D are

in counterclockwise order. Let s_1 , s_2 , and d denote the lengths of the line segments AB , CD , and OE , where E is the point of intersection of the diagonals of T , and O is the center of the circle. Determine the least upper bound of $\frac{s_1 - s_2}{d}$ over all such T for which $d \neq 0$, and describe all cases, if any, in which it is attained.

1993 年 B5 · 几何与多变量微积分

年份 1993 题号 B5 入口积分

Show there do not exist four points in the Euclidean plane such that the pairwise distances between the points are all odd integers.

2003 年 B5 · 几何与多变量微积分

年份 2003 题号 B5 入口积分

Let A, B , and C be equidistant points on the circumference of a circle of unit radius centered at O , and let P be any point in the circle's interior. Let a, b, c be the distance from P to A, B, C , respectively. Show that there is a triangle with side lengths a, b, c , and that the area of this triangle depends only on the distance from P to O .

2017 年 B5 · 几何与多变量微积分

年份 2017 题号 B5 入口积分

A line in the plane of a triangle T is called an *equalizer* if it divides T into two regions having equal area and equal perimeter. Find positive integers $a > b > c$, with a as small as possible, such that there exists a triangle with side lengths a, b, c that has exactly two distinct equalizers.

1990 年 B6 · 几何与多变量微积分

年份 1990 题号 B6 入口积分

Let S be a nonempty closed bounded convex set in the plane. Let K be a line and t a positive number. Let L_1 and L_2 be support lines for S parallel to K , and let $\bar{L}$ be the line parallel to K and midway between L_1 and L_2 . Let $B_S(K, t)$ be the band of points whose distance from $\bar{L}$ is at most $(t/2)w$, where w is the distance between L_1 and L_2 . What is the smallest t such that

$$S \cap \bigcap_K B_S(K, t) \neq \emptyset$$

for all S ? (K runs over all lines in the plane.)

1996 年 B6 · 几何与多变量微积分

年份 1996 题号 B6 入口积分

Let $(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n)$ be the vertices of a convex polygon which contains the origin in its interior. Prove that there exist positive real numbers x and y such that

$$(a_1, b_1)x^{a_1}y^{b_1} + (a_2, b_2)x^{a_2}y^{b_2} + \dots + (a_n, b_n)x^{a_n}y^{b_n} = (0, 0).$$

1997 年 B6 · 几何与多变量微积分

年份 1997 题号 B6 入口整除与同余

The dissection of the 3–4–5 triangle shown below (into four congruent right triangles similar to the original) has diameter $5/2$. Find the least diameter of a dissection of this triangle into four parts. (The diameter of a dissection is the least upper bound of the distances between pairs of points belonging to the same part.)

2000 年 B6 · 几何与多变量微积分

年份 2000 题号 B6 入口积分

Let B be a set of more than $2^{n+1}/n$ distinct points with coordinates of the form $(\pm 1, \pm 1, \dots, \pm 1)$ in n -dimensional space with $n \geq 3$. Show that there are three distinct points in B which are the vertices of an equilateral triangle.

2013 年 B6 · 几何与多变量微积分

年份 2013 题号 B6 入口积分

Let $n \geq 1$ be an odd integer. Alice and Bob play the following game, taking alternating turns, with Alice playing first. The playing area consists of n spaces, arranged in a line. Initially all spaces are empty. At each turn, a player either

- places a stone in an empty space, or
- removes a stone from a nonempty space s , places a stone in the nearest empty space to the left of s (if such a space exists), and places a stone in the nearest empty space to the right of s (if such a space exists).

Furthermore, a move is permitted only if the resulting position has not occurred previously in the game. A player loses if he or she is unable to move. Assuming that both players play optimally throughout the game, what moves may Alice make on her first turn?

2019 年 B6 · 几何与多变量微积分

年份 2019 题号 B6 入口积分

Let $\mathbb{Z}^n$ be the integer lattice in $\mathbb{R}^n$. Two points in $\mathbb{Z}^n$ are called *neighbors* if they differ by exactly 1 in one coordinate and are equal in all other coordinates. For which integers $n \geq 1$ does there exist a set of points $S \subset \mathbb{Z}^n$ satisfying the following two conditions?

- (1) If p is in S , then none of the neighbors of p is in S .
- (2) If $p \in \mathbb{Z}^n$ is not in S , then exactly one of the neighbors of p is in S .

第 7 章

概率

范畴要点：概率、期望、随机变量、组合概率

2012 年 A1 · 概率

年份 2012 题号 A1 入口积分

Let $d_1, d_2, \dots, d_{12}$ be real numbers in the open interval $(1, 12)$. Show that there exist distinct indices i, j, k such that d_i, d_j, d_k are the side lengths of an acute triangle.

2001 年 A2 · 概率

年份 2001 题号 A2 入口概率与期望

You have coins $C_1, C_2, \dots, C_n$. For each k , C_k is biased so that, when tossed, it has probability $1/(2k+1)$ of falling heads. If the n coins are tossed, what is the probability that the number of heads is odd? Express the answer as a rational function of n .

2007 年 A3 · 概率

年份 2007 题号 A3 入口环/形式幂级数

Let k be a positive integer. Suppose that the integers $1, 2, 3, \dots, 3k+1$ are written down in random order. What is the probability that at no time during this process, the sum of the integers that have been written up to that time is a positive integer divisible by 3? Your answer should be in closed form, but may include factorials.

2008 年 A3 · 概率

年份 2008 题号 A3 入口整除与同余

Start with a finite sequence $a_1, a_2, \dots, a_n$ of positive integers. If possible, choose two indices $j < k$ such that a_j does not divide a_k , and replace a_j and a_k by $\gcd(a_j, a_k)$ and $\text{lcm}(a_j, a_k)$, respectively. Prove that if this process is repeated, it must eventually stop and the final sequence does not depend on the choices made. (Note: $\gcd$ means greatest common divisor and lcm means least common multiple.)

1989 年 A4 · 概率

年份 1989 题号 A4 入口概率与期望

If α is an irrational number, $0 < \alpha < 1$, is there a finite game with an honest coin such that the probability of one player winning the game is α ? (An honest coin is one for which the probability of heads and the probability of tails are both $\frac{1}{2}$. A game is finite if with probability 1 it must end in a finite number of moves.)

2014 年 A4 · 概率

年份 2014 题号 A4 入口积分

Suppose X is a random variable that takes on only nonnegative integer values, with $E[X] = 1$, $E[X^2] = 2$, and $E[X^3] = 5$. (Here $E[y]$ denotes the expectation of the random variable Y .) Determine the smallest possible value of the probability of the event $X = 0$.

2020 年 A4 · 概率

年份 2020 题号 A4 入口环/形式幂级数

Consider a horizontal strip of $N+2$ squares in which the first and the last square are black and the remaining N squares are all white. Choose a white square uniformly at random, choose one of its two neighbors with equal probability, and color this neighboring square black if it is not already black. Repeat this process until all the remaining white squares have only black neighbors. Let $w(N)$ be the expected number of white squares remaining. Find

$$\lim_{N \rightarrow \infty} \frac{w(N)}{N}.$$

2022 年 A4 · 概率

年份 2022 题号 A4 入口积分

Suppose that $X_1, X_2, \dots$ are real numbers between 0 and 1 that are chosen independently and uniformly at random. Let $S = \sum_{i=1}^k X_i/2^i$, where k is the least positive integer such that $X_k < X_{k+1}$, or $k = \infty$ if there is no such integer. Find the expected value of S .

1989 年 A5 · 概率

年份 1989 题号 A5 入口积分

Let m be a positive integer and let $\mathcal{G}$ be a regular $(2m+1)$ -gon inscribed in the unit circle. Show that there is a positive constant A , independent of m , with the following property. For any points p inside $\mathcal{G}$ there are two distinct vertices v_1 and v_2 of $\mathcal{G}$ such that

$$||p - v_1| - |p - v_2|| < \frac{1}{m} - \frac{A}{m^3}.$$

Here $|s - t|$ denotes the distance between the points s and t .

2004 年 A5 · 概率

年份 2004 题号 A5 入口概率与期望

An $m \times n$ checkerboard is colored randomly: each square is independently assigned red or black with probability $1/2$. We say that two squares, p and q , are in the same connected monochromatic region if there is a sequence of squares, all of the same color, starting at p and ending at q , in which successive squares in the sequence share a common side. Show that the expected number of connected monochromatic regions is greater than $mn/8$.

2017 年 A5 · 概率

年份 2017 题号 A5 入口积分

Each of the integers from 1 to n is written on a separate card, and then the cards are combined into a deck and shuffled. Three players, A , B , and C , take turns in the order $A, B, C, A, \dots$ choosing one card at random from the deck. (Each card in the deck is equally likely to be chosen.) After a card is chosen, that card and all higher-numbered cards are removed from the deck, and the remaining cards are reshuffled before the next turn. Play continues until one of the three players wins the game by drawing the card numbered 1.

Show that for each of the three players, there are arbitrarily large values of n for which that player has the highest probability among the three players of winning the game.

2024 年 A5 · 概率

年份 2024 题号 A5 入口积分

Consider a circle Ω with radius 9 and center at the origin $(0, 0)$, and a disc Δ with radius 1 and center at $(r, 0)$, where $0 \leq r \leq 8$. Two points P and Q are chosen independently and uniformly at random on Ω . Which value(s) of r minimize the probability that the chord $\overline{PQ}$ intersects Δ ?

1986 年 A6 · 概率

年份 1986 题号 A6 入口多项式

Let $a_1, a_2, \dots, a_n$ be real numbers, and let $b_1, b_2, \dots, b_n$ be distinct positive integers. Suppose that there is a polynomial $f(x)$ satisfying the identity

$$(1 - x)^n f(x) = 1 + \sum_{i=1}^n a_i x^{b_i}.$$

Find a simple expression (not involving any sums) for $f(1)$ in terms of $b_1, b_2, \dots, b_n$ and n (but independent of $a_1, a_2, \dots, a_n$).

1992 年 A6 · 概率

年份 1992 题号 A6 入口积分

Four points are chosen at random on the surface of a sphere. What is the probability that the center of the sphere lies inside the tetrahedron whose vertices are at the four points? (It is understood that each point is independently chosen relative to a uniform distribution on the sphere.)

2005 年 A6 · 概率

年份 2005 题号 A6 入口积分

Let n be given, $n \geq 4$, and suppose that $P_1, P_2, \dots, P_n$ are n randomly, independently and uniformly, chosen points on a circle. Consider the convex n -gon whose vertices are the P_i . What is the probability that at least one of the vertex angles of this polygon is acute?

2006 年 A6 · 概率

年份 2006 题号 A6 入口积分

Four points are chosen uniformly and independently at random in the interior of a given circle. Find the probability that they are the vertices of a convex quadrilateral.

1989 年 B1 · 概率

年份 1989 题号 B1 入口积分

A dart, thrown at random, hits a square target. Assuming that any two parts of the target of equal area are equally likely to be hit, find the probability that the point hit is nearer to the center than to any edge. Express your answer in the form $\frac{a\sqrt{b} + c}{d}$, where a, b, c, d are integers.

2002 年 B1 · 概率

年份 2002 题号 B1 入口概率与期望

Shanille O'Keal shoots free throws on a basketball court. She hits the first and misses the second, and thereafter the probability that she hits the next shot is equal to the proportion of shots she has hit so far. What is the probability she hits exactly 50 of her first 100 shots?

2021 年 B1 · 概率

年份 2021 题号 B1 入口概率与期望

Suppose that the plane is tiled with an infinite checkerboard of unit squares. If another unit square is dropped on the plane at random with position and orientation independent of the checkerboard tiling, what is the probability that it does not cover any of the corners of the squares of the checkerboard?

2023 年 B1 · 概率

年份 2023 题号 B1 入口递推/序列

Consider an m -by- n grid of unit squares, indexed by (i, j) with $1 \leq i \leq m$ and $1 \leq j \leq n$. There are $(m-1)(n-1)$ coins, which are initially placed in the squares (i, j) with $1 \leq i \leq m-1$ and $1 \leq j \leq n-1$. If a coin occupies the square (i, j) with $i \leq m-1$ and $j \leq n-1$ and the squares $(i+1, j)$, $(i, j+1)$, and $(i+1, j+1)$ are unoccupied, then a legal move is to slide the coin from (i, j) to $(i+1, j+1)$. How many distinct configurations of coins can be reached starting from the initial configuration by a (possibly empty) sequence of legal moves?

1993 年 B2 · 概率

年份 1993 题号 B2 入口整除与同余

Consider the following game played with a deck of $2n$ cards numbered from 1 to $2n$. The deck is randomly shuffled and n cards are dealt to each of two players. Beginning with A , the players take turns discarding one of their remaining cards and announcing its number. The game ends as soon as the sum of the numbers on the discarded cards is divisible by $2n+1$. The last person to discard wins the game. Assuming optimal strategy by both A and B , what is the probability that A wins?

1993 年 B3 · 概率

年份 1993 题号 B3 入口积分

Two real numbers x and y are chosen at random in the interval $(0,1)$ with respect to the uniform distribution. What is the probability that the closest integer to x/y is even? Express the answer in the form $r + s\pi$, where r and s are rational numbers.

2020 年 B3 · 概率

年份 2020 题号 B3 入口积分

Let $x_0 = 1$, and let δ be some constant satisfying $0 < \delta < 1$. Iteratively, for $n = 0, 1, 2, \dots$, a point x_{n+1} is chosen uniformly from the interval $[0, x_n]$. Let Z be the smallest value of n for which $x_n < \delta$. Find the expected value of Z , as a function of δ .

2023 年 B3 · 概率

年份 2023 题号 B3 入口积分

A sequence $y_1, y_2, \dots, y_k$ of real numbers is called *zigzag* if $k = 1$, or if $y_2 - y_1, y_3 - y_2, \dots, y_k - y_{k-1}$ are nonzero and alternate in sign. Let $X_1, X_2, \dots, X_n$ be chosen independently from the uniform distribution on $[0, 1]$. Let $a(X_1, X_2, \dots, X_n)$ be the largest value of k for which there exists an increasing sequence of integers $i_1, i_2, \dots, i_k$ such that $X_{i_1}, X_{i_2}, \dots, X_{i_k}$ is zigzag. Find the expected value of $a(X_1, X_2, \dots, X_n)$ for $n \geq 2$.

1985 年 B4 · 概率

年份 1985 题号 B4 入口积分

Let C be the unit circle $x^2 + y^2 = 1$. A point p is chosen randomly on the circumference C and another point q is chosen randomly from the interior of C (these points are chosen independently and uniformly over their domains). Let R be the rectangle with sides parallel to the x and y -axes with diagonal pq . What is the probability that no point of R lies outside of C ?

2002 年 B4 · 概率

年份 2002 题号 B4 入口积分

An integer n , unknown to you, has been randomly chosen in the interval $[1, 2002]$ with uniform probability. Your objective is to select n in an **odd** number of guesses. After each incorrect guess, you are informed whether n is higher or lower, and you **must** guess an integer on your next turn among the numbers that are still feasibly correct. Show that you have a strategy so that the chance of winning is greater than $2/3$.

2019 年 B4 · 概率

年份 2019 题号 B4 入口导数/可微

Let $\mathcal{F}$ be the set of functions $f(x, y)$ that are twice continuously differentiable for $x \geq 1, y \geq 1$ and that satisfy the following two equations (where subscripts denote partial derivatives):

$$\begin{aligned} x f_x + y f_y &= xy \ln(xy), \\ x^2 f_{xx} + y^2 f_{yy} &= xy. \end{aligned}$$

For each $f \in \mathcal{F}$, let

$$m(f) = \min_{s \geq 1} (f(s+1, s+1) - f(s+1, s) - f(s, s+1) + f(s, s)).$$

Determine $m(f)$, and show that it is independent of the choice of f .

2024 年 B4 · 概率

年份 2024 题号 B4 入口积分

Let n be a positive integer. Set $a_{n,0} = 1$. For $k \geq 0$, choose an integer $m_{n,k}$ uniformly at random from the set $\{1, \dots, n\}$, and let

$$a_{n,k+1} = \begin{cases} a_{n,k} + 1, & \text{if } m_{n,k} > a_{n,k}; \\ a_{n,k}, & \text{if } m_{n,k} = a_{n,k}; \\ a_{n,k} - 1, & \text{if } m_{n,k} < a_{n,k}. \end{cases}$$

Let $E(n)$ be the expected value of $a_{n,n}$. Determine $\lim_{n \rightarrow \infty} E(n)/n$.

2022 年 B5 · 概率

年份 2022 题号 B5 入口积分

For $0 \leq p \leq 1/2$, let $X_1, X_2, \dots$ be independent random variables such that

$$X_i = \begin{cases} 1 & \text{with probability } p, \\ -1 & \text{with probability } p, \\ 0 & \text{with probability } 1 - 2p, \end{cases}$$

for all $i \geq 1$. Given a positive integer n and integers $b, a_1, \dots, a_n$, let $P(b, a_1, \dots, a_n)$ denote the probability that $a_1 X_1 + \dots + a_n X_n = b$. For which values of p is it the case that

$$P(0, a_1, \dots, a_n) \geq P(b, a_1, \dots, a_n)$$

for all positive integers n and all integers $b, a_1, \dots, a_n$?

1989 年 B6 · 概率

年份 1989 题号 B6 入口多项式

Let $(x_1, x_2, \dots, x_n)$ be a point chosen at random from the n -dimensional region defined by $0 < x_1 < x_2 < \dots < x_n < 1$. Let f be a continuous function on $[0, 1]$ with $f(1) = 0$. Set $x_0 = 0$ and $x_{n+1} = 1$. Show that the expected value of the Riemann sum

$$\sum_{i=0}^n (x_{i+1} - x_i) f(x_{i+1})$$

is $\int_0^1 f(t) P(t) dt$, where P is a polynomial of degree n , independent of f , with $0 \leq P(t) \leq 1$ for $0 \leq t \leq 1$.

2007 年 B6 · 概率

年份 2007 题号 B6 入口积分

For each positive integer n , let $f(n)$ be the number of ways to make $n!$ cents using an unordered collection of coins, each worth $k!$ cents for some k , $1 \leq k \leq n$. Prove that for some constant C , independent of n ,

$$n^{n^2/2 - Cn} e^{-n^2/4} \leq f(n) \leq n^{n^2/2 + Cn} e^{-n^2/4}.$$

第 8 章

高等组合与离散结构

范畴要点：极值组合、图论、生成函数、复杂递推、编码、组合博弈

1997 年 A2 · 高等组合与离散结构

年份 1997 题号 A2 入口三角函数

Players $1, 2, 3, \dots, n$ are seated around a table, and each has a single penny. Player 1 passes a penny to player 2, who then passes two pennies to player 3. Player 3 then passes one penny to Player 4, who passes two pennies to Player 5, and so on, players alternately passing one penny or two to the next player who still has some pennies. A player who runs out of pennies drops out of the game and leaves the table. Find an infinite set of numbers n for which some player ends up with all n pennies.

2023 年 A4 · 高等组合与离散结构

年份 2023 题号 A4 入口积分

Let $v_1, \dots, v_{12}$ be unit vectors in $\mathbb{R}^3$ from the origin to the vertices of a regular icosahedron. Show that for every vector $v \in \mathbb{R}^3$ and every $\varepsilon > 0$, there exist integers $a_1, \dots, a_{12}$ such that $\|a_1 v_1 + \dots + a_{12} v_{12} - v\| < \varepsilon$.

2022 年 A5 · 高等组合与离散结构

年份 2022 题号 A5 入口组合博弈

Alice and Bob play a game on a board consisting of one row of 2022 consecutive squares. They take turns placing tiles that cover two adjacent squares, with Alice going first. By rule, a tile must not cover a square that is already covered by another tile. The game ends when no tile can be placed according to this rule. Alice's goal is to maximize the number of uncovered squares when the game ends; Bob's goal is to minimize it. What is the greatest number of uncovered squares that Alice can ensure at the end of the game, no matter how Bob plays?

2025 年 A5 · 高等组合与离散结构

年份 2025 题号 A5 入口积分

Let n be an integer with $n \geq 2$. For a sequence $s = (s_1, \dots, s_{n-1})$ where each $s_i = \pm 1$, let $f(s)$ be the number of permutations $(a_1, \dots, a_n)$ of $\{1, 2, \dots, n\}$ such that $s_i(a_{i+1} - a_i) > 0$ for all i . For each n , determine the sequences s for which $f(s)$

is maximal.

1990 年 A6 · 高等组合与离散结构

年份 1990 题号 A6 入口集合/构造

If X is a finite set, let $|X|$ denote the number of elements in X . Call an ordered pair (S, T) of subsets of $\{1, 2, \dots, n\}$ *admissible* if $s > |T|$ for each $s \in S$, and $t > |S|$ for each $t \in T$. How many admissible ordered pairs of subsets of $\{1, 2, \dots, 10\}$ are there? Prove your answer.

1994 年 A6 · 高等组合与离散结构

年份 1994 题号 A6 入口积分

Let $f_1, \dots, f_{10}$ be bijections of the set of integers such that for each integer n , there is some composition $f_{i_1} \circ f_{i_2} \circ \dots \circ f_{i_m}$ of these functions (allowing repetitions) which maps 0 to n . Consider the set of 1024 functions

$$\mathcal{F} = \{f_1^{e_1} \circ f_2^{e_2} \circ \dots \circ f_{10}^{e_{10}}\},$$

$e_i = 0$ or 1 for $1 \leq i \leq 10$. (f_i^0 is the identity function and $f_i^1 = f_i$.) Show that if A is any nonempty finite set of integers, then at most 512 of the functions in $\mathcal{F}$ map A to itself.

2003 年 A6 · 高等组合与离散结构

年份 2003 题号 A6 入口积分

For a set S of nonnegative integers, let $r_S(n)$ denote the number of ordered pairs (s_1, s_2) such that $s_1 \in S$, $s_2 \in S$, $s_1 \neq s_2$, and $s_1 + s_2 = n$. Is it possible to partition the nonnegative integers into two sets A and B in such a way that $r_A(n) = r_B(n)$ for all n ?

2017 年 A6 · 高等组合与离散结构

年份 2017 题号 A6 入口积分

The 30 edges of a regular icosahedron are distinguished by labeling them $1, 2, \dots, 30$. How many different ways are there to paint each edge red, white, or blue such that each of the 20 triangular faces of the icosahedron has two edges of the same color and a third edge of a different color? [Note: the top matter on each exam paper included the logo of the Mathematical Association of America, which is itself an icosahedron.]

2023 年 A6 · 高等组合与离散结构

年份 2023 题号 A6 入口积分

Alice and Bob play a game in which they take turns choosing integers from 1 to n . Before any integers are chosen, Bob selects a goal of “odd” or “even”. On the first turn, Alice chooses one of the n integers. On the second turn, Bob chooses one of the remaining integers. They continue alternately choosing one of the integers that has not yet been chosen, until the n th turn, which is forced and ends the game. Bob wins if the parity of $\{k: \text{the number } k \text{ was chosen on the } k\text{th turn}\}$ matches his goal. For which values of n does Bob have a winning strategy?

1995 年 B1 · 高等组合与离散结构

年份 1995 题号 B1 入口积分

For a partition π of $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, let $\pi(x)$ be the number of elements in the part containing x . Prove that for any two partitions π and π' , there are two distinct numbers x and y in $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ such that $\pi(x) = \pi(y)$ and $\pi'(x) = \pi'(y)$. [A *partition* of a set S is a collection of disjoint subsets (parts) whose union is S .]

1996 年 B1 · 高等组合与离散结构

年份 1996 题号 B1 入口集合/构造

Define a **selfish** set to be a set which has its own cardinality (number of elements) as an element. Find, with proof, the number of subsets of $\{1, 2, \dots, n\}$ which are *minimal* selfish sets, that is, selfish sets none of whose proper subsets is selfish.

2001 年 B1 · 高等组合与离散结构

年份 2001 题号 B1 入口环/形式幂级数

Let n be an even positive integer. Write the numbers $1, 2, \dots, n^2$ in the squares of an $n \times n$ grid so that the k -th row, from left to right, is

$$(k-1)n+1, (k-1)n+2, \dots, (k-1)n+n.$$

Color the squares of the grid so that half of the squares in each row and in each column are red and the other half are black (a checkerboard coloring is one possibility). Prove that for each coloring, the sum of the numbers on the red squares is equal to the sum of the numbers on the black squares.

2002 年 B2 · 高等组合与离散结构

年份 2002 题号 B2 入口组合博弈

Consider a polyhedron with at least five faces such that exactly three edges emerge from each of its vertices. Two players play the following game:

Each player, in turn, signs his or her name on a previously unsigned face. The winner is the player who first succeeds in signing three faces that share a common vertex.

Show that the player who signs first will always win by playing as well as possible.

2009 年 B2 · 高等组合与离散结构

年份 2009 题号 B2 入口三角函数

A game involves jumping to the right on the real number line. If a and b are real numbers and $b > a$, the cost of jumping from a to b is $b^3 - ab^2$. For what real numbers c can one travel from 0 to 1 in a finite number of jumps with total cost exactly c ?

2020 年 B2 · 高等组合与离散结构

年份 2020 题号 B2 入口积分

Let k and n be integers with $1 \leq k < n$. Alice and Bob play a game with k pegs in a line of n holes. At the beginning of the game, the pegs occupy the k leftmost holes. A legal move consists of moving a single peg to any vacant hole that is further to the right. The players alternate moves, with Alice playing first. The game ends when the pegs are in the k rightmost holes, so whoever is next to play cannot move and therefore loses. For what values of n and k does Alice have a winning strategy?

2009 年 B3 · 高等组合与离散结构

年份 2009 题号 B3 入口积分

Call a subset S of $\{1, 2, \dots, n\}$ *mediocre* if it has the following property: Whenever a and b are elements of S whose average is an integer, that average is also an element of S . Let $A(n)$ be the number of mediocre subsets of $\{1, 2, \dots, n\}$. [For instance, every subset of $\{1, 2, 3\}$ except $\{1, 3\}$ is mediocre, so $A(3) = 7$.] Find all positive integers n such that $A(n+2) - 2A(n+1) + A(n) = 1$.

2012 年 B3 · 高等组合与离散结构

年份 2012 题号 B3 入口三角函数

A round-robin tournament of $2n$ teams lasted for $2n - 1$ days, as follows. On each day, every team played one game against another team, with one team winning and one team losing in each of the n games. Over the course of the tournament, each team played every other team exactly once. Can one necessarily choose one winning

team from each day without choosing any team more than once?

2013 年 B3 · 高等组合与离散结构

年份 2013 题号 B3 入口级数/求和

Let $\mathcal{P}$ be a nonempty collection of subsets of $\{1, \dots, n\}$ such that:

- (i) if $S, S' \in \mathcal{P}$, then $S \cup S' \in \mathcal{P}$ and $S \cap S' \in \mathcal{P}$, and
- (ii) if $S \in \mathcal{P}$ and $S \neq \emptyset$, then there is a subset $T \subset S$ such that $T \in \mathcal{P}$ and T contains exactly one fewer element than S .

Suppose that $f : \mathcal{P} \rightarrow \mathbb{R}$ is a function such that $f(\emptyset) = 0$ and

$$f(S \cup S') = f(S) + f(S') - f(S \cap S') \text{ for all } S, S' \in \mathcal{P}.$$

Must there exist real numbers $f_1, \dots, f_n$ such that

$$f(S) = \sum_{i \in S} f_i$$

for every $S \in \mathcal{P}$?

2022 年 B3 · 高等组合与离散结构

年份 2022 题号 B3 入口环/形式幂级数

Assign to each positive real number a color, either red or blue. Let D be the set of all distances $d > 0$ such that there are two points of the same color at distance d apart. Recolor the positive reals so that the numbers in D are red and the numbers not in D are blue. If we iterate this recoloring process, will we always end up with all the numbers red after a finite number of steps?

1989 年 B4 · 高等组合与离散结构

年份 1989 题号 B4 入口积分

Can a countably infinite set have an uncountable collection of non-empty subsets such that the intersection of any two of them is finite?

1995 年 B5 · 高等组合与离散结构

年份 1995 题号 B5 入口组合博弈

A game starts with four heaps of beans, containing 3,4,5 and 6 beans. The two players move alternately. A move consists of taking **either**

- a) one bean from a heap, provided at least two beans are left behind in that heap,
- or**

b) a complete heap of two or three beans.

The player who takes the last heap wins. To win the game, do you want to move first or second? Give a winning strategy.

2000 年 B5 · 高等组合与离散结构

年份 2000 题号 B5 入口积分

Let S_0 be a finite set of positive integers. We define finite sets $S_1, S_2, \dots$ of positive integers as follows: the integer a is in S_{n+1} if and only if exactly one of $a-1$ or a is in S_n . Show that there exist infinitely many integers N for which $S_N = S_0 \cup \{N+a : a \in S_0\}$.

2015 年 B5 · 高等组合与离散结构

年份 2015 题号 B5 入口排列组合

Let P_n be the number of permutations π of $\{1, 2, \dots, n\}$ such that

$$|i - j| = 1 \text{ implies } |\pi(i) - \pi(j)| \leq 2$$

for all i, j in $\{1, 2, \dots, n\}$. Show that for $n \geq 2$, the quantity

$$P_{n+5} - P_{n+4} - P_{n+3} + P_n$$

does not depend on n , and find its value.

2008 年 B6 · 高等组合与离散结构

年份 2008 题号 B6 入口积分

Let n and k be positive integers. Say that a permutation σ of $\{1, 2, \dots, n\}$ is k -limited if $|\sigma(i) - i| \leq k$ for all i . Prove that the number of k -limited permutations of $\{1, 2, \dots, n\}$ is odd if and only if $n \equiv 0$ or $1 \pmod{2k+1}$.

第 9 章

实分析

范畴要点：严格极限、连续/可微函数、级数收敛、紧性、测度风格、函数方程

1994 年 A1 · 实分析

年份 1994 题号 A1 入口级数/求和

Suppose that a sequence $a_1, a_2, a_3, \dots$ satisfies $0 < a_n \leq a_{2n} + a_{2n+1}$ for all $n \geq 1$. Prove that the series $\sum_{n=1}^{\infty} a_n$ diverges.

1988 年 A2 · 实分析

年份 1988 题号 A2 入口积分

A not uncommon calculus mistake is to believe that the product rule for derivatives says that $(fg)' = f'g'$. If $f(x) = e^{x^2}$, determine, with proof, whether there exists an open interval (a, b) and a nonzero function g defined on (a, b) such that this wrong product rule is true for x in (a, b) .

1995 年 A2 · 实分析

年份 1995 题号 A2 入口积分

For what pairs (a, b) of positive real numbers does the improper integral

$$\int_b^{\infty} \left(\sqrt{\sqrt{x+a} - \sqrt{x}} - \sqrt{\sqrt{x} - \sqrt{x-b}} \right) dx$$

converge?

2009 年 A2 · 实分析

年份 2009 题号 A2 入口积分

Functions f, g, h are differentiable on some open interval around 0 and satisfy the equations and initial conditions

$$\begin{aligned} f' &= 2f^2gh + \frac{1}{gh}, & f(0) &= 1, \\ g' &= fg^2h + \frac{4}{fh}, & g(0) &= 1, \\ h' &= 3fgh^2 + \frac{1}{fg}, & h(0) &= 1. \end{aligned}$$

Find an explicit formula for $f(x)$, valid in some open interval around 0.

2011 年 A2 · 实分析

年份 2011 题号 A2 入口级数/求和

Let $a_1, a_2, \dots$ and $b_1, b_2, \dots$ be sequences of positive real numbers such that $a_1 = b_1 = 1$ and $b_n = b_{n-1}a_n - 2$ for $n = 2, 3, \dots$. Assume that the sequence (b_j) is bounded. Prove that

$$S = \sum_{n=1}^{\infty} \frac{1}{a_1 \dots a_n}$$

converges, and evaluate S .

1987 年 A3 · 实分析

年份 1987 题号 A3 入口函数与不等式

For all real x , the real-valued function $y = f(x)$ satisfies

$$y'' - 2y' + y = 2e^x.$$

- (a) If $f(x) > 0$ for all real x , must $f'(x) > 0$ for all real x ? Explain.
- (b) If $f'(x) > 0$ for all real x , must $f(x) > 0$ for all real x ? Explain.

1988 年 A3 · 实分析

年份 1988 题号 A3 入口级数/求和

Determine, with proof, the set of real numbers x for which

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} \csc \frac{1}{n} - 1 \right)^x$$

converges.

2012 年 A3 · 实分析

年份 2012 题号 A3 入口函数与不等式

Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function such that

- (i) $f(x) = \frac{2-x^2}{2} f\left(\frac{x^2}{2-x^2}\right)$ for every x in $[-1, 1]$,
 - (ii) $f(0) = 1$, and
 - (iii) $\lim_{x \rightarrow 1^-} \frac{f(x)}{\sqrt{1-x}}$ exists and is finite.
- Prove that f is unique, and express $f(x)$ in closed form.

2016 年 A3 · 实分析

年份 2016 题号 A3 入口积分

Suppose that f is a function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) + f\left(1 - \frac{1}{x}\right) = \arctan x$$

for all real $x \neq 0$. (As usual, $y = \arctan x$ means $-\pi/2 < y < \pi/2$ and $\tan y = x$.) Find

$$\int_0^1 f(x) dx.$$

2017 年 A3 · 实分析

年份 2017 题号 A3 入口积分

Let a and b be real numbers with $a < b$, and let f and g be continuous functions from $[a, b]$ to $(0, \infty)$ such that $\int_a^b f(x) dx = \int_a^b g(x) dx$ but $f \neq g$. For every positive integer n , define

$$I_n = \int_a^b \frac{(f(x))^{n+1}}{(g(x))^n} dx.$$

Show that $I_1, I_2, I_3, \dots$ is an increasing sequence with $\lim_{n \rightarrow \infty} I_n = \infty$.

2020 年 A3 · 实分析

年份 2020 题号 A3 入口级数/求和

Let $a_0 = \pi/2$, and let $a_n = \sin(a_{n-1})$ for $n \geq 1$. Determine whether

$$\sum_{n=1}^{\infty} a_n^2$$

converges.

2023 年 A3 · 实分析

年份 2023 题号 A3 入口导数/可微

Determine the smallest positive real number r such that there exist differentiable functions $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

- (a) $f(0) > 0$,
- (b) $g(0) = 0$,
- (c) $|f'(x)| \leq |g(x)|$ for all x ,
- (d) $|g'(x)| \leq |f(x)|$ for all x , and
- (e) $f(r) = 0$.

1992 年 A4 · 实分析

年份 1992 题号 A4 入口导数/可微

Let f be an infinitely differentiable real-valued function defined on the real numbers. If

$$f\left(\frac{1}{n}\right) = \frac{n^2}{n^2 + 1}, \quad n = 1, 2, 3, \dots,$$

compute the values of the derivatives $f^{(k)}(0)$, $k = 1, 2, 3, \dots$

2000 年 A4 · 实分析

年份 2000 题号 A4 入口积分

Show that the improper integral

$$\lim_{B \rightarrow \infty} \int_0^B \sin(x) \sin(x^2) dx$$

converges.

2011 年 A5 · 实分析

年份 2011 题号 A5 入口导数/可微

Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable functions with the following properties:

- $F(u, u) = 0$ for every $u \in \mathbb{R}$;
- for every $x \in \mathbb{R}$, $g(x) > 0$ and $x^2 g(x) \leq 1$;
- for every $(u, v) \in \mathbb{R}^2$, the vector $\nabla F(u, v)$ is either $\mathbf{0}$ or parallel to the vector $\langle g(u), -g(v) \rangle$.

Prove that there exists a constant C such that for every $n \geq 2$ and any $x_1, \dots, x_{n+1} \in \mathbb{R}$, we have

$$\min_{i \neq j} |F(x_i, x_j)| \leq \frac{C}{n}.$$

2018 年 A5 · 实分析

年份 2018 题号 A5 入口积分

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function satisfying $f(0) = 0$, $f(1) = 1$, and $f(x) \geq 0$ for all $x \in \mathbb{R}$. Show that there exist a positive integer n and a real number x such that $f^{(n)}(x) < 0$.

1996 年 A6 · 实分析

年份 1996 题号 A6 入口三角函数

Let $c > 0$ be a constant. Give a complete description, with proof, of the set of all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = f(x^2 + c)$ for all $x \in \mathbb{R}$. Note that $\mathbb{R}$ denotes the set of real numbers.

2004 年 A6 · 实分析

年份 2004 题号 A6 入口积分

Suppose that $f(x, y)$ is a continuous real-valued function on the unit square $0 \leq x \leq 1, 0 \leq y \leq 1$. Show that

$$\begin{aligned} & \int_0^1 \left(\int_0^1 f(x, y) dx \right)^2 dy + \int_0^1 \left(\int_0^1 f(x, y) dy \right)^2 dx \\ & \leq \left(\int_0^1 \int_0^1 f(x, y) dx dy \right)^2 + \int_0^1 \int_0^1 [f(x, y)]^2 dx dy. \end{aligned}$$

2009 年 A6 · 实分析

年份 2009 题号 A6 入口积分

Let $f : [0, 1]^2 \rightarrow \mathbb{R}$ be a continuous function on the closed unit square such that $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist and are continuous on the interior $(0, 1)^2$. Let $a = \int_0^1 f(0, y) dy$, $b = \int_0^1 f(1, y) dy$, $c = \int_0^1 f(x, 0) dx$, $d = \int_0^1 f(x, 1) dx$. Prove or disprove: There must be a point (x_0, y_0) in $(0, 1)^2$ such that

$$\frac{\partial f}{\partial x}(x_0, y_0) = b - a \quad \text{and} \quad \frac{\partial f}{\partial y}(x_0, y_0) = d - c.$$

2010 年 A6 · 实分析

年份 2010 题号 A6 入口积分

Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a strictly decreasing continuous function such that $\lim_{x \rightarrow \infty} f(x) = 0$. Prove that $\int_0^\infty \frac{f(x) - f(x+1)}{f(x)} dx$ diverges.

2019 年 A6 · 实分析

年份 2019 题号 A6 入口积分

Let g be a real-valued function that is continuous on the closed interval $[0, 1]$ and twice differentiable on the open interval $(0, 1)$. Suppose that for some real number $r > 1$,

$$\lim_{x \rightarrow 0^+} \frac{g(x)}{x^r} = 0.$$

Prove that either

$$\lim_{x \rightarrow 0^+} g'(x) = 0 \quad \text{or} \quad \limsup_{x \rightarrow 0^+} x^r |g''(x)| = \infty.$$

1990 年 B1 · 实分析

年份 1990 题号 B1 入口积分

Find all real-valued continuously differentiable functions f on the real line such that for all x ,

$$(f(x))^2 = \int_0^x [(f(t))^2 + (f'(t))^2] dt + 1990.$$

2012 年 B1 · 实分析

年份 2012 题号 B1 入口函数与不等式

Let S be a class of functions from $[0, \infty)$ to $[0, \infty)$ that satisfies:

- (i) The functions $f_1(x) = e^x - 1$ and $f_2(x) = \ln(x + 1)$ are in S ;
- (ii) If $f(x)$ and $g(x)$ are in S , the functions $f(x) + g(x)$ and $f(g(x))$ are in S ;
- (iii) If $f(x)$ and $g(x)$ are in S and $f(x) \geq g(x)$ for all $x \geq 0$, then the function $f(x) - g(x)$ is in S .

Prove that if $f(x)$ and $g(x)$ are in S , then the function $f(x)g(x)$ is also in S .

2015 年 B1 · 实分析

年份 2015 题号 B1 入口导数/可微

Let f be a three times differentiable function (defined on $\mathbb{R}$ and real-valued) such that f has at least five distinct real zeros. Prove that $f + 6f' + 12f'' + 8f'''$ has at least two distinct real zeros.

2016 年 B1 · 实分析

年份 2016 题号 B1 入口级数/求和

Let $x_0, x_1, x_2, \dots$ be the sequence such that $x_0 = 1$ and for $n \geq 0$,

$$x_{n+1} = \ln(e^{x_n} - x_n)$$

(as usual, the function $\ln$ is the natural logarithm). Show that the infinite series

$$x_0 + x_1 + x_2 + \cdots$$

converges and find its sum.

1991 年 B2 · 实分析

年份 1991 题号 B2 入口导数/可微

Suppose f and g are non-constant, differentiable, real-valued functions defined on $(-\infty, \infty)$. Furthermore, suppose that for each pair of real numbers x and y ,

$$\begin{aligned} f(x+y) &= f(x)f(y) - g(x)g(y), \\ g(x+y) &= f(x)g(y) + g(x)f(y). \end{aligned}$$

If $f'(0) = 0$, prove that $(f(x))^2 + (g(x))^2 = 1$ for all x .

1997 年 B2 · 实分析

年份 1997 题号 B2 入口导数/可微

Let f be a twice-differentiable real-valued function satisfying

$$f(x) + f''(x) = -xg(x)f'(x),$$

where $g(x) \geq 0$ for all real x . Prove that $|f(x)|$ is bounded.

2007 年 B2 · 实分析

年份 2007 题号 B2 入口积分

Suppose that $f : [0, 1] \rightarrow \mathbb{R}$ has a continuous derivative and that $\int_0^1 f(x) dx = 0$. Prove that for every $\alpha \in (0, 1)$,

$$\left| \int_0^\alpha f(x) dx \right| \leq \frac{1}{8} \max_{0 \leq x \leq 1} |f'(x)|.$$

1989 年 B3 · 实分析

年份 1989 题号 B3 入口积分

Let f be a function on $[0, \infty)$, differentiable and satisfying

$$f'(x) = -3f(x) + 6f(2x)$$

for $x > 0$. Assume that $|f(x)| \leq e^{-\sqrt{x}}$ for $x \geq 0$ (so that $f(x)$ tends rapidly to 0 as x increases). For n a non-negative integer, define

$$\mu_n = \int_0^\infty x^n f(x) dx$$

(sometimes called the n th moment of f).

- Express μ_n in terms of μ_0 .
- Prove that the sequence $\{\mu_n \frac{3^n}{n!}\}$ always converges, and that the limit is 0 only if $\mu_0 = 0$.

1994 年 B3 · 实分析

年份 1994 题号 B3 入口导数/可微

Find the set of all real numbers k with the following property: For any positive, differentiable function f that satisfies $f'(x) > f(x)$ for all x , there is some number N such that $f(x) > e^{kx}$ for all $x > N$.

2005 年 B3 · 实分析

年份 2005 题号 B3 入口导数/可微

Find all differentiable functions $f : (0, \infty) \rightarrow (0, \infty)$ for which there is a positive real number a such that

$$f' \left(\frac{a}{x} \right) = \frac{x}{f(x)}$$

for all $x > 0$.

2011 年 B3 · 实分析

年份 2011 题号 B3 入口积分

Let f and g be (real-valued) functions defined on an open interval containing 0, with g nonzero and continuous at 0. If fg and f/g are differentiable at 0, must f be differentiable at 0?

1988 年 B4 · 实分析

年份 1988 题号 B4 入口级数/求和

Prove that if $\sum_{n=1}^{\infty} a_n$ is a convergent series of positive real numbers, then so is $\sum_{n=1}^{\infty} (a_n)^{n/(n+1)}$.

2000 年 B4 · 实分析

年份 2000 题号 B4 入口函数与不等式

Let $f(x)$ be a continuous function such that $f(2x^2 - 1) = 2xf(x)$ for all x . Show that $f(x) = 0$ for $-1 \leq x \leq 1$.

2013 年 B4 · 实分析

年份 2013 题号 B4 入口积分

For any continuous real-valued function f defined on the interval $[0, 1]$, let

$$\mu(f) = \int_0^1 f(x) dx, \quad \text{Var}(f) = \int_0^1 (f(x) - \mu(f))^2 dx,$$

$$M(f) = \max_{0 \leq x \leq 1} |f(x)|.$$

Show that if f and g are continuous real-valued functions defined on the interval $[0, 1]$, then

$$\text{Var}(fg) \leq 2\text{Var}(f)M(g)^2 + 2\text{Var}(g)M(f)^2.$$

2023 年 B4 · 实分析

年份 2023 题号 B4 入口环/形式幂级数

For a nonnegative integer n and a strictly increasing sequence of real numbers $t_0, t_1, \dots, t_n$, let $f(t)$ be the corresponding real-valued function defined for $t \geq t_0$ by the following properties:

- (a) $f(t)$ is continuous for $t \geq t_0$, and is twice differentiable for all $t > t_0$ other than $t_1, \dots, t_n$;
- (b) $f(t_0) = 1/2$;
- (c) $\lim_{t \rightarrow t_k^+} f'(t) = 0$ for $0 \leq k \leq n$;
- (d) For $0 \leq k \leq n-1$, we have $f''(t) = k+1$ when $t_k < t < t_{k+1}$, and $f''(t) = n+1$ when $t > t_n$.

Considering all choices of n and $t_0, t_1, \dots, t_n$ such that $t_k \geq t_{k-1} + 1$ for $1 \leq k \leq n$, what is the least possible value of T for which $f(t_0 + T) = 2023$?

2001 年 B5 · 实分析

年份 2001 题号 B5 入口积分

Let a and b be real numbers in the interval $(0, 1/2)$, and let g be a continuous real-valued function such that $g(g(x)) = ag(x) + bx$ for all real x . Prove that $g(x) = cx$ for some constant c .

2006 年 B5 · 实分析

年份 2006 题号 B5 入口积分

For each continuous function $f : [0, 1] \rightarrow \mathbb{R}$, let $I(f) = \int_0^1 x^2 f(x) dx$ and $J(x) = \int_0^1 x (f(x))^2 dx$. Find the maximum value of $I(f) - J(f)$ over all such functions f .

2008 年 B5 · 实分析

年份 2008 题号 B5 入口整除与同余

Find all continuously differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for every rational number q , the number $f(q)$ is rational and has the same denominator as q . (The denominator of a rational number q is the unique positive integer b such that $q = a/b$ for some integer a with $\gcd(a, b) = 1$.) (Note: $\gcd$ means greatest common divisor.)

2009 年 B5 · 实分析

年份 2009 题号 B5 入口导数/可微

Let $f : (1, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that

$$f'(x) = \frac{x^2 - f(x)^2}{x^2(f(x)^2 + 1)} \quad \text{for all } x > 1.$$

Prove that $\lim_{x \rightarrow \infty} f(x) = \infty$.

2012 年 B5 · 实分析

年份 2012 题号 B5 入口函数与不等式

Prove that, for any two bounded functions $g_1, g_2 : \mathbb{R} \rightarrow [1, \infty)$, there exist functions $h_1, h_2 : \mathbb{R} \rightarrow \mathbb{R}$ such that, for every $x \in \mathbb{R}$,

$$\sup_{s \in \mathbb{R}} (g_1(s)^x g_2(s)) = \max_{t \in \mathbb{R}} (x h_1(t) + h_2(t)).$$

2018 年 B5 · 实分析

年份 2018 题号 B5 入口导数/可微

Let $f = (f_1, f_2)$ be a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ with continuous partial derivatives $\frac{\partial f_i}{\partial x_j}$ that are positive everywhere. Suppose that

$$\frac{\partial f_1}{\partial x_1} \frac{\partial f_2}{\partial x_2} - \frac{1}{4} \left(\frac{\partial f_1}{\partial x_2} + \frac{\partial f_2}{\partial x_1} \right)^2 > 0$$

everywhere. Prove that f is one-to-one.

2003 年 B6 · 实分析

年份 2003 题号 B6 入口积分

Let $f(x)$ be a continuous real-valued function defined on the interval $[0, 1]$. Show that

$$\int_0^1 \int_0^1 |f(x) + f(y)| dx dy \geq \int_0^1 |f(x)| dx.$$

2004 年 B6 · 实分析

年份 2004 题号 B6 入口积分

Let $\mathcal{A}$ be a non-empty set of positive integers, and let $N(x)$ denote the number of elements of $\mathcal{A}$ not exceeding x . Let $\mathcal{B}$ denote the set of positive integers b that can be written in the form $b = a - a'$ with $a \in \mathcal{A}$ and $a' \in \mathcal{A}$. Let $b_1 < b_2 < \cdots$ be the members of $\mathcal{B}$, listed in increasing order. Show that if the sequence $b_{i+1} - b_i$ is

unbounded, then

$$\lim_{x \rightarrow \infty} N(x)/x = 0.$$

2014 年 B6 · 实分析

年份 2014 题号 B6 入口积分

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a function for which there exists a constant $K > 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in [0, 1]$. Suppose also that for each rational number $r \in [0, 1]$, there exist integers a and b such that $f(r) = a + br$. Prove that there exist finitely many intervals $I_1, \dots, I_n$ such that f is a linear function on each I_i and $[0, 1] = \bigcup_{i=1}^n I_i$.

2022 年 B6 · 实分析

年份 2022 题号 B6 入口函数与不等式

Find all continuous functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(xf(y)) + f(yf(x)) = 1 + f(x + y)$$

for all $x, y > 0$.

第 10 章

抽象代数与高等代数

范畴要点：群、环、域、多项式结构、有限域、代数数

1999 年 A1 · 抽象代数与高等代数

年份 1999 题号 A1 入口多项式

Find polynomials $f(x), g(x)$, and $h(x)$, if they exist, such that for all x ,

$$|f(x)| - |g(x)| + h(x) = \begin{cases} -1 & \text{if } x < -1 \\ 3x + 2 & \text{if } -1 \leq x \leq 0 \\ -2x + 2 & \text{if } x > 0. \end{cases}$$

2001 年 A1 · 抽象代数与高等代数

年份 2001 题号 A1 入口级数/求和

Consider a set S and a binary operation $*$, i.e., for each $a, b \in S$, $a * b \in S$. Assume $(a * b) * a = b$ for all $a, b \in S$. Prove that $a * (b * a) = b$ for all $a, b \in S$.

2002 年 A1 · 抽象代数与高等代数

年份 2002 题号 A1 入口多项式

Let k be a fixed positive integer. The n -th derivative of $\frac{1}{x^k-1}$ has the form $\frac{P_n(x)}{(x^k-1)^{n+1}}$ where $P_n(x)$ is a polynomial. Find $P_n(1)$.

1999 年 A2 · 抽象代数与高等代数

年份 1999 题号 A2 入口多项式

Let $p(x)$ be a polynomial that is nonnegative for all real x . Prove that for some k , there are polynomials $f_1(x), \dots, f_k(x)$ such that

$$p(x) = \sum_{j=1}^k (f_j(x))^2.$$

2012 年 A2 · 抽象代数与高等代数

年份 2012 题号 A2 入口级数/求和

Let $*$ be a commutative and associative binary operation on a set S . Assume that for every x and y in S , there exists z in S such that $x * z = y$. (This z may depend on x and y .) Show that if a, b, c are in S and $a * c = b * c$, then $a = b$.

2022 年 A2 · 抽象代数与高等代数

年份 2022 题号 A2 入口多项式

Let n be an integer with $n \geq 2$. Over all real polynomials $p(x)$ of degree n , what is the largest possible number of negative coefficients of $p(x)^2$?

2023 年 A2 · 抽象代数与高等代数

年份 2023 题号 A2 入口多项式

Let n be an even positive integer. Let p be a monic, real polynomial of degree $2n$; that is to say, $p(x) = x^{2n} + a_{2n-1}x^{2n-1} + \cdots + a_1x + a_0$ for some real coefficients $a_0, \dots, a_{2n-1}$. Suppose that $p(1/k) = k^2$ for all integers k such that $1 \leq |k| \leq n$. Find all other real numbers x for which $p(1/x) = x^2$.

2024 年 A2 · 抽象代数与高等代数

年份 2024 题号 A2 入口多项式

For which real polynomials p is there a real polynomial q such that

$$p(p(x)) - x = (p(x) - x)^2 q(x)$$

for all real x ?

1991 年 A3 · 抽象代数与高等代数

年份 1991 题号 A3 入口多项式

Find all real polynomials $p(x)$ of degree $n \geq 2$ for which there exist real numbers $r_1 < r_2 < \cdots < r_n$ such that

1. $p(r_i) = 0$, $i = 1, 2, \dots, n$, and
2. $p'(\frac{r_i + r_{i+1}}{2}) = 0$ $i = 1, 2, \dots, n-1$,

where $p'(x)$ denotes the derivative of $p(x)$.

1997 年 A4 · 抽象代数与高等代数

年份 1997 题号 A4 入口群论

Let G be a group with identity e and $\phi : G \rightarrow G$ a function such that

$$\phi(g_1)\phi(g_2)\phi(g_3) = \phi(h_1)\phi(h_2)\phi(h_3)$$

whenever $g_1g_2g_3 = e = h_1h_2h_3$. Prove that there exists an element $a \in G$ such that $\psi(x) = a\phi(x)$ is a homomorphism (i.e. $\psi(xy) = \psi(x)\psi(y)$ for all $x, y \in G$).

2017 年 A4 · 抽象代数与高等代数

年份 2017 题号 A4 入口群论

A class with $2N$ students took a quiz, on which the possible scores were $0, 1, \dots, 10$. Each of these scores occurred at least once, and the average score was exactly 7.4. Show that the class can be divided into two groups of N students in such a way that the average score for each group was exactly 7.4.

2018 年 A4 · 抽象代数与高等代数

年份 2018 题号 A4 入口群论

Let m and n be positive integers with $\gcd(m, n) = 1$, and let

$$a_k = \left\lfloor \frac{mk}{n} \right\rfloor - \left\lfloor \frac{m(k-1)}{n} \right\rfloor$$

for $k = 1, 2, \dots, n$. Suppose that g and h are elements in a group G and that

$$gh^{a_1}gh^{a_2} \cdots gh^{a_n} = e,$$

where e is the identity element. Show that $gh = hg$. (As usual, $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x .)

1999 年 A5 · 抽象代数与高等代数

年份 1999 题号 A5 入口多项式

Prove that there is a constant C such that, if $p(x)$ is a polynomial of degree 1999, then

$$|p(0)| \leq C \int_{-1}^1 |p(x)| dx.$$

2007 年 A5 · 抽象代数与高等代数

年份 2007 题号 A5 入口群论

Suppose that a finite group has exactly n elements of order p , where p is a prime. Prove that either $n = 0$ or p divides $n + 1$.

2008 年 A5 · 抽象代数与高等代数

年份 2008 题号 A5 入口多项式

Let $n \geq 3$ be an integer. Let $f(x)$ and $g(x)$ be polynomials with real coefficients such that the points $(f(1), g(1)), (f(2), g(2)), \dots, (f(n), g(n))$ in $\mathbb{R}^2$ are the vertices of a regular n -gon in counterclockwise order. Prove that at least one of $f(x)$ and $g(x)$ has degree greater than or equal to $n - 1$.

2009 年 A5 · 抽象代数与高等代数

年份 2009 题号 A5 入口群论

Is there a finite abelian group G such that the product of the orders of all its elements is 2^{2009} ?

2010 年 A5 · 抽象代数与高等代数

年份 2010 题号 A5 入口群论

Let G be a group, with operation $*$. Suppose that

- (i) G is a subset of $\mathbb{R}^3$ (but $*$ need not be related to addition of vectors);
- (ii) For each $\mathbf{a}, \mathbf{b} \in G$, either $\mathbf{a} \times \mathbf{b} = \mathbf{a} * \mathbf{b}$ or $\mathbf{a} \times \mathbf{b} = 0$ (or both), where $\times$ is the usual cross product in $\mathbb{R}^3$.

Prove that $\mathbf{a} \times \mathbf{b} = 0$ for all $\mathbf{a}, \mathbf{b} \in G$.

2012 年 A5 · 抽象代数与高等代数

年份 2012 题号 A5 入口域/有限域

Let $\mathbb{F}_p$ denote the field of integers modulo a prime p , and let n be a positive integer. Let v be a fixed vector in $\mathbb{F}_p^n$, let M be an $n \times n$ matrix with entries of $\mathbb{F}_p$, and define $G : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ by $G(x) = v + Mx$. Let $G^{(k)}$ denote the k -fold composition of G with itself, that is, $G^{(1)}(x) = G(x)$ and $G^{(k+1)}(x) = G(G^{(k)}(x))$. Determine all pairs p, n for which there exist v and M such that the p^n vectors $G^{(k)}(0)$, $k = 1, 2, \dots, p^n$ are distinct.

2016 年 A5 · 抽象代数与高等代数

年份 2016 题号 A5 入口群论

Suppose that G is a finite group generated by the two elements g and h , where the order of g is odd. Show that every element of G can be written in the form

$$g^{m_1} h^{n_1} g^{m_2} h^{n_2} \dots g^{m_r} h^{n_r}$$

with $1 \leq r \leq |G|$ and $m_1, n_1, m_2, n_2, \dots, m_r, n_r \in \{-1, 1\}$. (Here $|G|$ is the number of elements of G .)

2019 年 A5 · 抽象代数与高等代数

年份 2019 题号 A5 入口域/有限域

Let p be an odd prime number, and let $\mathbb{F}_p$ denote the field of integers modulo p . Let $\mathbb{F}_p[x]$ be the ring of polynomials over $\mathbb{F}_p$, and let $q(x) \in \mathbb{F}_p[x]$ be given by

$$q(x) = \sum_{k=1}^{p-1} a_k x^k,$$

where

$$a_k = k^{(p-1)/2} \pmod{p}.$$

Find the greatest nonnegative integer n such that $(x-1)^n$ divides $q(x)$ in $\mathbb{F}_p[x]$.

1985 年 A6 · 抽象代数与高等代数

年份 1985 题号 A6 入口多项式

If $p(x) = a_0 + a_1x + \cdots + a_mx^m$ is a polynomial with real coefficients a_i , then set

$$\Gamma(p(x)) = a_0^2 + a_1^2 + \cdots + a_m^2.$$

Let $F(x) = 3x^2 + 7x + 2$. Find, with proof, a polynomial $g(x)$ with real coefficients such that

(i) $g(0) = 1$, and

(ii) $\Gamma(f(x)^n) = \Gamma(g(x)^n)$

for every integer $n \geq 1$.

1989 年 A6 · 抽象代数与高等代数

年份 1989 题号 A6 入口域/有限域

Let $\alpha = 1 + a_1x + a_2x^2 + \cdots$ be a formal power series with coefficients in the field of two elements. Let

$$a_n = \begin{cases} 1 & \text{if every block of zeros in the} \\ & \text{binary expansion of } n \text{ has an} \\ & \text{even number of zeros in the} \\ & \text{block} \\ 0 & \text{otherwise.} \end{cases}$$

(For example, $a_{36} = 1$ because $36 = 100100_2$ and $a_{20} = 0$ because $20 = 10100_2$.) Prove that $\alpha^3 + x\alpha + 1 = 0$.

2008 年 A6 · 抽象代数与高等代数

年份 2008 题号 A6 入口群论

Prove that there exists a constant $c > 0$ such that in every nontrivial finite group G there exists a sequence of length at most $c \log |G|$ with the property that each element of G equals the product of some subsequence. (The elements of G in the sequence are not required to be distinct. A *subsequence* of a sequence is obtained by selecting some of the terms, not necessarily consecutive, without reordering them; for example, 4, 4, 2 is a subsequence of 2, 4, 6, 4, 2, but 2, 2, 4 is not.)

2011 年 A6 · 抽象代数与高等代数

年份 2011 题号 A6 入口群论

Let G be an abelian group with n elements, and let

$$\{g_1 = e, g_2, \dots, g_k\} \subsetneq G$$

be a (not necessarily minimal) set of distinct generators of G . A special die, which randomly selects one of the elements $g_1, g_2, \dots, g_k$ with equal probability, is rolled m times and the selected elements are multiplied to produce an element $g \in G$. Prove that there exists a real number $b \in (0, 1)$ such that

$$\lim_{m \rightarrow \infty} \frac{1}{b^{2m}} \sum_{x \in G} \left(\text{Prob}(g = x) - \frac{1}{n} \right)^2$$

is positive and finite.

2016 年 A6 · 抽象代数与高等代数

年份 2016 题号 A6 入口多项式

Find the smallest constant C such that for every real polynomial $P(x)$ of degree 3 that has a root in the interval $[0, 1]$,

$$\int_0^1 |P(x)| dx \leq C \max_{x \in [0, 1]} |P(x)|.$$

1985 年 B1 · 抽象代数与高等代数

年份 1985 题号 B1 入口多项式

Let k be the smallest positive integer for which there exist distinct integers m_1, m_2, m_3, m_4, m_5 such that the polynomial

$$p(x) = (x - m_1)(x - m_2)(x - m_3)(x - m_4)(x - m_5)$$

has exactly k nonzero coefficients. Find, with proof, a set of integers m_1, m_2, m_3, m_4, m_5 for which this minimum k is achieved.

2003 年 B1 · 抽象代数与高等代数

年份 2003 题号 B1 入口多项式

Do there exist polynomials $a(x), b(x), c(y), d(y)$ such that

$$1 + xy + x^2y^2 = a(x)c(y) + b(x)d(y)$$

holds identically?

2005 年 B1 · 抽象代数与高等代数

年份 2005 题号 B1 入口多项式

Find a nonzero polynomial $P(x, y)$ such that $P(\lfloor a \rfloor, \lfloor 2a \rfloor) = 0$ for all real numbers a . (Note: $\lfloor \nu \rfloor$ is the greatest integer less than or equal to ν .)

1989 年 B2 · 抽象代数与高等代数

年份 1989 题号 B2 入口群论

Let S be a non-empty set with an associative operation that is left and right cancellative ($xy = xz$ implies $y = z$, and $yx = zx$ implies $y = z$). Assume that for every a in S the set $\{a^n : n = 1, 2, 3, \dots\}$ is finite. Must S be a group?

1999 年 B2 · 抽象代数与高等代数

年份 1999 题号 B2 入口多项式

Let $P(x)$ be a polynomial of degree n such that $P(x) = Q(x)P''(x)$, where $Q(x)$ is a quadratic polynomial and $P''(x)$ is the second derivative of $P(x)$. Show that if $P(x)$ has at least two distinct roots then it must have n distinct roots.

2013 年 B2 · 抽象代数与高等代数

年份 2013 题号 B2 入口多项式

Let $C = \bigcup_{N=1}^{\infty} C_N$, where C_N denotes the set of those 'cosine polynomials' of the form

$$f(x) = 1 + \sum_{n=1}^N a_n \cos(2\pi nx)$$

for which:

- (i) $f(x) \geq 0$ for all real x , and
- (ii) $a_n = 0$ whenever n is a multiple of 3.

Determine the maximum value of $f(0)$ as f ranges through C , and prove that this maximum is attained.

1987 年 B3 · 抽象代数与高等代数

年份 1987 题号 B3 入口域/有限域

Let F be a field in which $1 + 1 \neq 0$. Show that the set of solutions to the equation $x^2 + y^2 = 1$ with x and y in F is given by $(x, y) = (1, 0)$ and

$$(x, y) = \left(\frac{r^2 - 1}{r^2 + 1}, \frac{2r}{r^2 + 1} \right)$$

where r runs through the elements of F such that $r^2 \neq -1$.

1990 年 B4 · 抽象代数与高等代数

年份 1990 题号 B4 入口群论

Let G be a finite group of order n generated by a and b . Prove or disprove: there is a sequence

$$g_1, g_2, g_3, \dots, g_{2n}$$

such that

- (1) every element of G occurs exactly twice, and
- (2) g_{i+1} equals $g_i a$ or $g_i b$ for $i = 1, 2, \dots, 2n$. (Interpret g_{2n+1} as g_1 .)

1992 年 B4 · 抽象代数与高等代数

年份 1992 题号 B4 入口多项式

Let $p(x)$ be a nonzero polynomial of degree less than 1992 having no nonconstant factor in common with $x^3 - x$. Let

$$\frac{d^{1992}}{dx^{1992}} \left(\frac{p(x)}{x^3 - x} \right) = \frac{f(x)}{g(x)}$$

for polynomials $f(x)$ and $g(x)$. Find the smallest possible degree of $f(x)$.

2007 年 B4 · 抽象代数与高等代数

年份 2007 题号 B4 入口多项式

Let n be a positive integer. Find the number of pairs P, Q of polynomials with real coefficients such that

$$(P(X))^2 + (Q(X))^2 = X^{2n} + 1$$

and $\deg P > \deg Q$.

2010 年 B4 · 抽象代数与高等代数

年份 2010 题号 B4 入口多项式

Find all pairs of polynomials $p(x)$ and $q(x)$ with real coefficients for which

$$p(x)q(x+1) - p(x+1)q(x) = 1.$$

2014 年 B4 · 抽象代数与高等代数

年份 2014 题号 B4 入口多项式

Show that for each positive integer n , all the roots of the polynomial

$$\sum_{k=0}^n 2^{k(n-k)} x^k$$

are real numbers.

1986 年 B5 · 抽象代数与高等代数

年份 1986 题号 B5 入口多项式

Let $f(x, y, z) = x^2 + y^2 + z^2 + xyz$. Let $p(x, y, z), q(x, y, z), r(x, y, z)$ be polynomials with real coefficients satisfying

$$f(p(x, y, z), q(x, y, z), r(x, y, z)) = f(x, y, z).$$

Prove or disprove the assertion that the sequence p, q, r consists of some permutation of $\pm x, \pm y, \pm z$, where the number of minus signs is 0 or 2.

1990 年 B5 · 抽象代数与高等代数

年份 1990 题号 B5 入口多项式

Is there an infinite sequence $a_0, a_1, a_2, \dots$ of nonzero real numbers such that for $n = 1, 2, 3, \dots$ the polynomial

$$p_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

has exactly n distinct real roots?

1991 年 B5 · 抽象代数与高等代数

年份 1991 题号 B5 入口域/有限域

Let p be an odd prime and let $\mathbb{Z}_p$ denote (the field of) integers modulo p . How many elements are in the set

$$\{x^2 : x \in \mathbb{Z}_p\} \cap \{y^2 + 1 : y \in \mathbb{Z}_p\}?$$

2005 年 B5 · 抽象代数与高等代数

年份 2005 题号 B5 入口多项式

Let $P(x_1, \dots, x_n)$ denote a polynomial with real coefficients in the variables $x_1, \dots, x_n$, and suppose that

$$\left(\frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_n^2} \right) P(x_1, \dots, x_n) = 0 \quad (\text{identically})$$

and that

$$x_1^2 + \cdots + x_n^2 \text{ divides } P(x_1, \dots, x_n).$$

Show that $P = 0$ identically.

2007 年 B5 · 抽象代数与高等代数

年份 2007 题号 B5 入口多项式

Let k be a positive integer. Prove that there exist polynomials $P_0(n), P_1(n), \dots, P_{k-1}(n)$ (which may depend on k) such that for any integer n ,

$$\left\lfloor \frac{n}{k} \right\rfloor^k = P_0(n) + P_1(n) \left\lfloor \frac{n}{k} \right\rfloor + \cdots + P_{k-1}(n) \left\lfloor \frac{n}{k} \right\rfloor^{k-1}.$$

($\lfloor a \rfloor$ means the largest integer $\leq a$.)

2014 年 B5 · 抽象代数与高等代数

年份 2014 题号 B5 入口群论

In the 75th annual Putnam Games, participants compete at mathematical games. Patniss and Keeta play a game in which they take turns choosing an element from the group of invertible $n \times n$ matrices with entries in the field $\mathbb{Z}/p\mathbb{Z}$ of integers modulo p , where n is a fixed positive integer and p is a fixed prime number. The rules of the game are:

- (1) A player cannot choose an element that has been chosen by either player on any previous turn.
- (2) A player can only choose an element that commutes with all previously chosen elements.
- (3) A player who cannot choose an element on his/her turn loses the game.

Patniss takes the first turn. Which player has a winning strategy? (Your answer may depend on n and p .)

2019 年 B5 · 抽象代数与高等代数

年份 2019 题号 B5 入口多项式

Let F_m be the m th Fibonacci number, defined by $F_1 = F_2 = 1$ and $F_m = F_{m-1} + F_{m-2}$ for all $m \geq 3$. Let $p(x)$ be the polynomial of degree 1008 such that $p(2n+1) = F_{2n+1}$ for $n = 0, 1, 2, \dots, 1008$. Find integers j and k such that $p(2019) = F_j - F_k$.

2024 年 B5 · 抽象代数与高等代数

年份 2024 题号 B5 入口多项式

Let k and m be positive integers. For a positive integer n , let $f(n)$ be the number of integer sequences $x_1, \dots, x_k, y_1, \dots, y_m, z$ satisfying $1 \leq x_1 \leq \dots \leq x_k \leq z \leq n$ and $1 \leq y_1 \leq \dots \leq y_m \leq z \leq n$. Show that $f(n)$ can be expressed as a polynomial in n with nonnegative coefficients.

1985 年 B6 · 抽象代数与高等代数

年份 1985 题号 B6 入口群论

Let G be a finite set of real $n \times n$ matrices $\{M_i\}$, $1 \leq i \leq r$, which form a group under matrix multiplication. Suppose that $\sum_{i=1}^r \text{tr}(M_i) = 0$, where $\text{tr}(A)$ denotes the trace of the matrix A . Prove that $\sum_{i=1}^r M_i$ is the $n \times n$ zero matrix.

1986 年 B6 · 抽象代数与高等代数

年份 1986 题号 B6 入口域/有限域

Suppose A, B, C, D are $n \times n$ matrices with entries in a field F , satisfying the conditions that AB^T and CD^T are symmetric and $AD^T - BC^T = I$. Here I is the $n \times n$ identity matrix, and if M is an $n \times n$ matrix, M^T is its transpose. Prove that $A^T D - C^T B = I$.

1987 年 B6 · 抽象代数与高等代数

年份 1987 题号 B6 入口域/有限域

Let F be the field of p^2 elements, where p is an odd prime. Suppose S is a set of $(p^2 - 1)/2$ distinct nonzero elements of F with the property that for each $a \neq 0$ in F , exactly one of a and $-a$ is in S . Let N be the number of elements in the intersection $S \cap \{2a : a \in S\}$. Prove that N is even.

2021 年 B6 · 抽象代数与高等代数

年份 2021 题号 B6 入口群论

Given an ordered list of $3N$ real numbers, we can *trim* it to form a list of N numbers as follows: We divide the list into N groups of 3 consecutive numbers, and within

each group, discard the highest and lowest numbers, keeping only the median.

Consider generating a random number X by the following procedure: Start with a list of 3^{2021} numbers, drawn independently and uniformly at random between 0 and 1. Then trim this list as defined above, leaving a list of 3^{2020} numbers. Then trim again repeatedly until just one number remains; let X be this number. Let μ be the expected value of $|X - \frac{1}{2}|$. Show that

$$\mu \geq \frac{1}{4} \left(\frac{2}{3} \right)^{2021}.$$

第 11 章

复分析

范畴要点：复数函数、复平面、多项式复根、单位圆、解析思想

1998 年 A2 · 复分析

年份 1998 题号 A2 入口平面几何

Let s be any arc of the unit circle lying entirely in the first quadrant. Let A be the area of the region lying below s and above the x -axis and let B be the area of the region lying to the right of the y -axis and to the left of s . Prove that $A + B$ depends only on the arc length, and not on the position, of s .

1989 年 A3 · 复分析

年份 1989 题号 A3 入口复数函数

Prove that if

$$11z^{10} + 10iz^9 + 10iz - 11 = 0,$$

then $|z| = 1$. (Here z is a complex number and $i^2 = -1$.)

2005 年 A3 · 复分析

年份 2005 题号 A3 入口复数函数

Let $p(z)$ be a polynomial of degree n all of whose zeros have absolute value 1 in the complex plane. Put $g(z) = p(z)/z^{n/2}$. Show that all zeros of $g'(z) = 0$ have absolute value 1.

2015 年 A3 · 复分析

年份 2015 题号 A3 入口复数函数

Compute

$$\log_2 \left(\prod_{a=1}^{2015} \prod_{b=1}^{2015} (1 + e^{2\pi i ab/2015}) \right)$$

Here i is the imaginary unit (that is, $i^2 = -1$).

2019 年 A3 · 复分析

年份 2019 题号 A3 入口复数函数

Given real numbers $b_0, b_1, \dots, b_{2019}$ with $b_{2019} \neq 0$, let $z_1, z_2, \dots, z_{2019}$ be the roots in the complex plane of the polynomial

$$P(z) = \sum_{k=0}^{2019} b_k z^k.$$

Let $\mu = (|z_1| + \dots + |z_{2019}|)/2019$ be the average of the distances from $z_1, z_2, \dots, z_{2019}$ to the origin. Determine the largest constant M such that $\mu \geq M$ for all choices of $b_0, b_1, \dots, b_{2019}$ that satisfy

$$1 \leq b_0 < b_1 < b_2 < \dots < b_{2019} \leq 2019.$$

1987 年 A4 · 复分析

年份 1987 题号 A4 入口复数函数

Let P be a polynomial, with real coefficients, in three variables and F be a function of two variables such that

$$P(ux, uy, uz) = u^2 F(y - x, z - x) \quad \text{for all real } x, y, z, u,$$

and such that $P(1, 0, 0) = 4$, $P(0, 1, 0) = 5$, and $P(0, 0, 1) = 6$. Also let A, B, C be complex numbers with $P(A, B, C) = 0$ and $|B - A| = 10$. Find $|C - A|$.

2023 年 A5 · 复分析

年份 2023 题号 A5 入口复数函数

For a nonnegative integer k , let $f(k)$ be the number of ones in the base 3 representation of k . Find all complex numbers z such that

$$\sum_{k=0}^{3^{1010}-1} (-2)^{f(k)} (z + k)^{2023} = 0.$$

1986 年 B2 · 复分析

年份 1986 题号 B2 入口复数函数

Prove that there are only a finite number of possibilities for the ordered triple $T = (x - y, y - z, z - x)$, where x, y, z are complex numbers satisfying the simultaneous equations

$$x(x - 1) + 2yz = y(y - 1) + 2zx = z(z - 1) + 2xy,$$

and list all such triples T .

2012 年 B2 · 复分析

年份 2012 题号 B2 入口级数/求和

Let P be a given (non-degenerate) polyhedron. Prove that there is a constant $c(P) > 0$ with the following property: If a collection of n balls whose volumes sum to V contains the entire surface of P , then $n > c(P)/V^2$.

2018 年 B2 · 复分析

年份 2018 题号 B2 入口复数函数

Let n be a positive integer, and let $f_n(z) = n + (n-1)z + (n-2)z^2 + \cdots + z^{n-1}$. Prove that f_n has no roots in the closed unit disk $\{z \in \mathbb{C} : |z| \leq 1\}$.

2020 年 B5 · 复分析

年份 2020 题号 B5 入口复数函数

For $j \in \{1, 2, 3, 4\}$, let z_j be a complex number with $|z_j| = 1$ and $z_j \neq 1$. Prove that

$$3 - z_1 - z_2 - z_3 - z_4 + z_1 z_2 z_3 z_4 \neq 0.$$

2012 年 B6 · 复分析

年份 2012 题号 B6 入口整除与同余

Let p be an odd prime number such that $p \equiv 2 \pmod{3}$. Define a permutation π of the residue classes modulo p by $\pi(x) \equiv x^3 \pmod{p}$. Show that π is an even permutation if and only if $p \equiv 3 \pmod{4}$.

第 12 章

拓扑、泛函分析与高级分析

范畴要点： 拓扑性质、度量空间、函数空间、算子、范数、高级不等式

2005 年 B6 · 拓扑、泛函分析与高级分析

年份 2005 题号 B6 入口积分

Let S_n denote the set of all permutations of the numbers $1, 2, \dots, n$. For $\pi \in S_n$, let $\sigma(\pi) = 1$ if π is an even permutation and $\sigma(\pi) = -1$ if π is an odd permutation. Also, let $\nu(\pi)$ denote the number of fixed points of π . Show that

$$\sum_{\pi \in S_n} \frac{\sigma(\pi)}{\nu(\pi) + 1} = (-1)^{n+1} \frac{n}{n+1}.$$

第 A 章

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1997	B2	实分析	411
1997	B3	初等数论	56
1997	B4	单变量微积分与初等分析	188
1997	B5	初等数论	61
1997	B6	几何与多变量微积分	326
1998	A1	几何与多变量微积分	263
1998	A2	复分析	480
1998	A3	单变量微积分与初等分析	102
1998	A4	初等数论	32
1998	A5	单变量微积分与初等分析	128
1998	A6	几何与多变量微积分	292
1998	B1	单变量微积分与初等分析	148
1998	B2	几何与多变量微积分	305
1998	B3	几何与多变量微积分	311
1998	B4	单变量微积分与初等分析	189
1998	B5	初等数论	62
1998	B6	单变量微积分与初等分析	209

年份	题号	范畴	本书顺序
1999	A1	抽象代数与高等代数	431
1999	A2	抽象代数与高等代数	434
1999	A3	单变量微积分与初等分析	103
1999	A4	单变量微积分与初等分析	111
1999	A5	抽象代数与高等代数	443
1999	A6	单变量微积分与初等分析	137
1999	B1	几何与多变量微积分	298
1999	B2	抽象代数与高等代数	460
1999	B3	单变量微积分与初等分析	175
1999	B4	单变量微积分与初等分析	190
1999	B5	线性代数	252
1999	B6	初等数论	67
2000	A1	单变量微积分与初等分析	75
2000	A2	单变量微积分与初等分析	91
2000	A3	几何与多变量微积分	280
2000	A4	实分析	398
2000	A5	几何与多变量微积分	289
2000	A6	初等数论	41
2000	B1	单变量微积分与初等分析	149
2000	B2	初等数论	52
2000	B3	单变量微积分与初等分析	176
2000	B4	实分析	418
2000	B5	高等组合与离散结构	382
2000	B6	几何与多变量微积分	327
2001	A1	抽象代数与高等代数	432
2001	A2	概率	331
2001	A3	初等数论	27
2001	A4	几何与多变量微积分	285
2001	A5	单变量微积分与初等分析	129
2001	A6	几何与多变量微积分	293
2001	B1	高等组合与离散结构	372
2001	B2	单变量微积分与初等分析	160
2001	B3	单变量微积分与初等分析	177
2001	B4	基础证明、集合与初等逻辑	7
2001	B5	实分析	421
2001	B6	单变量微积分与初等分析	210
2002	A1	抽象代数与高等代数	433
2002	A2	几何与多变量微积分	273
2002	A3	初等组合与离散数学	12
2002	A4	线性代数	227
2002	A5	单变量微积分与初等分析	130
2002	A6	初等数论	42
2002	B1	概率	347
2002	B2	高等组合与离散结构	373

年份	题号	范畴	本书顺序
2002	B3	单变量微积分与初等分析	178
2002	B4	概率	355
2002	B5	初等数论	63
2002	B6	线性代数	255
2003	A1	单变量微积分与初等分析	76
2003	A2	单变量微积分与初等分析	92
2003	A3	单变量微积分与初等分析	104
2003	A4	单变量微积分与初等分析	112
2003	A5	几何与多变量微积分	290
2003	A6	高等组合与离散结构	367
2003	B1	抽象代数与高等代数	457
2003	B2	单变量微积分与初等分析	161
2003	B3	初等数论	57
2003	B4	单变量微积分与初等分析	191
2003	B5	几何与多变量微积分	322
2003	B6	实分析	427
2004	A1	基础证明、集合与初等逻辑	1
2004	A2	几何与多变量微积分	274
2004	A3	线性代数	223
2004	A4	单变量微积分与初等分析	113
2004	A5	概率	339
2004	A6	实分析	402
2004	B1	初等数论	45
2004	B2	单变量微积分与初等分析	162
2004	B3	几何与多变量微积分	312
2004	B4	几何与多变量微积分	317
2004	B5	基础证明、集合与初等逻辑	8
2004	B6	实分析	428
2005	A1	单变量微积分与初等分析	77
2005	A2	几何与多变量微积分	275
2005	A3	复分析	482
2005	A4	线性代数	228
2005	A5	单变量微积分与初等分析	131
2005	A6	概率	344
2005	B1	抽象代数与高等代数	458
2005	B2	单变量微积分与初等分析	163
2005	B3	实分析	415
2005	B4	单变量微积分与初等分析	192
2005	B5	抽象代数与高等代数	471
2005	B6	拓扑、泛函分析与高级分析	492
2006	A1	几何与多变量微积分	264
2006	A2	初等数论	22
2006	A3	初等数论	28
2006	A4	初等组合与离散数学	15

年份	题号	范畴	本书顺序
2006	A5	单变量微积分与初等分析	132
2006	A6	概率	345
2006	B1	几何与多变量微积分	299
2006	B2	单变量微积分与初等分析	164
2006	B3	几何与多变量微积分	313
2006	B4	几何与多变量微积分	318
2006	B5	实分析	422
2006	B6	单变量微积分与初等分析	211
2007	A1	单变量微积分与初等分析	78
2007	A2	几何与多变量微积分	276
2007	A3	概率	332
2007	A4	初等数论	33
2007	A5	抽象代数与高等代数	444
2007	A6	几何与多变量微积分	294
2007	B1	初等数论	46
2007	B2	实分析	412
2007	B3	单变量微积分与初等分析	179
2007	B4	抽象代数与高等代数	465
2007	B5	抽象代数与高等代数	472
2007	B6	概率	360
2008	A1	单变量微积分与初等分析	79
2008	A2	线性代数	220
2008	A3	概率	333
2008	A4	单变量微积分与初等分析	114
2008	A5	抽象代数与高等代数	445
2008	A6	抽象代数与高等代数	453
2008	B1	几何与多变量微积分	300
2008	B2	单变量微积分与初等分析	165
2008	B3	几何与多变量微积分	314
2008	B4	初等数论	60
2008	B5	实分析	423
2008	B6	高等组合与离散结构	384
2009	A1	几何与多变量微积分	265
2009	A2	实分析	388
2009	A3	线性代数	224
2009	A4	基础证明、集合与初等逻辑	5
2009	A5	抽象代数与高等代数	446
2009	A6	实分析	403
2009	B1	初等数论	47
2009	B2	高等组合与离散结构	374
2009	B3	高等组合与离散结构	376
2009	B4	线性代数	245
2009	B5	实分析	424
2009	B6	单变量微积分与初等分析	212

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2010	A1	单变量微积分与初等分析	80
2010	A2	单变量微积分与初等分析	93
2010	A3	单变量微积分与初等分析	105
2010	A4	初等数论	34
2010	A5	抽象代数与高等代数	447
2010	A6	实分析	404
2010	B1	单变量微积分与初等分析	150
2010	B2	几何与多变量微积分	306
2010	B3	单变量微积分与初等分析	180
2010	B4	抽象代数与高等代数	466
2010	B5	单变量微积分与初等分析	202
2010	B6	线性代数	256
2011	A1	几何与多变量微积分	266
2011	A2	实分析	389
2011	A3	单变量微积分与初等分析	106
2011	A4	线性代数	229
2011	A5	实分析	399
2011	A6	抽象代数与高等代数	454
2011	B1	单变量微积分与初等分析	151
2011	B2	初等数论	53
2011	B3	实分析	416
2011	B4	线性代数	246
2011	B5	单变量微积分与初等分析	203
2011	B6	初等数论	68
2012	A1	概率	330
2012	A2	抽象代数与高等代数	435
2012	A3	实分析	392
2012	A4	单变量微积分与初等分析	115
2012	A5	抽象代数与高等代数	448
2012	A6	几何与多变量微积分	295
2012	B1	实分析	407
2012	B2	复分析	488
2012	B3	高等组合与离散结构	377
2012	B4	单变量微积分与初等分析	193
2012	B5	实分析	425
2012	B6	复分析	491
2013	A1	几何与多变量微积分	267
2013	A2	单变量微积分与初等分析	94
2013	A3	单变量微积分与初等分析	107
2013	A4	几何与多变量微积分	286
2013	A5	几何与多变量微积分	291
2013	A6	单变量微积分与初等分析	138
2013	B1	单变量微积分与初等分析	152
2013	B2	抽象代数与高等代数	461

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2013	B3	高等组合与离散结构	378
2013	B4	实分析	419
2013	B5	基础证明、集合与初等逻辑	9
2013	B6	几何与多变量微积分	328
2014	A1	初等数论	17
2014	A2	线性代数	221
2014	A3	基础证明、集合与初等逻辑	4
2014	A4	概率	335
2014	A5	初等数论	37
2014	A6	线性代数	234
2014	B1	初等数论	48
2014	B2	单变量微积分与初等分析	166
2014	B3	线性代数	240
2014	B4	抽象代数与高等代数	467
2014	B5	抽象代数与高等代数	473
2014	B6	实分析	429
2015	A1	几何与多变量微积分	268
2015	A2	初等数论	23
2015	A3	复分析	483
2015	A4	单变量微积分与初等分析	116
2015	A5	初等数论	38
2015	A6	线性代数	235
2015	B1	实分析	408
2015	B2	初等数论	54
2015	B3	线性代数	241
2015	B4	几何与多变量微积分	319
2015	B5	高等组合与离散结构	383
2015	B6	初等数论	69
2016	A1	初等数论	18
2016	A2	单变量微积分与初等分析	95
2016	A3	实分析	393
2016	A4	几何与多变量微积分	287
2016	A5	抽象代数与高等代数	449
2016	A6	抽象代数与高等代数	455
2016	B1	实分析	409
2016	B2	单变量微积分与初等分析	167
2016	B3	几何与多变量微积分	315
2016	B4	线性代数	247
2016	B5	单变量微积分与初等分析	204
2016	B6	单变量微积分与初等分析	213
2017	A1	单变量微积分与初等分析	81
2017	A2	初等数论	24
2017	A3	实分析	394
2017	A4	抽象代数与高等代数	441

年份	题号	范畴	本书顺序
2017	A5	概率	340
2017	A6	高等组合与离散结构	368
2017	B1	几何与多变量微积分	301
2017	B2	单变量微积分与初等分析	168
2017	B3	单变量微积分与初等分析	181
2017	B4	单变量微积分与初等分析	194
2017	B5	几何与多变量微积分	323
2017	B6	初等数论	70
2018	A1	单变量微积分与初等分析	82
2018	A2	线性代数	222
2018	A3	单变量微积分与初等分析	108
2018	A4	抽象代数与高等代数	442
2018	A5	实分析	400
2018	A6	几何与多变量微积分	296
2018	B1	线性代数	237
2018	B2	复分析	489
2018	B3	单变量微积分与初等分析	182
2018	B4	单变量微积分与初等分析	195
2018	B5	实分析	426
2018	B6	单变量微积分与初等分析	214
2019	A1	单变量微积分与初等分析	83
2019	A2	几何与多变量微积分	277
2019	A3	复分析	484
2019	A4	几何与多变量微积分	288
2019	A5	抽象代数与高等代数	450
2019	A6	实分析	405
2019	B1	几何与多变量微积分	302
2019	B2	单变量微积分与初等分析	169
2019	B3	线性代数	242
2019	B4	概率	356
2019	B5	抽象代数与高等代数	474
2019	B6	几何与多变量微积分	329
2020	A1	初等数论	19
2020	A2	单变量微积分与初等分析	96
2020	A3	实分析	395
2020	A4	概率	336
2020	A5	单变量微积分与初等分析	133
2020	A6	单变量微积分与初等分析	139
2020	B1	初等数论	49
2020	B2	高等组合与离散结构	375
2020	B3	概率	352
2020	B4	单变量微积分与初等分析	196
2020	B5	复分析	490
2020	B6	单变量微积分与初等分析	215

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2021	A1	几何与多变量微积分	269
2021	A2	单变量微积分与初等分析	97
2021	A3	几何与多变量微积分	281
2021	A4	单变量微积分与初等分析	117
2021	A5	初等数论	39
2021	A6	初等数论	43
2021	B1	概率	348
2021	B2	单变量微积分与初等分析	170
2021	B3	几何与多变量微积分	316
2021	B4	单变量微积分与初等分析	197
2021	B5	线性代数	253
2021	B6	抽象代数与高等代数	479
2022	A1	单变量微积分与初等分析	84
2022	A2	抽象代数与高等代数	436
2022	A3	初等数论	29
2022	A4	概率	337
2022	A5	高等组合与离散结构	363
2022	A6	单变量微积分与初等分析	140
2022	B1	初等数论	50
2022	B2	单变量微积分与初等分析	171
2022	B3	高等组合与离散结构	379
2022	B4	单变量微积分与初等分析	198
2022	B5	概率	358
2022	B6	实分析	430
2023	A1	单变量微积分与初等分析	85
2023	A2	抽象代数与高等代数	437
2023	A3	实分析	396
2023	A4	高等组合与离散结构	362
2023	A5	复分析	486
2023	A6	高等组合与离散结构	369
2023	B1	概率	349
2023	B2	单变量微积分与初等分析	172
2023	B3	概率	353
2023	B4	实分析	420
2023	B5	初等数论	64
2023	B6	线性代数	257
2024	A1	单变量微积分与初等分析	86
2024	A2	抽象代数与高等代数	438
2024	A3	初等组合与离散数学	13
2024	A4	初等数论	35
2024	A5	概率	341
2024	A6	线性代数	236
2024	B1	单变量微积分与初等分析	153
2024	B2	几何与多变量微积分	307

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2024	B3	单变量微积分与初等分析	183
2024	B4	概率	357
2024	B5	抽象代数与高等代数	475
2024	B6	单变量微积分与初等分析	216
2025	A1	初等数论	20
2025	A2	单变量微积分与初等分析	98
2025	A3	初等数论	30
2025	A4	线性代数	230
2025	A5	高等组合与离散结构	364
2025	A6	初等数论	44
2025	B1	几何与多变量微积分	303
2025	B2	几何与多变量微积分	308
2025	B3	初等数论	58
2025	B4	线性代数	248
2025	B5	初等数论	65
2025	B6	单变量微积分与初等分析	217